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**A NEAR ZERO FIELD MAGNETORESISTANCE STUDY OF SPIN-DEPENDENT
CURRENTS IN SEMICONDUCTOR AND DIELECTRIC MATERIALS**

A Dissertation in

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by

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ABSTRACT

Near zero field magnetoresistance (NZFMR) is a new spin-based spectroscopic technique that offers some of the analytical power of electrically detected magnetic resonance (EDMR) when studying performance limiting defects in semiconductors and dielectrics. NZFMR measurements are far simpler and considerably less complex than other spin-based spectroscopic techniques. NZFMR is based upon the fact that spin-based conductivity of materials can change as an externally applied magnetic field is swept through zero. This results in a spectrum somewhat similar to those observed in electron paramagnetic resonance measurements (EPR). Attempts to interpret this phenomenon are currently being developed. This work focuses on EDMR and NZFMR measurements on three model systems, amorphous hydrogenated carbon (a-C:H), amorphous diamond-like carbon (DLC), and amorphous hydrogenated silicon carbide (a-SiC:H). A major focus of this work is the study of stress induced leakage current (SILC) and time dependent dielectric breakdown (TDDB). These measurements were then used to inform the development of the computational models in order to determine what physical information can be extracted from NZFMR spectra and to provide insight into TDDB. The current in the amorphous thin film samples is primarily due to variable range hopping (VRH) or trap assisted tunneling (TAT). The amorphous carbon samples (both a-C:H and DLC) exhibited unstable electrical characteristics under constant biasing conditions over time. This suggests that there are physical changes occurring in these samples over time. The a-C:H samples exhibit a behavior indicative of TDDB with a decrease in current over time along with a reduction in EDMR and NZFMR signals and trapped space charge. The DLC shows an increase in current due to stress along with a change in the shape of the EDMR and NZFMR signals, likely indicating the introduction of new defects. Some low field EDMR and NZFMR results are also reported in

SiGe alloy p-MOSFETs. A new NZFMR spectrometer was built into a wafer probing station to allow for non-destructive measurements to be made on larger samples, rather than dicing wafers and mounting samples to custom made holders, allowing for a higher throughput of EDMR and NZFMR measurements.

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Chapter 1

Introduction

Near zero field magneto-resistance (NZFMR) is a new spin based spectroscopic technique that may offer much of the analytical power of electrically detected magnetic resonance (EDMR). This technique is simpler and more cost effective than EDMR and may be utilized to study electrically active point defects in a wide variety of semiconductor and dielectric materials that may contribute to reliability problems such as stress induced leakage currents (SILC) and time dependent dielectric breakdown (TDDB). NZFMR also offers the opportunity to study three-dimensional device architectures that block the microwave or radio frequency radiation required for EDMR. The theoretical aspects of NZFMR are only starting to be understood as new computational models are being developed to analyze NZFMR spectra. Well-developed computational models would allow for the extraction of certain physical parameters related to the defects in the material such as hyperfine coupling constants and site to site hopping rates, as well as dissociation rates in a wide variety of material systems.

Several model systems, amorphous carbon and amorphous silicon carbide, were studied in order to learn about the electronic transport as well as SILC and TDDB processes in the materials. Comparisons to a relatively crude and newly developed NZFMR computational model allowed for the extraction of reasonable values for the parameters discussed previously given the limitations of the models and the compositions of the samples. The parameters involved in the model, such as hopping and dissociation rates, have been linked to physical phenomena involved in transport in the structures. This work shows that a relatively crude model, based upon the stochastic quantum Liouville equation (SLE) can generate NZFMR simulations in fairly close agreement with the observed NZFMR spectra. The rudimentary SLE

model results are in reasonable agreement with the observed NZFMR spectra. The results of this study will inform further development of the computational NZFMR models in order to analyze other systems.

Chapter 1.1 Reliability Problems; Variable Range Hopping, Stress Induced Leakage Currents, and Time Dependent Dielectric Breakdown in Semiconductors

Variable range hopping (VRH) [1], [2], also known as trap assisted tunneling (TAT), is a phenomenon in which a dielectric or semiconductor material has a high enough defect density to allow for electronic transport through the dielectric or semiconductor via site to site tunneling. Electrons and/or holes tunnel through the dielectric or semiconductor from site to site until reaching the other side. Dielectric and insulator materials are typically utilized to prevent the flow of charge carriers across them, so a large hopping current can severely inhibit the performance of a device in which they are a part. The band diagram of such a material and the transport path through it can be idealized to look like Figure 1.1 below.

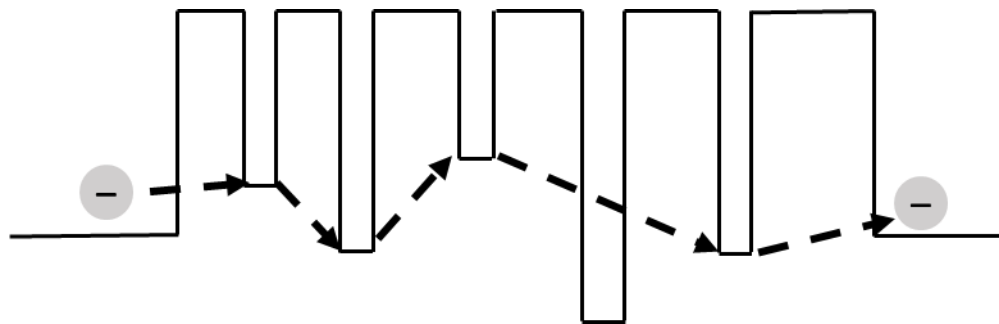


Figure 1.1: Idealized band diagram of potential wells (traps) in a dielectric which allow charge carriers to tunnel from site to site. Only defects within certain energy levels of the electron can be accessed.

VRH is a tunneling process which depends on the depth and separation of the wells in the band structure as well as upon the temperature. The primary defects of concern are those within approximately $2kT$ of the Fermi level, where k is the Boltzmann constant and T is the absolute temperature. Sites well below that level will have too high a probability of being occupied and sites well above will require the electron to gain too much energy to jump. At low voltages, this process is fairly ohmic and scales with an exponential of the reciprocal of the one-fourth power of the temperature in the early model proposed by Mott that is of the form $\exp(-\frac{A}{T^{\frac{1}{4}}})$, where A is a constant. [1], [2] The $\exp(-\frac{A}{T^{\frac{1}{4}}})$ dependency is not always observed. Efros and Shklovskii [3] have shown that the temperature dependent conductivity scales with $\exp(-\frac{A}{T^{\frac{1}{2}}})$ when accounting for a ‘soft’ gap which arises from the Coulomb interaction between localized electrons.

Thin dielectric materials can begin to degrade under stress if the electric fields in the dielectric approach breakdown causing leakage currents to increase over time. One aspect of this phenomenon is typically referred to as stress induced leakage current (SILC) [4]–[8]; it is one of the primary reliability issues with device scaling as the effect is greater for thinner oxides [5]–[8]. Studies have shown SILC to be caused by an increase in trap assisted tunneling as localized defects are created within the oxide [4]–[7] as well as potentially near the interface [5]. Recent models dictate that inelastic tunneling through the oxide is required to fit the data from SILC experiments [8]–[10]. Most studies agree that trapping defect densities increase with increased tunneling current in a positive feedback loop, making SILC a bigger problem for thinner oxides that are more susceptible to tunneling in the first place. [5]–[8]

Another reliability problem in devices with dielectric materials is known as time dependent dielectric breakdown (TDDB), a phenomenon in which dielectric materials operating

at electric fields below their breakdown field will suddenly and catastrophically fail with little to no warning. TDDB is not fully understood but it is certain that charge carriers tunneling into and out of the dielectric may play a role. This phenomenon has been studied extensively and has been found in a variety of material systems [11]–[15]. A curious aspect of this problem is the frequent observation of an initial reduction in leakage current through the dielectric before the material breaks down and becomes conductive. Devices that undergo TDDB go through several types of breakdown during stressing, soft and hard breakdown.[15] In soft breakdown, leakage through the oxide increases slightly and somewhat erratically; in hard breakdown the insulating material essentially becomes conductive [15].

There are several widely utilized models in the literature describing TDDB. One model for this phenomenon has been known as the “E-model”, because the model predicts that the log of time to failure is linearly proportional to the electric field across the dielectric, as shown in equation 1.1.1. [16]–[18]

$$\ln(TF) \propto \frac{\Delta H_0}{k_B T} - \gamma E_{ox} \quad (1.1.1)$$

where ΔH_0 is the enthalpy of activation for oxide breakdown, E_{ox} is the electric field in the oxide, k_B is Boltzmann’s constant, T is the absolute temperature, and γ is the field acceleration parameter. In this model, the electric field across the oxide generates traps, where overlap between the wave functions of trapped charges eventually create a conduction subband. The increased current and subsequent Joule heating cause the material to break down [17]. A competing theory for the mechanisms involved in TDDB is known as the “current-based” or “1/E model” [18]–[20]. This model suggests that breakdown is due to hole injection into the oxide and the log of time to failure is proportional to the inverse of the electric field. Recently it

has been proposed that the best explanation of TDDB comes from a combination of both the electric field model and the current-based model [11], [21].

Another potential explanation for TDDB is that electrons tunnel into deep level defects near the oxide/conductor interface, raising the effective barrier height for other electrons attempting to tunnel through, temporarily reducing leakage while simultaneously increasing the electric field in the rest of the oxide. This model provides an explanation for the often-observed initial reduction in dielectric leakage current over time, preceding the eventual catastrophic failure of TDDB. More electrons may continue to tunnel into the dielectric over time, raising the barrier further and increasing the local electric field to levels capable of breaking, for example, Si-O bonds [22].

Chapter 1.2 Electrical Measurements

Many electrically active defects that contribute to these reliability problems in metal-oxide-semiconductor (MOS) devices can be detected using purely electrical measurements. These measurements include, but are not limited to, capacitance measurements [23], [24], the Fitzgerald-Grove gated diode or DCIV measurement [25], and charge pumping [26]–[29]. Capacitance versus voltage measurements on MOS systems can determine the number of interface states and, with assumptions, also determine the amount of oxide charge present in a material by comparing the measured curve to a calculated ideal curve (with no defects). In silicon dioxide on silicon structures, shifts in the curve corresponding to mid-silicon band gap fermi energy are due to oxide charge while interface states cause the curve to stretch [23].

Gated diode direct current versus voltage (DCIV) measurements are fairly simple way to evaluate stressing damage in metal-oxide-semiconductor field effect transistors (MOSFET). In DCIV measurements, the source and drain are forward biased relative to the body and the gate is biased to keep the device in depletion. A recombination current will occur at the interface and will be at a maximum value when the number of electrons and holes are equal, achieved when the quasi-Fermi levels are symmetrically located about the mid-gap at the interface. The increase in current due to recombination provides information regarding the number of interface defects although the method is only sensitive to deep level defects which facilitate recombination [25].

Interface states can also be measured using charge pumping which is sensitive to defects with a wider range of energy levels than DCIV [26]–[29]. In this measurement, the source and drain of a MOSFET are grounded or slightly reversed biased with respect to the channel. Current is measured through the substrate or body as a trapezoidal waveform is applied to the gate to bring the device from accumulation to inversion and back again in quick succession. Interface states are filled and emptied when the Fermi level is swept through the bandgap resulting in a somewhat steady state current through the body due to recombination events at the interface. The recombination current is maximized when the traps are filled and emptied completely each cycle. This occurs when the trapezoidal gate pulse cycles beyond the flat band and threshold voltages. The relationship between the charge pumping current and the defect density when this condition is met is given as [28]:

$$I_{CP}^{max} = qf_{CP}A_{eff}\overline{D_{it}}\Delta E_{CP} \quad (1.2.1)$$

where q is the electric charge, f_{CP} is the frequency at which the gate is switched from accumulation to inversion and back, A_{eff} is the effective gate area, $\overline{D_{it}}$ is the interface trap

density per unit energy, and ΔE_{CP} is the energy window probed by the measurement (typically a large portion of the bandgap). It is important that the time between high and low voltage levels is sufficient to allow free carriers to diffuse away from the interface. In order for equation 1.2.1 to be correct, enough time must also be provided at inversion for the charge carriers to diffuse from the source and drain of the MOSFET to reach and recombine with the trapped charges at the middle of the channel. Both DCIV and charge pumping measurements rely on recombination events which are inherently spin dependent.

Chapter 1.3 Electron Paramagnetic Resonance Family of Measurements

One of the most powerful tools to study defects in semiconductor and dielectric materials is electron paramagnetic resonance (EPR) [30]–[32]. EPR is sensitive to paramagnetic defects in a sample and is used to identify the physical and chemical nature of these defects. The simplest case of a free electron has a well understood magnetic resonance condition; however, an electron's resonance conditions in real materials will deviate from the simplest case mostly due to the effects of spin orbit coupling and electron-nuclear hyperfine interactions [30]–[32]. Spin orbit coupling alters the relationship between the resonance field and the resonance frequency. It can also cause broadening of signals at higher magnetic fields in amorphous and disordered systems. Hyperfine interactions are in effect, the response due to local magnetic fields from nearby magnetic nuclei which cause the measured spectrum of an EPR measurement to be split into two or more lines (or resonance peaks). Broadening of lines due to hyperfine interactions is, to zero order, a constant, independent of resonance field and frequency [33].

Conventional EPR requires a minimum of approximately 10^{10} defects in a sample. Most modern devices have many orders of magnitude fewer defects and, while EPR is sensitive to all

paramagnetic defects, not all defects are electrically active. A different measurement scheme was introduced to get around these limits known as electrically detected magnetic resonance (EDMR) [34], [35]. EDMR measures changes in a sample's current or voltage as defects within it undergoes magnetic resonance.

EDMR requires approximately 10^7 fewer defects in a sample compared to conventional EPR, allowing it to be utilized on fully processed devices, and is only sensitive to electrically active defects, or defects affecting the electrical properties of the device. The size of the EDMR response of a device is also essentially resonant field and frequency independent [35], [36] allowing for measurements to be made across a wide range of resonance fields. This allows one to separate the effects of spin orbit coupling from hyperfine interactions because the effects of spin orbit coupling diminish as the resonance field and frequency are lowered.

There is a response related to EDMR which may be observed as the magnetic field approaches zero [37]–[43]. This is known as the near zero field magnetoresistance (NZFMR) effect. NZFMR measurements are not as complex as EPR or EDMR because they do not require all of the hardware necessary for magnetic resonance. The overall size of the NZFMR signals are often larger than EDMR signals; however, the phenomenon is not yet fully understood. Several studies have measured the NZFMR response in a variety of materials [37]–[42], [44]. The majority of these studies have also utilized resonance measurements in conjunction with the NZFMR measurements. The NZFMR response has also been utilized to as the basis for a high sensitivity, self-calibrating magnetic field sensor [43].

Chapter 2 Electron Paramagnetic Resonance and Electrically Detected Magnetic Resonance

Chapter 2.1 Electron Paramagnetic Resonance

Chapter 2.1.1 Electron Spin

Electrons and some atomic nuclei have an intrinsic angular momentum known as a “spin”. Electrons’ spins are described by the quantum spin vector $\mathbf{S}$, which contains a magnitude and a direction. Electrons’ spins are said to be either ‘up’ or ‘down’, and are described by the quantum number $m_s = +\frac{1}{2}$ or $m_s = -\frac{1}{2}$, respectively. This intrinsic spin gives rise to an electron’s magnetic moment

$$\boldsymbol{\mu}_e = g_e \mu_B \mathbf{S} \quad (2.1.1)$$

where $\boldsymbol{\mu}_e$ is the magnetic moment of the electron, g_e is the Landé factor, $g = 2.002319..$ and the Bohr magneton is

$$\mu_B = \frac{|e|\hbar}{4\pi m_e} = 9.27400994 \dots \times 10^{-24} \frac{J}{T} \quad (2.1.2)$$

where e is the electron charge, Planck’s constant $\hbar = 6.626 \times 10^{-34} Js$, and m_e is the electron mass. The electron magnetic moment creates an energy splitting between the spin up and spin down state when the electron spin is placed in an applied magnetic field, $\mathbf{B}$. For the simplest case of an electron otherwise unperturbed by its surroundings, the individual energy levels can be computed from

$$E = -\boldsymbol{\mu}_e \cdot \mathbf{B} = g\mu_B \mathbf{S} \cdot \mathbf{B}. \quad (2.1.3)$$

We choose a reference frame that aligns the magnetic field, $\mathbf{B}$, with the z-axis to simplify the equation to

$$E = -\boldsymbol{\mu}_e \cdot \mathbf{B} = g_e \mu_B S_z B_0 \quad (2.1.4)$$

where S_z is the magnitude of the electron spin vector projected onto the z-axis and B_0 is the strength of the applied magnetic field in the z-axis direction. The magnitude of S_z is either $+\frac{1}{2}$ or $-\frac{1}{2}$ for the spin up and spin down states respectively. The energy difference between these up and down states is therefore described by the equation

$$\Delta E = g_e \mu_B B_0 \quad (2.1.5)$$

and the concept it illustrated in Figure 2.1.

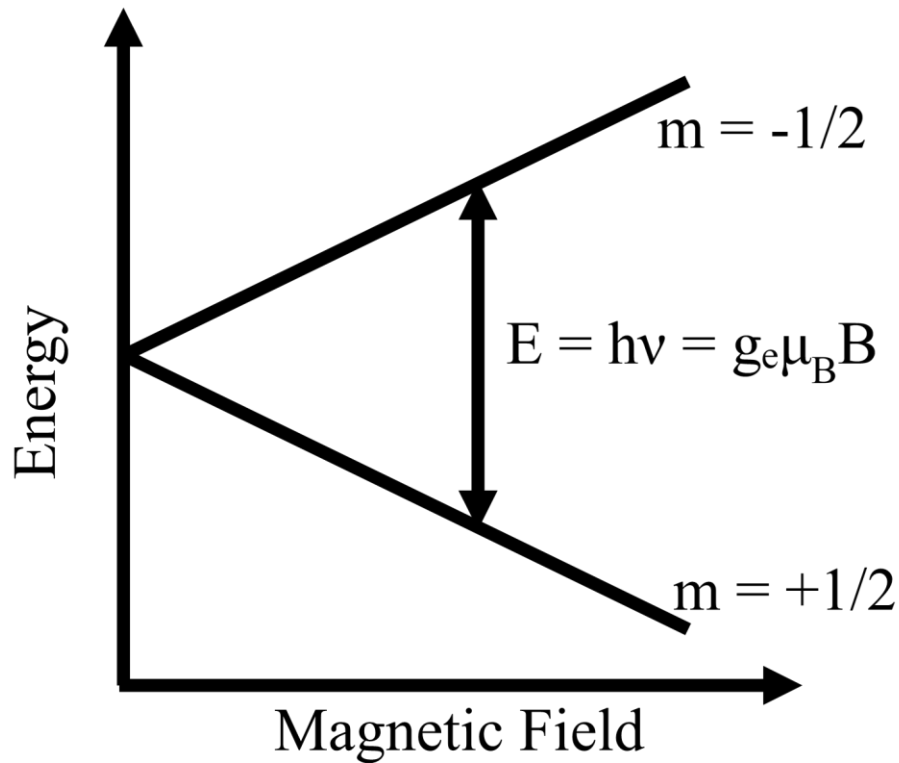


Figure 2.1: Schematic of energy separation between the spin up ($m = +\frac{1}{2}$) and spin down ($m = -\frac{1}{2}$) in an externally applied magnetic field, $\mathbf{B}$. This is typically referred to as Zeeman splitting.

The energy level splitting which occurs when an electron spin is placed in a magnetic field is called the Zeeman effect. The energy splitting is proportional to the applied magnetic field. Electrons in one of these states can absorb or release energy to switch, or ‘flip’, between states. In EPR, this is achieved by exposing the electron spin to electromagnetic radiation. The energy required to flip an isolated electron is described as

$$\Delta E = h\nu = g_e\mu_B B_0 \quad (2.1.6)$$

where h is Planck’s constant and ν is the frequency of the electromagnetic radiation. Equation 2.1.6 describes the simplest case of a free electron isolated from its surroundings, however, unpaired electrons in real materials will deviate from this idealized situation due to their surroundings. Measuring the deviations from this ideal situation gives EPR its analytical power to study the paramagnetic sites in real materials. If no magnetic nuclei are nearby, the EPR condition is given by

$$\Delta E = h\nu = g\mu_B B_0 \quad (2.1.7)$$

where the free electron g from equation 2.1.6 is replaced by an orientation dependent g factor which is best described as a second rank tensor and is one of the tools used to capture deviations from the simplest case of a free electron. More complex expressions can occur under some circumstances not relevant to the work described in this thesis.

In conventional EPR, a sample with paramagnetic centers is placed into a microwave cavity with a very high quality, or Q factor. A standing wave is set up in this cavity to provide an oscillating magnetic field, B_1 , with a frequency of ν . The cavity is placed in a large slowly varying magnetic field, typically provided by an electromagnet which sweeps the applied magnetic field through the resonant field. Resonance is detected when the applied microwave radiation is absorbed by the sample when the resonance condition is met, causing the Q factor of

the cavity to change. This change in Q causes a change in the reflected power back to a diode in the microwave bridge. An absorption spectrum is measured when the reflected microwave power is plotted against the slowly varying magnetic field as seen in Figure 2.2 a). A lock-in detection scheme is utilized to increase the signal to noise ratio by added low frequency modulation coils to the microwave cavity, if the absorption spectrum has a gaussian-like line shape, this results in a derivative like spectrum seen in Figure 2.2 b).

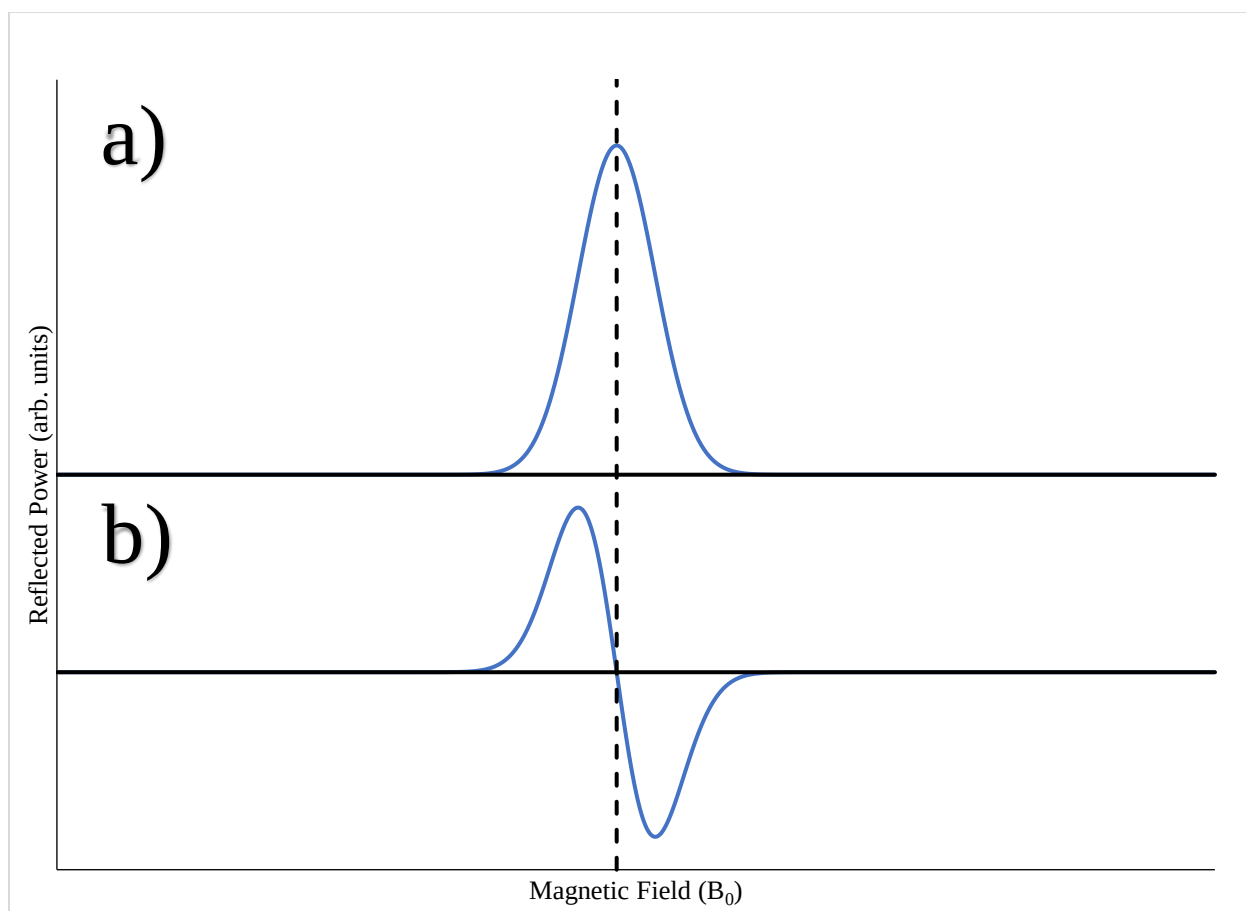


Figure 2.2: a) Simulated EPR absorption response of electron in a paramagnetic center with a fixed microwave frequency and swept magnetic field, B_0 . b) Simulated EPR response using lock-in detection scheme. Modulation and lock-in detection produces an approximate derivative of the absorption response.

It is important to understand how the spins respond to the microwave radiation in the cavity. Electrons in the low energy state can absorb a photon and flip to the high energy state, while electrons in the high energy state can release a photon and flip to the low energy state via stimulated emission. In a perfectly balanced system (infinite spin temperature) where there are an equal number of electrons in the spin up and spin down states, these two processes would occur at the same rate and would therefore cancel each other out, providing no net response. It is therefore clear to see how the magnitude of the EPR response is related to the spin polarization, that is, the difference in the number of electrons in the spin up and spin down states.

The ratio of the number of spins in the up state and the down state for free and isolated electrons can be approximated by

$$\frac{N_+}{N_-} = \exp\left(\frac{-g_e\mu_B B_0}{k_B T}\right) \quad (2.1.8)$$

where $k_B = 1.3806 \times 10^{-23} \text{ J/K}$ and T is the absolute temperature in Kelvin. X-band EPR measurements are typically made at magnetic fields of approximately 3500 Gauss and the simplest measurements are usually made at room temperature, approximately 300 K. Since the term in the exponent is much less than one, equation 2.1.8 can therefore be simplified to be approximately

$$\frac{N_+}{N_-} = 1 - \frac{g_e\mu_B B_0}{k_B T} \quad (2.1.9)$$

using a series expansion. Using the parameters described above with the magnetic field of 3500 Gauss and an absolute temperature of 300K, this ratio is approximately $\frac{N_+}{N_-} = 0.9984$ if the g for the defect is question is close to the free electron g value. This indicates that the spin populations are very similar at room temperature; however, this level of polarization is still high

enough to detect approximately 10^{10} total paramagnetic sites. It is clear in equation 2.1.9 that increased spin polarization can be achieved by increasing the strength of the applied magnetic field and reducing the temperature of the sample.

Chapter 2.1.2 Spin Relaxation

As the spins flip due to exposure to the electromagnetic radiation satisfying the resonance condition, they are also interacting with their local environment. Electrons spins can return to their original spin populations by interacting with their surroundings in material through a process called spin relaxation. The first method is spin lattice relaxation. The time it takes for a spin system out of equilibrium to relax back to its equilibrium magnetization is described by the spin-lattice relaxation time, T_1 . Increased coupling of the electron to the lattice results in decreased T_1 times. There is a second relaxation process involving interactions between electron spins. Electron spins can interact with each other causing them to precess at slightly different rates, becoming de-phased over a spin-spin relaxation time, T_2 .

When the electrons are not allowed to relax back to a polarized state, the EPR response will vanish, therefore it is important to ensure that the correct microwave power level is used in the measurements. Electron spin populations with long T_1 times will not be able to return to a polarized state if too much microwave power is applied to them, and the signal amplitude will shrink with increasing microwave power after the spin system has been saturated. Below this saturation point, at lower microwave powers, the size of the signal will increase roughly linearly with square root of applied power, and the EPR response is said to be in the linear region.

Chapter 2.1.3 Spin Orbit Coupling and Electron Nuclear Hyperfine Interactions

Chapter 2.1.3.1 Spin Orbit Coupling

As mentioned previously, electrons in real materials do not behave like the idealized simplest case of a free electron isolated from its environment. If we consider an unpaired electron bound to an atomic nucleus, the electron will have contributions to its magnetic moment from its intrinsic quantum spin and from the motion of the electron around the nucleus. The electron's magnetic moment is described by

$$\boldsymbol{\mu}_e = g\mu_B\mathbf{S} + \mu_B\mathbf{L} \quad (2.1.10)$$

where $\mathbf{L}$ is the orbital angular momentum of the electron. The electron 'orbiting' the nucleus acts as a small current loop which in turn generates its own magnetic field around the loop. From the perspective of an observer on the electron, the nucleus appears to be travelling around it, providing an effective magnetic field which alters the magnetic moment of the electron.

This interaction between the orbital angular momentum of the electron spin and the nearby nucleus is referred to as 'spin-orbit coupling'. The amount of spin-orbit coupling experienced by an electron is measured as a change in the g factor. The deviation from the free electron g_e is more prominent in samples with larger nuclei containing more nuclear charge as well as electron spins with higher orbital angular momentum. The g factor is best described as a second rank tensor and structural information can also be gained by measuring g for different orientations in space. The g tensor is usually anisotropic in solid state materials and takes the form

$$\mathbf{g} = \begin{pmatrix} g_{xx} & g_{xy} & g_{xz} \\ g_{yx} & g_{yy} & g_{yz} \\ g_{zx} & g_{zy} & g_{zz} \end{pmatrix}. \quad (2.1.11)$$

In the principal axis system of the $\mathbf{g}$ tensor it takes the form of

$$\mathbf{g} = \begin{pmatrix} g_{xx} & 0 & 0 \\ 0 & g_{yy} & 0 \\ 0 & 0 & g_{zz} \end{pmatrix}. \quad (2.1.12)$$

It is these values of g_{xx} , g_{yy} , and g_{zz} in which the symmetry of a defect can be determined; as defects with axial symmetry will have two of these values be the same.

Chapter 2.1.3.2 Electron Nuclear Hyperfine Interactions

Atomic nuclei can also possess spin angular momentum, though not all do. The spin angular momentum, $\mathbf{I}$, will alter the resonance condition of an electron in a paramagnetic site. Unlike electrons, which all have spin $\frac{1}{2}$, the nuclear spin depends on the number of protons and neutrons in the nucleus. A nucleus with an even number of both protons and neutrons will have no nuclear spin moment, $\mathbf{I} = 0$. A nucleus with an odd number of protons, neutrons, or both will have a magnetic moment. An odd number of nucleons will result in a half integer nuclear spin ($I = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots$) while an even number of nucleons will result in an integer nuclear spin ($I = 0, 1, 2, 3, \dots$). The magnetic moment of the nuclear spin angular momentum is described by

$$\boldsymbol{\mu}_N = g_N \mu_N \mathbf{I} \quad (2.1.13)$$

where g_N is the nuclear g factor and μ_N is the nuclear Bohr magneton. The values of g_N are known for every stable atomic isotope.

Nearby nuclei with magnetic moments can align in certain orientations with a large externally applied magnetic field. A paramagnetic electron close to a nucleus with a nuclear magnetic moment will therefore experience a local magnetic field from a magnetic nucleus that differs from the externally applied field. The interaction between electrons and nearby magnetic nuclei is known as electron nuclear hyperfine interactions and is characterized by a splitting of

the measured EPR spectrum into two or more lines. If the applied field is large, magnetic nuclei will align with the applied magnetic field in $2I + 1$ orientations, giving rise to $2I + 1$ lines in a measured EPR spectrum.

An example of a spin $\frac{1}{2}$ nucleus is shown below in Figure 2.3. Notice that for a spin-0 nucleus, the dotted lines would represent the energy splitting and a single line would appear at the magnetic field corresponding to the g value of the defect at the vertical dotted line. The local magnetic field from a spin- $\frac{1}{2}$ nucleus results in two lines centered about the magnetic field corresponding to the g value of the defect.

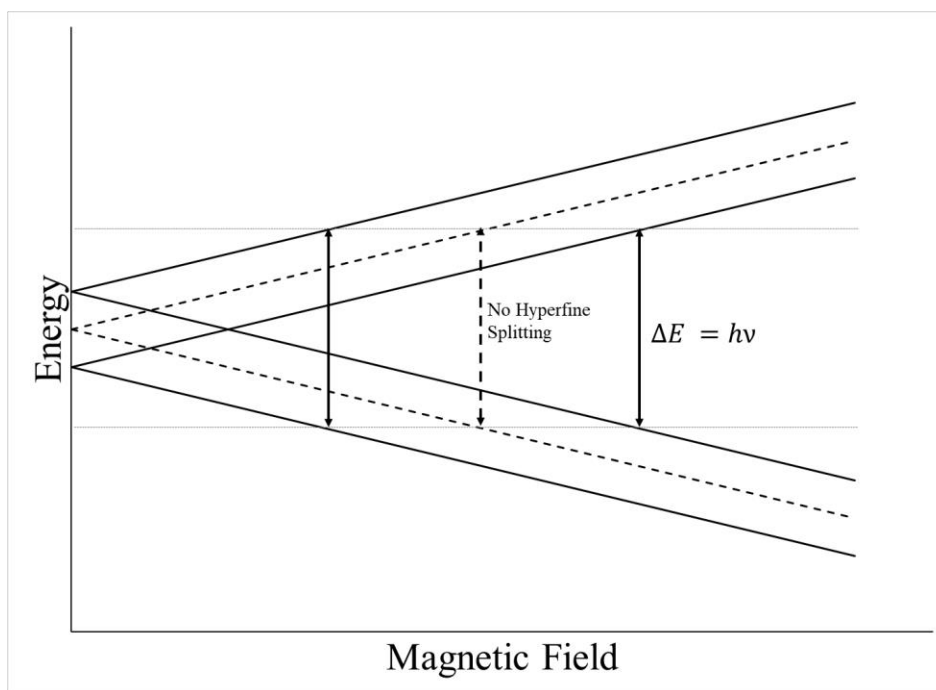


Figure 2.3: Illustration of how resonance fields are shifted by hyperfine interactions. With no hyperfine interactions (dotted line) a single line would appear at the resonance field corresponding to the g of the defect. With hyperfine interactions (solid lines) two lines appear centered about the g of the defect.

Taking these effects into account, the Hamiltonian shifts from the simple case of equation 2.1.6 to a more complex expression

$$H = \mu_B \mathbf{B} \cdot \mathbf{g} \cdot \mathbf{S} + \sum_i \mathbf{I}_i \cdot \mathbf{A}_i \cdot \mathbf{S} \quad (2.1.14)$$

where, in the second term, $\mathbf{I}_i$ is the nuclear spin operator for each nucleus, $\mathbf{A}_i$ is the hyperfine tensor for each nucleus, and $\mathbf{S}$ is, once again, the electron spin operator. Information about defect structure and chemistry can be gained by determining measured values for g and hyperfine parameters.

Chapter 2.2 Electrically Detected Magnetic Resonance

EDMR provides a method to study fully processed devices and smaller samples with magnetic resonance. EPR provides a strong analytical tool to study samples with at least 10^{10} total paramagnetic sites, however in most technologically relevant systems, there are orders of magnitude fewer defects. Alternate detection schemes can be used to get around this limitation of conventional EPR. A variant of EPR which bypasses this limitation is electrically detected magnetic resonance (EDMR). The most common EDMR experiments involve changes in spin dependent currents due to recombination or tunneling. This detection scheme has the advantage of having a higher sensitivity, usually around 10^3 total defects. The measurement can also be made on fully processed devices and is only sensitive to electrically active defects which affect electrical conduction in these devices and materials.

EDMR measurements made at X-band use most of the same equipment as the EPR measurements discussed before, with a few small additions. The detection scheme for EDMR relies on changes in the electrical characteristics of a sample as it undergoes magnetic resonance, therefore an electrical connection must be made to the sample in the microwave cavity and a current to voltage converter is used with a biasing circuit. A sample can be mounted and wire

bonded onto a specialized device holder with electrical leads allowing for the application of electrical bias and measuring sample current. Measured EDMR spectra are generated by plotting the change in the electrical current through the sample as a function of the slowly varied magnetic field, B_0 . Just like the EPR measurements discussed before, magnetic field modulation and lock-in detection are used to improve the signal to noise ratio. EDMR also has the advantage of having a sensitivity which is essentially magnetic field independent. The signal amplitude does not vary much over a wide range of resonant fields and frequencies, unlike EPR in which the signal amplitude is highly magnetic field dependent, as discussed previously. [35], [36] The two primary spin dependent currents measured are spin-dependent recombination (SDR) and spin-dependent trap assisted tunneling (SDTAT)

Chapter 2.2.1 Spin Dependent Recombination

Spin dependent recombination can be explained at a rudimentary level in terms of a slightly modified Shockley-Read-Hall model for recombination [34], [35], [45]–[47]. Electrons and holes can be captured by deep level defects within the bandgap, as seen in Figure 2.4. When an electron in the conduction band and the electron in the paramagnetic defect are polarized in the same direction, the conduction band electron cannot be captured due to the Pauli-Exclusion principle. The electron in the trap can be flipped during magnetic resonance, converting what was a triplet state into a singlet state, making the previously forbidden transition possible. The conduction electron can now be captured by the deep level defect and recombine with a hole from the valence band. An increased number of recombination events during magnetic resonance causes a measurable change in the steady state current.

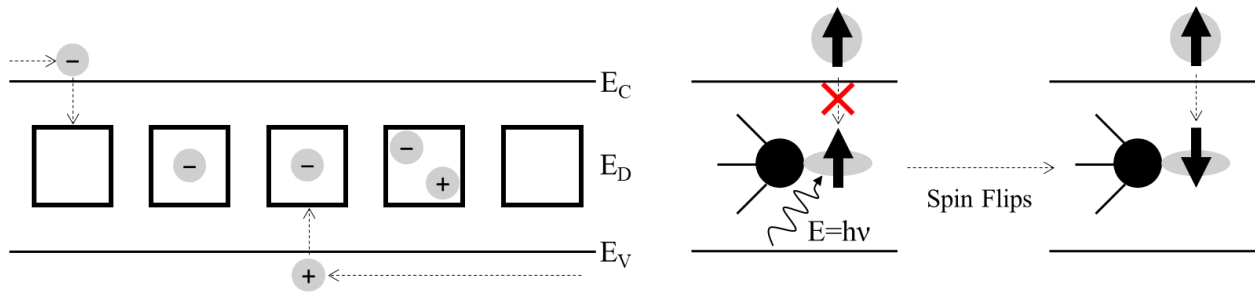


Figure 2.4: (Left) Schematic of recombination event mediated by a deep level defect. The initial electron capture is only possible if the spins of the electrons do not violate the Pauli-Exclusion principle. (Right) flipping the spin of the neutral defect can turn a forbidden transition into an allowed one.

This SDR mechanism is responsible for the signal in experiments such as the spin dependent gated diode measurements [46], [48], [49] and spin dependent charge pumping (SDCP) [50], [51]. In these experiments, the resonance signal arises from measuring the changes in the steady state currents from the electrical measurements detailed above due to magnetic resonance.

Chapter 2.2.2 Spin Dependent Trap Assisted Tunneling

Spin dependent trap assisted tunneling (SDTAT) is an EDMR measurement that relies on the spin dependent nature of VRH currents [33], [44], [49], [52]. Figure 2.5 shows that electrons cannot tunnel to a site that has an electron with the same spin orientation due to the Pauli-Exclusion principle, not unlike an electron being captured by a deep level in SDR. The electron can be flipped during magnetic resonance, allowing it to tunnel into the next site. This change in tunneling current through the material is measured during SDTAT and gives rise to the EDMR spectra associated with the defects involved.

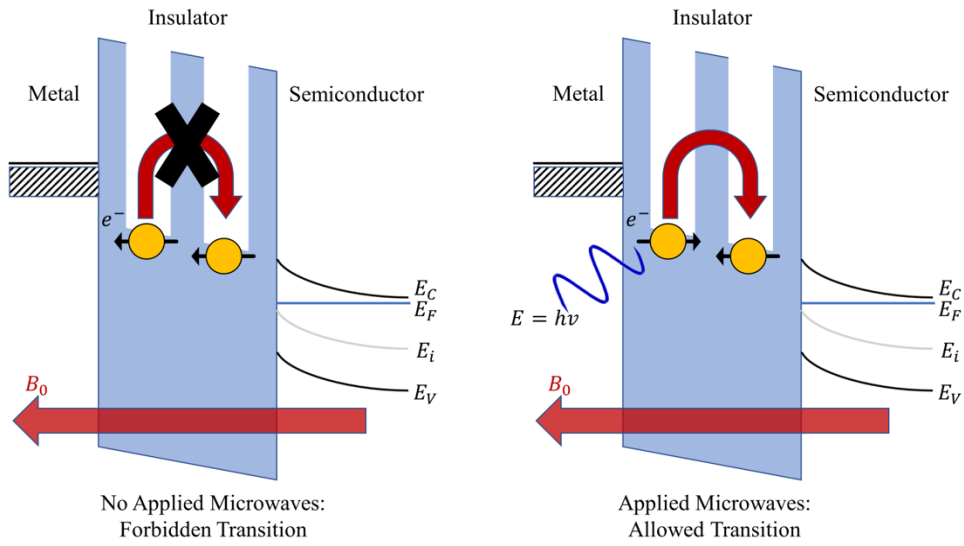


Figure 2.5: Cartoon illustration of (left) tunneling event forbidden due to Pauli-Exclusion principle and (right) allowed transition after spin flip due to magnetic resonance. Adapted from Myers et al.[53]

These different types of EDMR measurements allow for an approximate measurement of the physical region and energy levels probed during resonance measurements, as only defects that contribute to the device current will be detected while defects in the bulk and other regions that do not affect device performance will be neglected; these unimportant defects would be detected in a conventional EPR measurement assuming the quantity is high enough. The size and shape of the resonance signals (determined usually by spin orbit coupling and hyperfine interactions) provide information about physical and chemical nature of the defects in the device. EDMR has been utilized extensively to study defects in semiconductor devices [33], [37], [41], [44], [46], [48], [51]–[56].

Chapter 3

Near Zero Field Magnetoresistance in Inorganic Semiconductors and Dielectrics

All of the spin dependent measurements outlined above appear to give signals at near zero magnetic field as well as at resonance. Magnetoresistance is a change in the electrical conductivity of a material at low applied magnetic fields. It is not as well understood as conventional EPR and EDMR measurements; however, some models have been put forward to explain the phenomenon of organic magnetoresistance (OMAR) in positionally disordered organic semiconductors [57]–[62]. Unlike EDMR measurements, in NZFMR, the shift from a forbidden to an allowed transition is not caused by flipping spins at resonance, but by a mixing of states due to variations in local magnetic fields. The models for OMAR presented so far focus on unipolar transport through disordered organic materials, which is qualitatively very similar to SDTAT in inorganic semiconductors outlined above. These similarities make amorphous inorganic semiconductors an ideal choice to use to adapt the current OMAR models and expand our knowledge of NZFMR.

Chapter 3.1 Magnetoresistance in Organic Materials

Current proposed models for OMAR suggest that charge carriers attempt to move across a disordered material by hopping from site to site, like a network of random resistors similar to the model put forward by Miller and Abrahams [63]. In this model, the bulk conductivity is set by the distance between sites that a charge carrier must move between to get across the device. A critical distance, r_c is defined as the longest allowed hopping distance required to form a path through the material. Longer critical hopping distances lead to lower conductivity. Two localized charge carriers in adjacent sites are referred to as a polaron pair while a doubly

occupied site is referred to as a bipolaron. A polaron may only hop to another polaron's site, forming a bipolaron, if the polaron pair is in the singlet configuration because of the Pauli-Exclusion principle. In the presence of a magnetic field that aligns electron spins, certain hops can be blocked if the polaron attempts to hop to a site occupied by a polaron with the same spin orientation, or the polaron pair is in a triplet state. This is referred to by Harmon and Flatté as spin blocking [60]–[62] and is the foundation of magnetoresistance in disordered organic semiconductors.

The model proposed by Harmon and Flatté is similar to a two-site model proposed by Wagemans et al. [58] which was expanded upon by Schellekens et al. [59] in which the relative spin projection of charge carriers in adjacent sites can change due to local hyperfine fields. In the case of organic semiconductors, these hyperfine interactions primarily come from randomly oriented hydrogen nuclei. Each site has a magnetic field that is the vector sum of the local random hyperfine field and the externally applied field. At low and zero applied magnetic fields, the charge carriers will precess about their individual random hyperfine fields allowing a triplet state to gain more singlet character as the relative orientation of the spins change. A large applied magnetic field will suppress the random hyperfine fields and the polarons on different sites will experience almost identical local magnetic fields. When the local field is overwhelmed by the applied field, the polaron spins will precess about near identical magnetic fields, suppressing any spin mixing. In this case polaron pairs in the triplet state will remain that way and be unable form bipolarons, as illustrated in Figure 3.1.

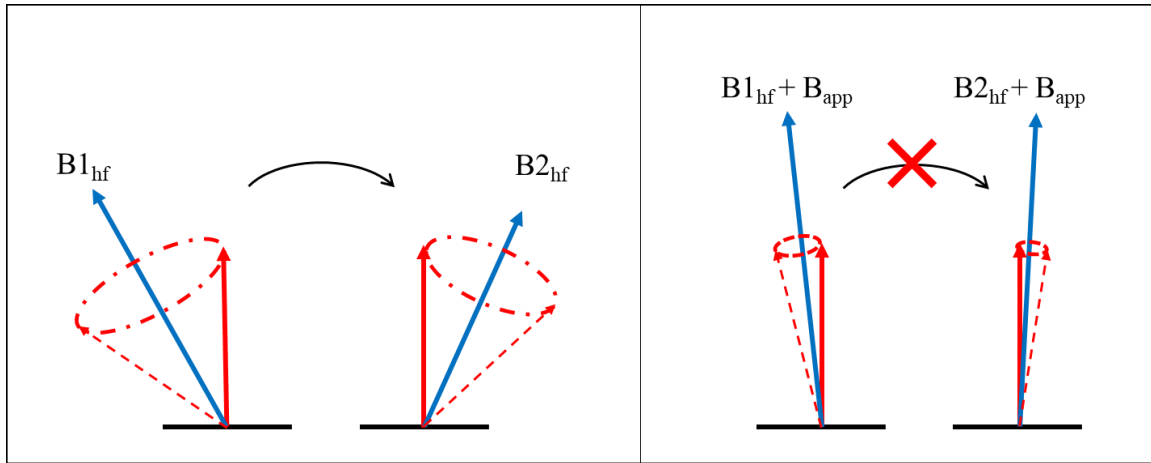


Figure 3.1: Spin mixing due to hyperfine fields. Left) Spins precess coherently about random hyperfine fields alternating between all triplet character (solid red line) and partial singlet character (dotted red line) allowing for hopping to occur. Right) Spins precess about similar magnetic fields maintaining more of their triplet character, preventing hopping. Recreated from Schellekens et al. [59]

Current through the material will increase as more hopping sites are accessible and it will decrease when there are fewer, therefore as previously forbidden hopping sites become available due to triplet to singlet transitions at low applied magnetic fields, the current will be larger than at high applied magnetic fields where such transitions do not occur and hopping distances through the material will be longer. This concept is illustrated in Figure 3.2. Formation of bipolarons leads to increased conduction.

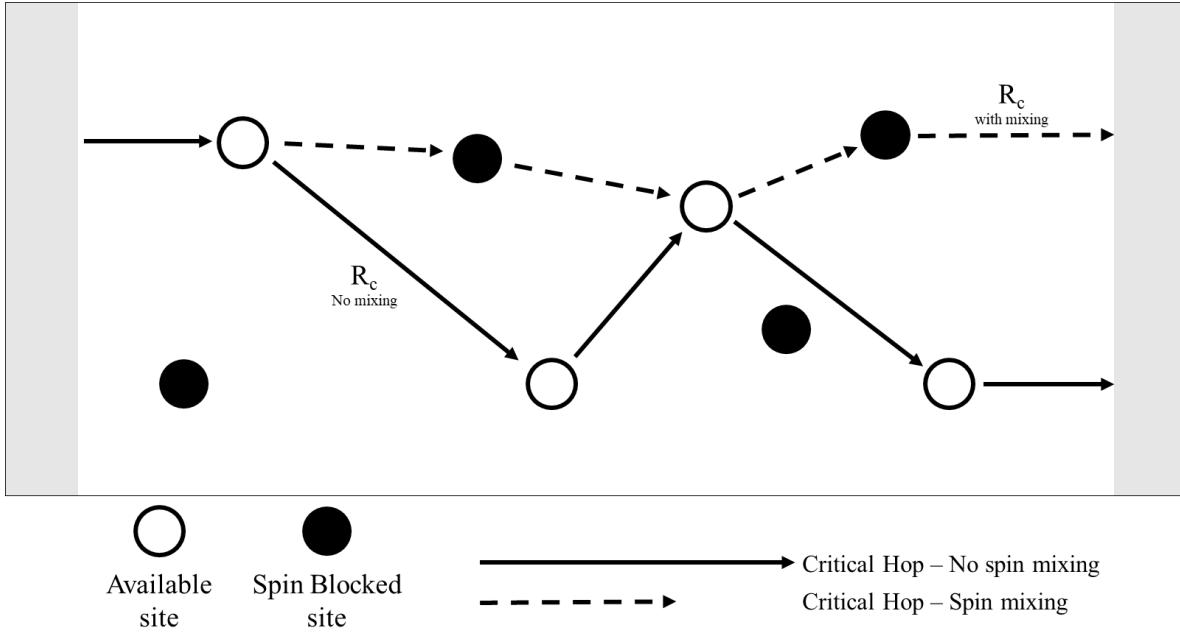


Figure 3.2: Illustration of spin blocking. The solid line represents the hopping path without spin mixing as spin blocked sites are inaccessible. The dotted line represents a possible hopping path with spin mixing as the spin blocked sites can transition to available sites as the spins precess in local hyperfine fields. The longer critical hopping distance due to blocked sites causes an increase in the resistance of the material. Recreated from Harmon and Flatté. [61]

Magnetoresistance is quantified by utilizing the density matrix formalism. The density operator for a simple two site model with an ensemble of N spins can be written as:

$$\rho = \frac{1}{N} \sum_{n=1}^N |\psi_n\rangle\langle\psi_n| \quad (3.1.1)$$

Choosing an orthonormal basis set φ_i allows us to define the density matrix as:

$$\rho_{i,j} = \langle \varphi_i | \rho | \varphi_j \rangle \quad (3.1.2)$$

The probability of finding the system in the corresponding basis states is given by the diagonal elements of the density matrix. The system will evolve coherently over time as the spins precess according to the time dependent Schrodinger equation and is described by the quantum Liouville equation:

$$\frac{\partial \rho}{\partial t} = -\frac{i}{\hbar} [H, \rho] \quad (3.1.3)$$

where H is the spin pair Hamiltonian [59], and $[H, \rho]$ is commutator of the Hamiltonian and the density matrix.

For a system in which spin pairs are formed and removed over time, it is necessary to include these phenomena in the equation governing the time evolution of the system which is achieved by utilizing the stochastic Liouville equation (SLE):

$$\frac{\partial \rho}{\partial t} = -\frac{i}{\hbar} [H, \rho] - \frac{1}{2} \{\Lambda, \rho\} + \Gamma \quad (3.1.4)$$

where $\{\Lambda, \rho\}$ is the anticommutator of the projection operator, Λ , and ρ , the first term is the coherent time evolution with no other interactions as seen in (3.1.3), the second is a dissipative term that removes spin pairs from the system, and the last is a generation term. OMAR is a steady state process and magnetoresistance values are determined by tracking the population of spins in the density matrix, ρ , for long time scales. In these unipolar models, increased bipolaron formation at low applied magnetic fields leads to increased hopping and therefore increased current.

Chapter 3.2 NZFMR Models in Solid State Materials

The models described above have been somewhat expanded in some material systems by Frantz et al. [64] and Harmon et al. [65]. The stochastic Liouville equation utilized in these models is the same as the previously described models.

Chapter 3.2.1 Electron-Hole Spin Dependent Recombination Model for NZFMR

This model [64] is based on a combination of the models put forward by Prigodin et al. [66] and Harmon and Flatté [62]. This model describes spin dependent recombination which occurs at deep level defects in inorganic materials. The model is similar to that shown in Figure 2.4 which illustrates spin dependent recombination due to resonance. An electron in the conduction band can approach a deep level defect with an unoccupied electron orbital near a nucleus with a nuclear spin and associated hyperfine field. This nuclear spin is represented by the blue arrows in Figure 3.3. The free electron can be temporarily captured in an intermediate state just below the conduction band where it can then attempt to enter the deep level defect. If the free electron and the defect electron are in the same spin orientation, a triplet state, the free electron will be unable to be captured by the trap. However, if the two electrons have opposite spin orientations, a singlet state, the free electron can then be captured by the deep level. Unlike in EMDR, the transition from a triplet state to a singlet state does not come from impinging electromagnetic radiation, but from spin mixing of the defect electron due to local magnetic fields, as described in Figure 3.1. If a singlet is formed due to spin mixing, the free electron will drop into the deep level and then recombine with a hole in the valence band, removing the electron and hole from the system. This essentially causes the spins to be annihilated. If the spin pair does not form a singlet and remains in the triplet state, then the free electron can dissociate from the intermediate state and return to the conduction band, causing no change to the recombination current. Increased spin mixing at low applied magnetic fields will cause the recombination current to increase as more spin pairs transition from triplet to singlet states. This process is illustrated in Figure 3.3.

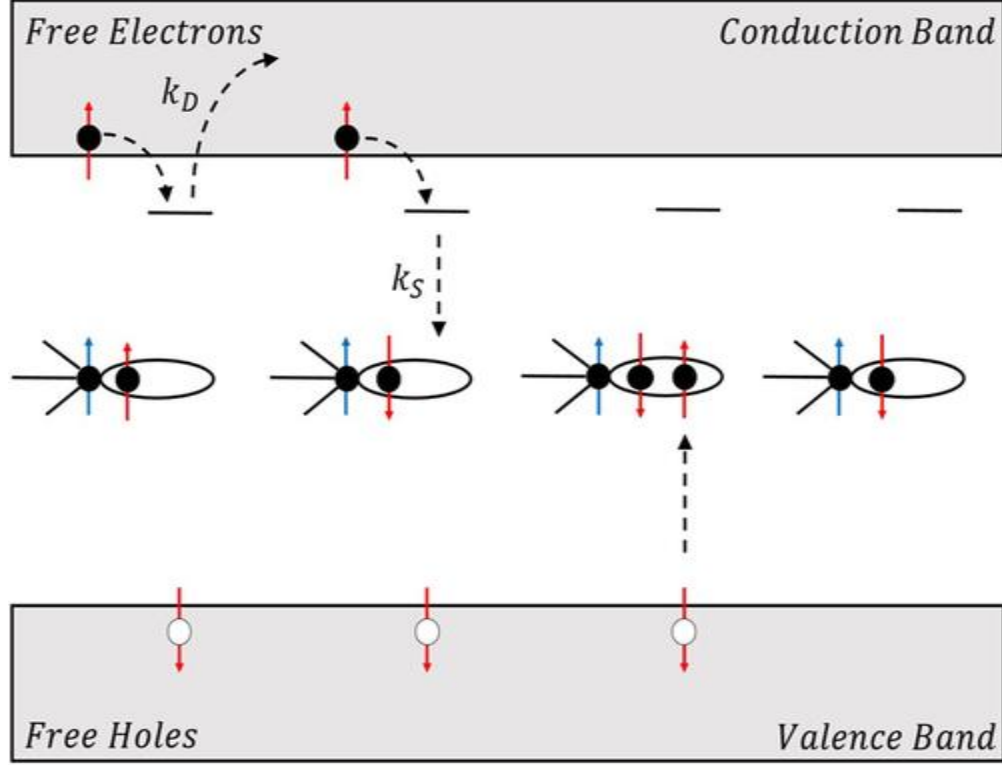


Figure 3.3: Model for near zero field magnetoresistance spin dependent recombination. Note the nuclear magnetic moment indicated by the blue arrows on the atomic nuclei. Adapted from Frantz et al. [64].

The density matrix in this model evolves in a manner described by an adapted form of the SLE first discussed by Hansen and Pendersen [67]

$$\frac{\partial \rho}{\partial t} = -\frac{i}{\hbar} [\mathbf{H}, \rho] - \frac{1}{2} (k_S + k_D) \{\mathbf{A}_S, \rho\} - \frac{k_D}{2} \{\mathbf{A}_T, \rho\} + \frac{1}{8} p \Gamma \quad (3.2.1)$$

where the first term represents the coherent evolution of the spin pairs due to local magnetic fields, the second term is a dissipative term that removes singlet electron hole pairs from the system at a rate of k_S through recombination, the third term is also dissipative which removes triplets in which the free electron has dissociated from the intermediate state back to the conduction band at a rate of k_D , $\mathbf{A}_S$ and $\mathbf{A}_T$ are the singlet and triplet projection operators, respectively, and the fourth term is a generation term that represents a source of carriers which

enter the system at a rate of p and $\mathbf{I}$ represents the spin populations of the generated pairs. For randomly generated spin pairs, $\mathbf{I}$ is the identity matrix. The last term has a factor of $\frac{1}{8}$ so that the total generation rate of all the spin pairs is equal to p since the identity matrix is 8x8. This is a steady state process in which spins are being generated and removed at constant rates so only the steady state form of equation 3.2.1 needs to be solved,

$$0 = -\frac{i}{\hbar} [\mathbf{H}, \boldsymbol{\rho}] - \frac{1}{2} (k_S + k_D) \{\mathbf{A}_S, \boldsymbol{\rho}\} - \frac{k_D}{2} \{\mathbf{A}_T, \boldsymbol{\rho}\} + \frac{1}{8} p \mathbf{I}. \quad (3.2.2)$$

Assuming that only the defect electron is affected by the local nuclear hyperfine interactions, the Hamiltonian for such a spin system is

$$\mathbf{H} = g\mu_B(\mathbf{S}_1 + \mathbf{S}_2) \cdot \mathbf{B} + a(\mathbf{S}_2 \cdot \mathbf{I}) \quad (3.2.3)$$

where $g\mu_B(\mathbf{S}_1 + \mathbf{S}_2)$ accounts for the Zeeman splitting due to the applied magnetic field, $\mathbf{B}$, of both the electron in the conduction band, $\mathbf{S}_1$, and of the electron in the defect, $\mathbf{S}_2$, and $a(\mathbf{S}_2 \cdot \mathbf{I})$ represents an isotropic electron-nuclear hyperfine interaction with strength a between the defect electron and the magnetic nucleus, $\mathbf{I}$. The solution to the steady state SLE for NZFMR SDR will be explored in Chapter 3.2.3.

Chapter 3.2.2 Spin Dependent Trap Assisted Tunneling Model for NZFMR

To model the NZFMR response in samples where the primary conduction method is trap assisted tunneling, the two-site model proposed by Wagemans et al. [58] was adapted by Frantz et al. [64]. Carriers that travel through a material via trap assisted tunneling will use many sites on the path across the material, but the conduction through the material can generally be limited most by the longest hop, like the two-site model for OMAR described above. It is not unreasonable to focus on this bottleneck spin pair when modeling the SDTAT NZFMR response.

The adapted two-site model is shown in Figure 3.4. The model consists of two sites around nuclei that electrons can occupy, sites α and β . The nuclei at both α and β sites have nuclear spins represented by the blue arrow resulting in hyperfine fields experienced by the electrons on that site. The hyperfine coupling constants for each site when occupied by an electron are a_α and a_β for the sites α and β respectively. An electron from the environment can hop to site α when it is unoccupied at a rate of $r_{e\alpha}$. Once the electron is on site α it will interact with the local hyperfine interaction by precessing about it, this is shown in Figure 3.4 with the red arrow and the dashed path that it follows around the hyperfine field. The electron trapped at site α can either jump directly to the environment at a rate of $r_{\alpha e}$ or it can hop to site β at a rate of $P_S r_{\alpha\beta}$, where P_S is the probability of the electrons at sites α and β to be in the singlet configuration. If the electron hops from site α to site β , the electron that is now at site β can then jump out of that site and into the environment at a rate of $r_{\beta e}$. All of these paths and rates are shown in Figure 3.4

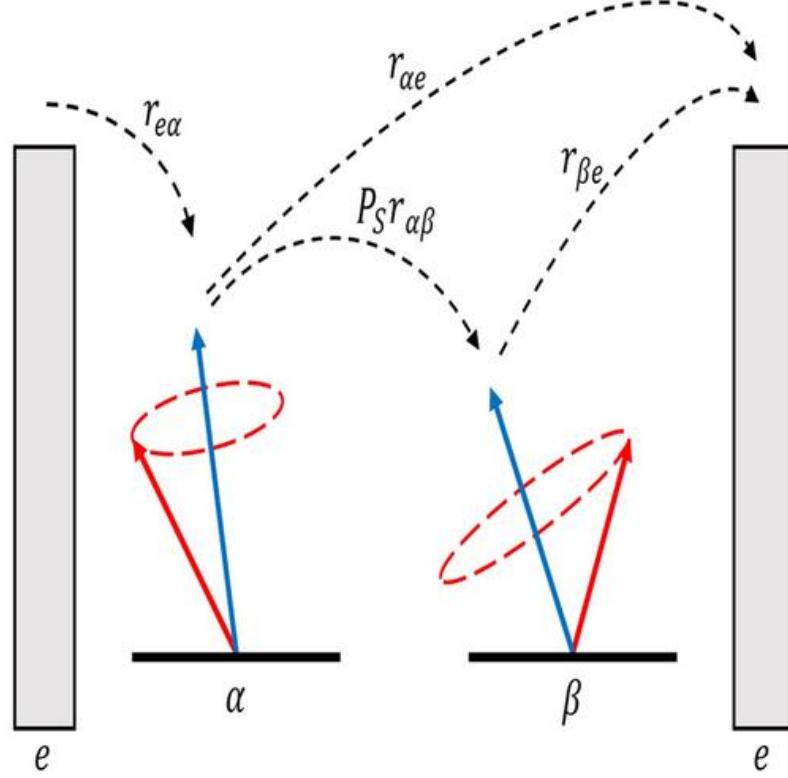


Figure 3.4: Two site model for spin mixing of electrons in NZFMR SDTAT. The blue arrows indicate the hyperfine fields from the atomic nuclei, the red arrows indicate the electron spin orientations. The red dashed lines indicate the precession of the electron spins as they evolve coherently due to the local hyperfine magnetic fields. Adapted from Frantz et al. [64]

The trap assisted tunneling current is defined as the flow of carriers from the environment on the left of Figure 3.4 to site α . There will almost always be a state available in the environment for an electron in site β to hop to, so $r_{\beta e}$ is set to ∞ in order to make the arithmetic simpler.[59]

With this simplification, the equation governing the density matrix of the spin system is the SLE adapted from [58] and has the form

$$\frac{\partial \rho}{\partial t} = -\frac{i}{\hbar} [\mathbf{H}, \rho] - \frac{1}{2} (r_{\alpha\beta} + r_{\alpha e}) \{\mathbf{A}_S, \rho\} - \frac{1}{2} r_{\alpha e} \{\mathbf{A}_T, \rho\} + \frac{1}{16} r_{e\alpha} (1 - \text{Tr}(\rho)) \mathbf{\Gamma} \quad (3.2.4)$$

where $\mathbf{A}_S$ and $\mathbf{A}_T$ are once again the singlet and triplet projection operators respectively and $\mathbf{I}$ is once again defined as the identity matrix for generating spin states in random orientations. If the trap states remain mostly empty during the hopping process (as is the case in this proposed model), then $Tr(\rho) \ll 1$, and can be ignored. Like the model for SDR NZFMR, the processes involved in SDTAT NZFMR are at the steady state and therefore only the steady state form of the simplified equation needs to be solved,

$$0 = -\frac{i}{\hbar}[\mathbf{H}, \rho] - \frac{1}{2}(r_{\alpha\beta} + r_{\alpha e})\{\mathbf{A}_S, \rho\} - \frac{1}{2}r_{\alpha e}\{\mathbf{A}_T, \rho\} + \frac{1}{16}r_{e\alpha}\mathbf{I}. \quad (3.2.5)$$

Unlike the SDR model, the SDTAT model accounts for two nuclear spins, raising the total number of spins in the system to four, even if one of the nuclei has a spin of 0. The four spins interacting with each other means that the identity matrix $\mathbf{I}$ must be a 16x16 matrix and a factor of $\frac{1}{16}$ must be included to account for it. The Hamiltonian for this spin system is described by

$$\mathbf{H} = g\mu_B(\mathbf{S}_1 + \mathbf{S}_2) \cdot \mathbf{B} + a_\alpha(\mathbf{S}_1 \cdot \mathbf{I}_\alpha) + a_\beta(\mathbf{S}_2 \cdot \mathbf{I}_\beta) \quad (3.2.6)$$

where the first term is once again the Zeeman splitting of electrons at sites α and β , the second term is the isotropic electron-nuclear hyperfine interaction between the electron and nuclear spin at site α , and the third term is the electron-nuclear hyperfine interaction between the electron and nuclear spin at site β .

Chapter 3.2.3 Solution of SLE in NZFMR Models

The models put forward by Frantz et al. [64] are governed by SLEs that can be algebraically manipulated into the form of the time-continuous Lyapunov equation

$$\mathcal{L}\rho + \rho\mathcal{L}^\dagger = g \quad (3.2.7)$$

where $\mathcal{L}^\dagger$ is the conjugate transpose of $\mathcal{L}$ which is given by

$$\mathcal{L} = \frac{i}{\hbar} \mathbf{H} + \frac{c_1 + c_2}{2} \mathbf{A}_S + \frac{c_2}{2} \mathbf{A}_T \quad (3.2.8)$$

and c_1 and c_2 represent the rate constants that depend on the form of the SLE being solved, either for SDR or SDTAT. The right side of equation 3.2.7, g , represents the rate that spins are generated. This equation has a known analytic solution of

$$\rho = \int_0^\infty e^{-i\mathcal{L}\tau} (g\mathbf{I}) e^{i\mathcal{L}^\dagger\tau} d\tau. \quad (3.2.9)$$

For the SDR NZFMR, the observable change in current occurs when singlets are formed and recombination occurs, therefore the change in current will be proportional to the product of the singlet formation probability and the trace of the singlet population of the density-matrix given by

$$I_{SDR} \propto \left(\frac{k_S}{p}\right) Tr(\mathbf{A}_S \rho) \quad (3.2.10)$$

where a scaling constant can be utilized when modeling a spectrum that arises from multiple conduction paths. The measurable change in SDTAT current will be proportional to the product of the rate at which sites are filled and the probability of the occupation of the sites, therefore the change in current will be given by

$$I_{SDTAT} \propto r_{e\alpha} (1 - Tr(\rho)). \quad (3.2.11)$$

These models utilize a least squares fitting to determine what the most likely interactions are between the spins in the system to produce the measured spectrum. The majority of the samples studied in this work were thin films subjected to SDTAT NZFMR and SDTAT EDMR measurements at X-band and low-magnetic-field.

Chapter 3.2.4 SDTAT NZFMR Simulations

The model for SDTAT NZFMR put forward by Frantz et al. [64] was utilized to simulate NZFMR spectra across a range of values for the variables $r_{e\alpha}$, $r_{\alpha e}$, $r_{\alpha\beta}$, a_α , and a_β in equations 3.2.5 and 3.2.6. The values used in these simulations are $r_{e\alpha} = 1 \text{ GHz}$, $r_{\alpha e} = 0.1 \text{ GHz}$, $r_{\alpha\beta} = 1 \text{ GHz}$, $a_\alpha = 1 \text{ mT}$ (10 Gauss), and $a_\beta = 1 \text{ mT}$ (10 Gauss) unless otherwise stated. The measurements made in the later chapters will be compared to these simulated spectra to identify how changing experimental parameters affect the variables from equations 3.2.5 and 3.2.6; for example, if the changes in a measured spectra as a function of bias or time match the trends seen in these simulations, we may be able to understand the physical processes that occur to cause the change in the experimentally measured spectra.

Chapter 3.2.4.1 Simulated effects of $r_{e\alpha}$

The generation rate at which carriers are injected from the environment, or a large pool of randomly oriented carriers, into the first hopping site is given by $r_{e\alpha}$ in Figure 3.4. This can essentially be thought of as how fast carriers arrive to the sites that form the rate limiting step in the hopping transport. A large value of $r_{e\alpha}$ indicates that there are a lot of charge carriers entering the first site in the adapted two site model. Smaller values of $r_{e\alpha}$ indicate that there are less carriers entering the first site. The value of $r_{e\alpha}$ will likely also affect the overall DC current through a sample, with higher values corresponding to higher currents. The effect of varying $r_{e\alpha}$ over a range of values is shown in Figure 3.5. The fixed value for $r_{\alpha\beta}$ was changed to 0.3 GHz for this plot.

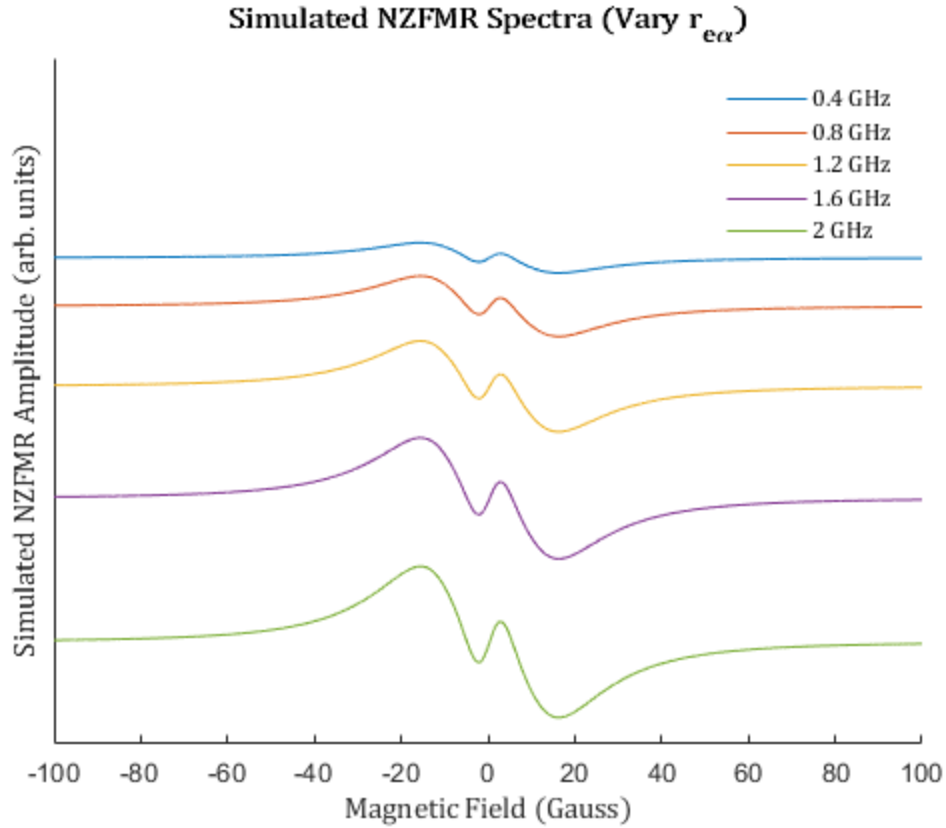


Figure 3.5: Raw simulated NZFMR spectra for different values of $r_{e\alpha}$. The order shown in the legend matches the order of the plots.

It is clear to see that increasing $r_{e\alpha}$ has the effect of increasing the NZFMR signal amplitude. The effect that $r_{e\alpha}$ has on the shape and width of the NZFMR signal can be seen more easily by normalizing these spectra. The normalized spectra are shown in Figure 3.6. In this case, the normalized spectra would overlap perfectly if not for the artificial offset. These two figures suggest that an NZFMR signal that changes only in amplitude, and not in shape, is most likely affected by a change in the rate, $r_{e\alpha}$, which primarily controls the signal amplitude of the spectrum.

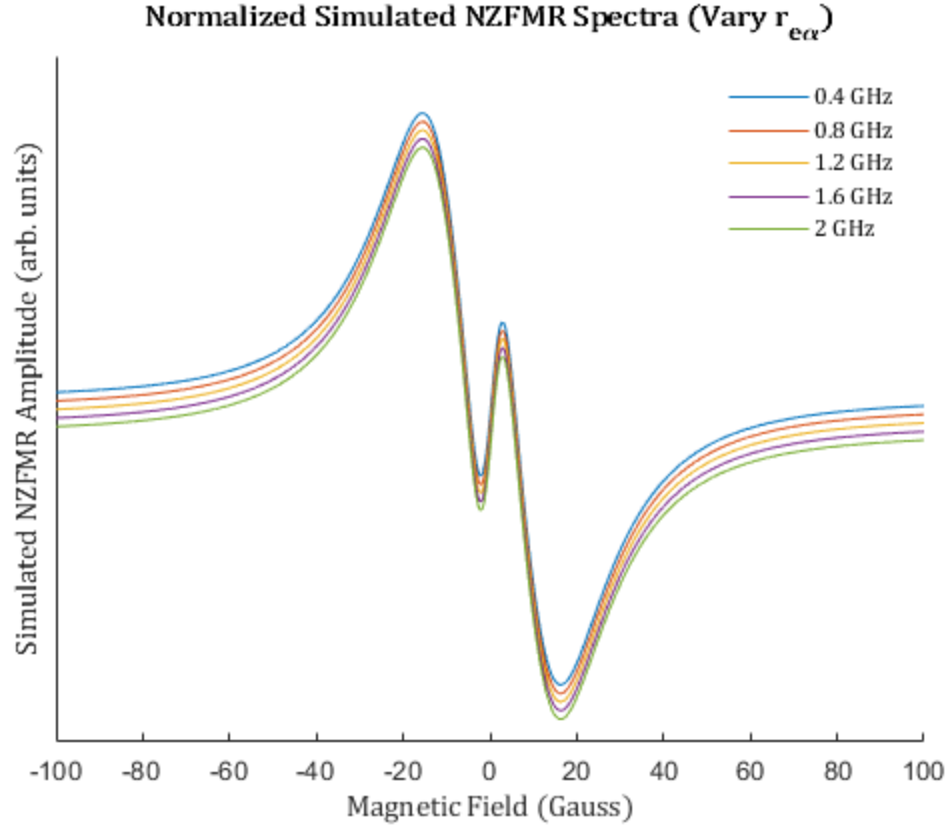


Figure 3.6: Normalized simulated NZFMR spectra for different values of $r_{e\alpha}$. The order shown in the legend matches the order of the plots. All spectra overlap perfectly without the added offset.

Chapter 3.2.4.2 Simulated effects of r_{ae}

Once a charge carrier reaches the rate limiting step in the hopping transport at site α , it can either dissociate by going directly to the environment at a rate of r_{ae} or it can hop to the next defect site, site β , at a rate of $P_S r_{\alpha\beta}$. The latter case will be discussed in detail in the next section. The rate r_{ae} can be thought of as the rate at which a charge carrier at site α is able to bypass the jump to site β and jump to larger pool of carriers. For the two-site model, this process is not spin dependent. The effects of changing r_{ae} are shown in Figure 3.7. The primary effect of increasing r_{ae} is a decrease in the amplitude of the center inflection line while the broader line remains almost constant.

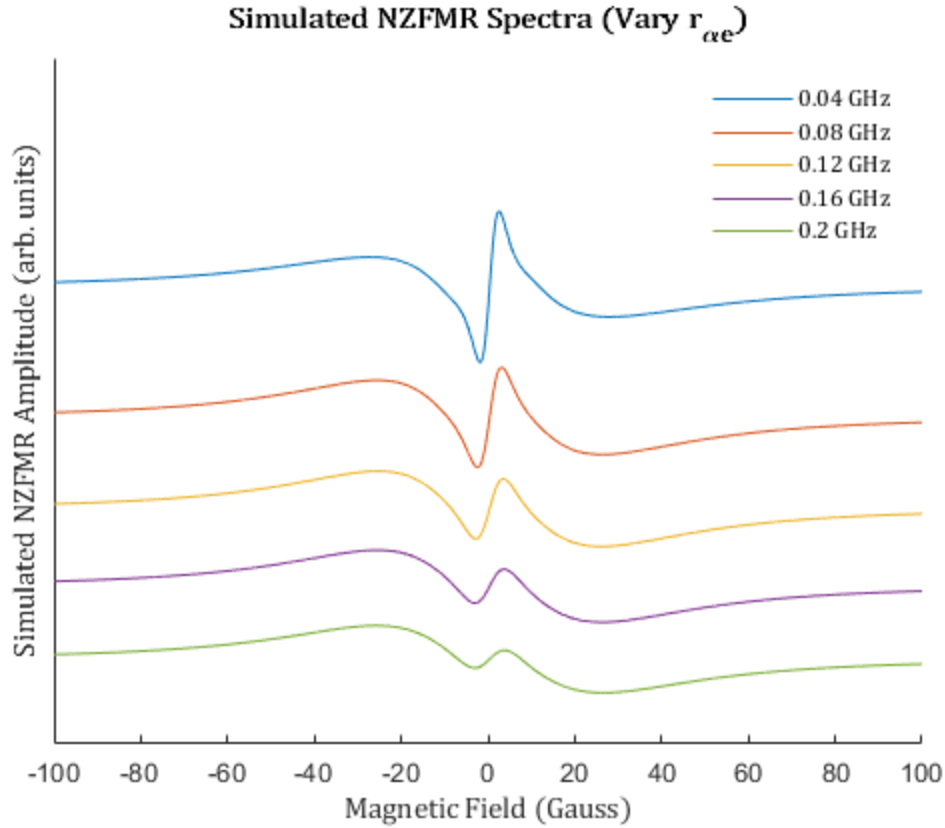


Figure 3.7: Raw simulated NZFMR spectra for different values of $r_{\alpha e}$. The order shown in the legend matches the order of the plots.

The effect of $r_{\alpha e}$ on the line shape is more easily seen when these spectra are normalized and superimposed on one another. Unlike the case of changing $r_{e\alpha}$, the normalized spectra do not overlap and so the simulated spectra were normalized twice. First, they were normalized to the broader signal in order to see how the width and shape of the broad signal change with $r_{\alpha e}$. This is shown in the top of Figure 3.8. There is very little change in the shape of the broader line due to changes in $r_{\alpha e}$. The spectra were then normalized to the narrow inflection signal in order to see how the width and shape of that signal change with $r_{\alpha e}$. This is shown in the bottom of Figure 3.8. Once again, the line width of the inflection line is almost constant as $r_{\alpha e}$ changes.

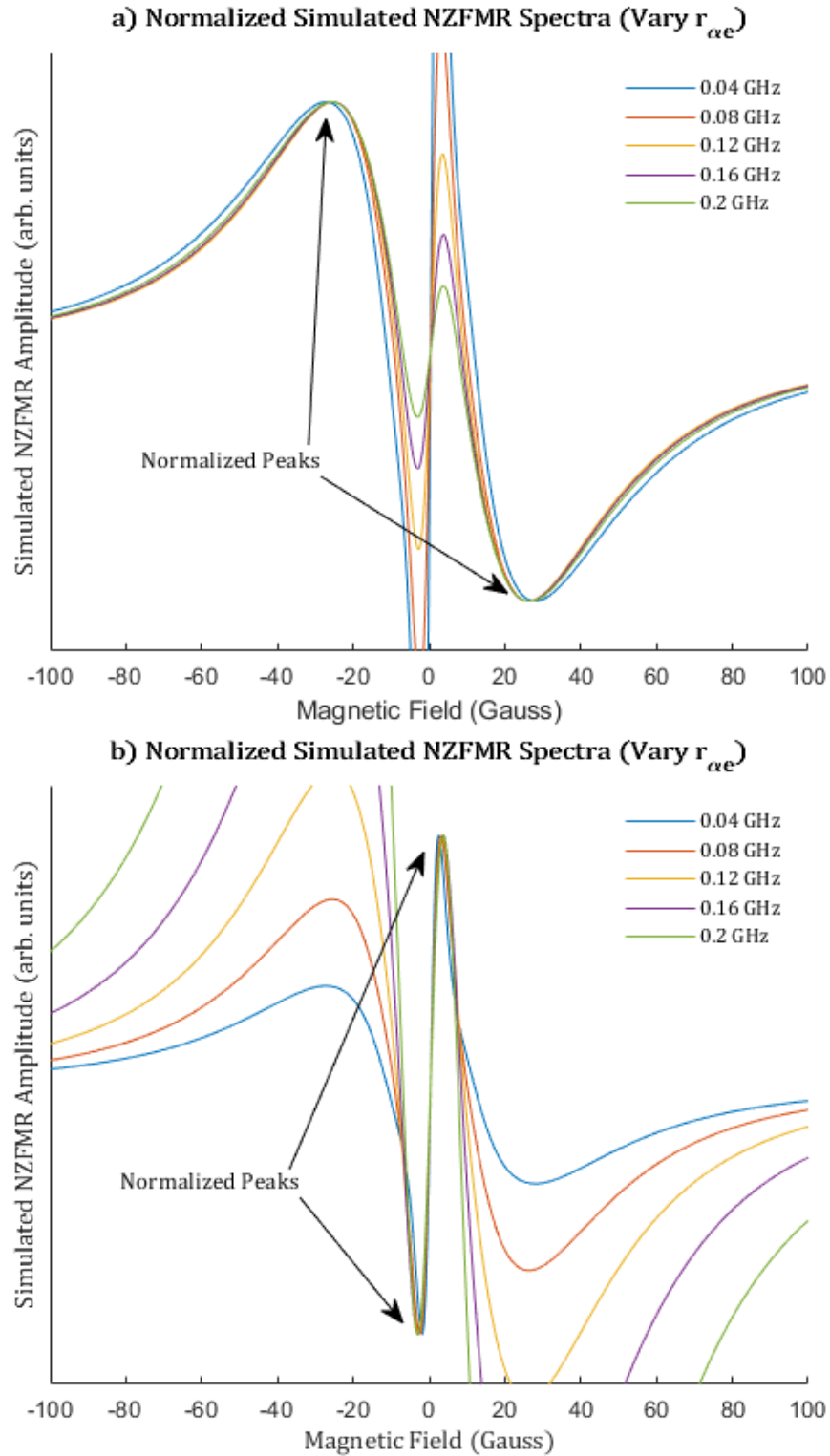


Figure 3.8: Normalized simulated spectra for different values of r_{ae} . a) Spectra normalized to the broader line to show changes in broad line shape. b) Spectra normalized to inflection line to show changes in inflection line shape. The shapes of both lines change little for different values of r_{ae} .

Chapter 3.2.4.3 Simulated Effects of $r_{\alpha\beta}$

The rate $P_S r_{\alpha\beta}$, is the singlet probability hopping rate, for which charge carriers hop from the initial site, site α , to the next site, site β . P_S is the singlet projection operator of one onto the other, as unlike the transport associated with $r_{\alpha e}$, the hops associated with $r_{\alpha\beta}$ are spin-dependent. P_S will change over time as the system evolves coherently and incoherently. The more singlet character between the charge carriers on site α and β , the higher the chance that the hopping event will occur due to the Pauli Exclusion principle. The effects of $r_{\alpha\beta}$ on the simulated spectra is shown in Figure 3.9. The overall amplitude of both the broad line and the inflection line decrease as $r_{\alpha\beta}$ is increased, however, the amplitude of the inflection line appears to decrease more slowly. These spectra were then normalized in order to see the effects of $r_{\alpha\beta}$ on the shape of the spectra. The plots in Figure 3.10 show the spectra normalized to the broad line (top) and the inflection line (bottom). It is clear to see that the broad line becomes wider for larger values of $r_{\alpha\beta}$, while the width of the narrow inflection line is essentially unchanged.

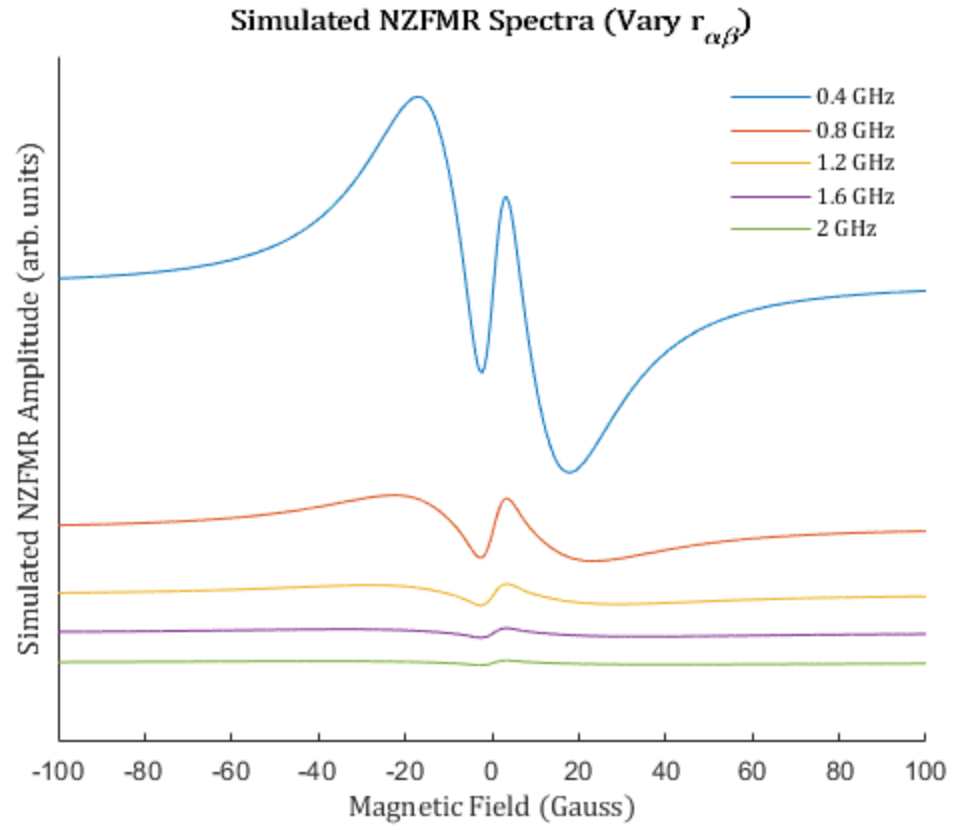


Figure 3.9: Raw simulated NZFMR spectra for different values of $r_{\alpha\beta}$. The order shown in the legend matches the order of the plots.

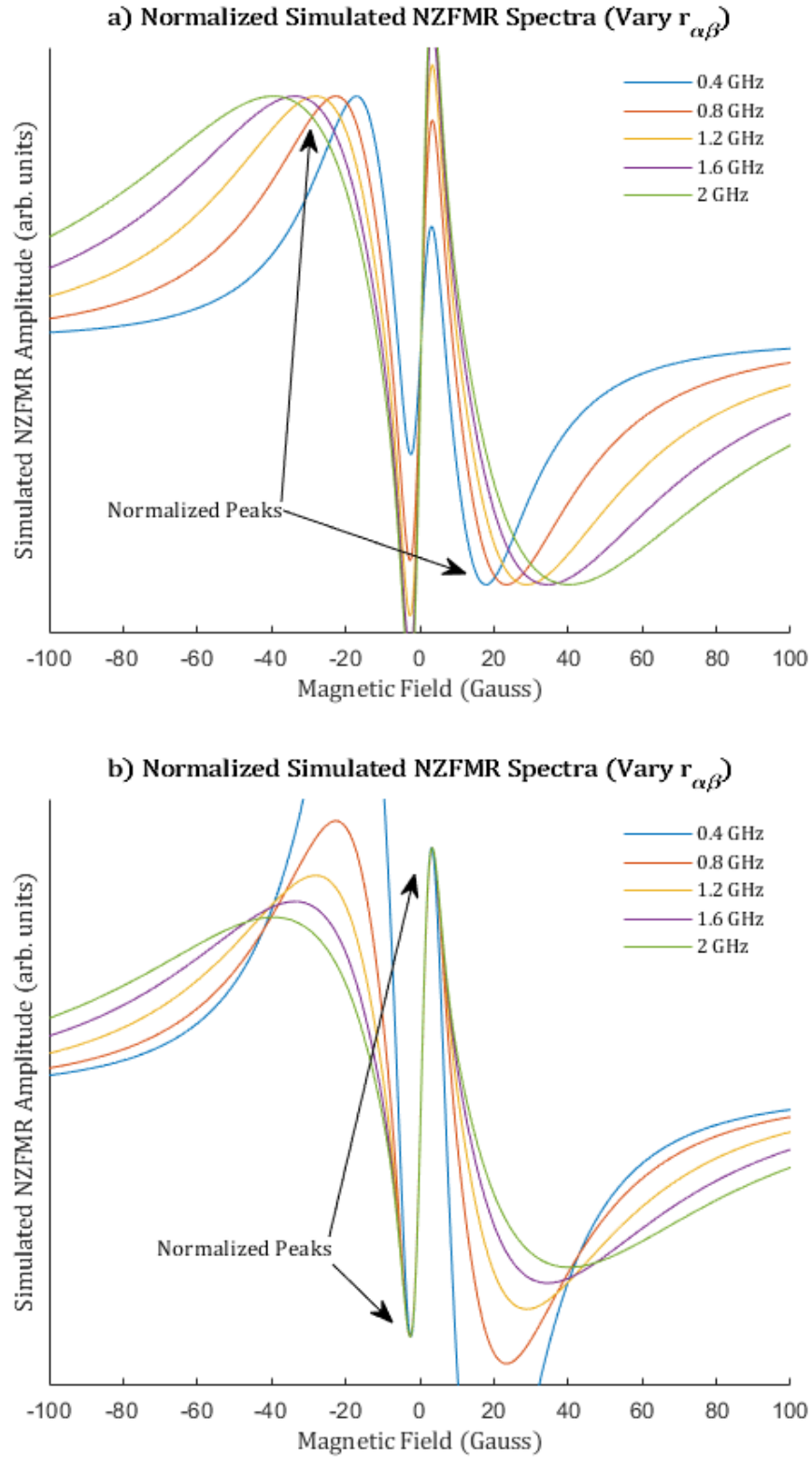


Figure 3.10: Normalized simulated spectra for different values of $r_{\alpha\beta}$. a) Spectra normalized to the broader line to show changes in broad line shape. b) Spectra normalized to inflection line to show changes in inflection line shape. The broad line width increases however the inflection line width is unchanged for increasing values of $r_{\alpha\beta}$.

Chapter 3.2.4.4 Simulated Effects of One Magnetic Nucleus

The hyperfine coupling constant between the magnetic nuclei at site α and a charge carrier at the same site is given as a_α . Large values of a_α indicates that a large magnetic field is experienced by the charge carrier due to the nearby magnetic nuclei is. This section discusses the effect of a single magnetic nucleus at site α and a non-magnetic nucleus at site β . The effect of the hyperfine coupling at site α on the simulated spectra is shown in Figure 3.11. Larger hyperfine interactions lead to more spin mixing and therefore larger signal amplitudes. The amplitude of the inflection line appears to become larger relative to the amplitude of the broad line as well. These spectra were then normalized to the broad line and the inflection line to see how the hyperfine interactions affect the shape and width of the lines. The spectra normalized to the broad line are shown in the top of Figure 3.12 and the spectra normalized to the inflection line are shown in the bottom of Figure 3.12. It is clear to see that larger hyperfine interactions cause both the broad lines and inflection lines to become wider. Hyperfine interactions are unlikely to change over the course of a study unless new defects with different interactions are introduced during the study.

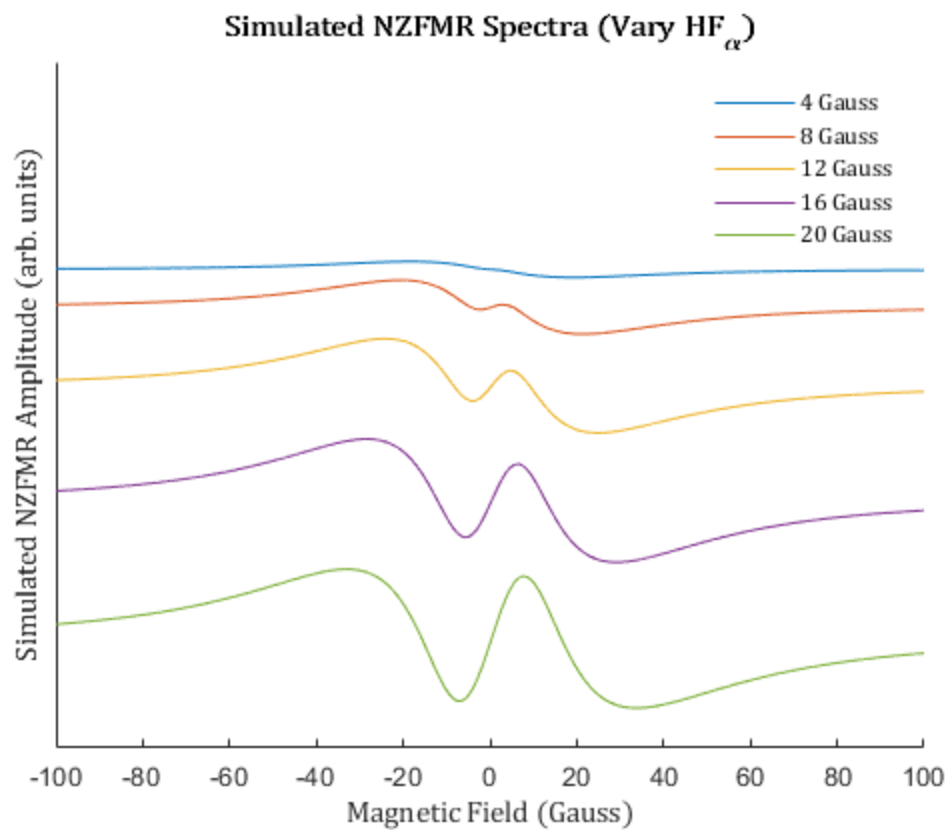


Figure 3.11: Raw simulated NZFMR spectra for different hyperfine values at site α with a non-magnetic nucleus at site β . The order shown in the legend matches the order of the plots.

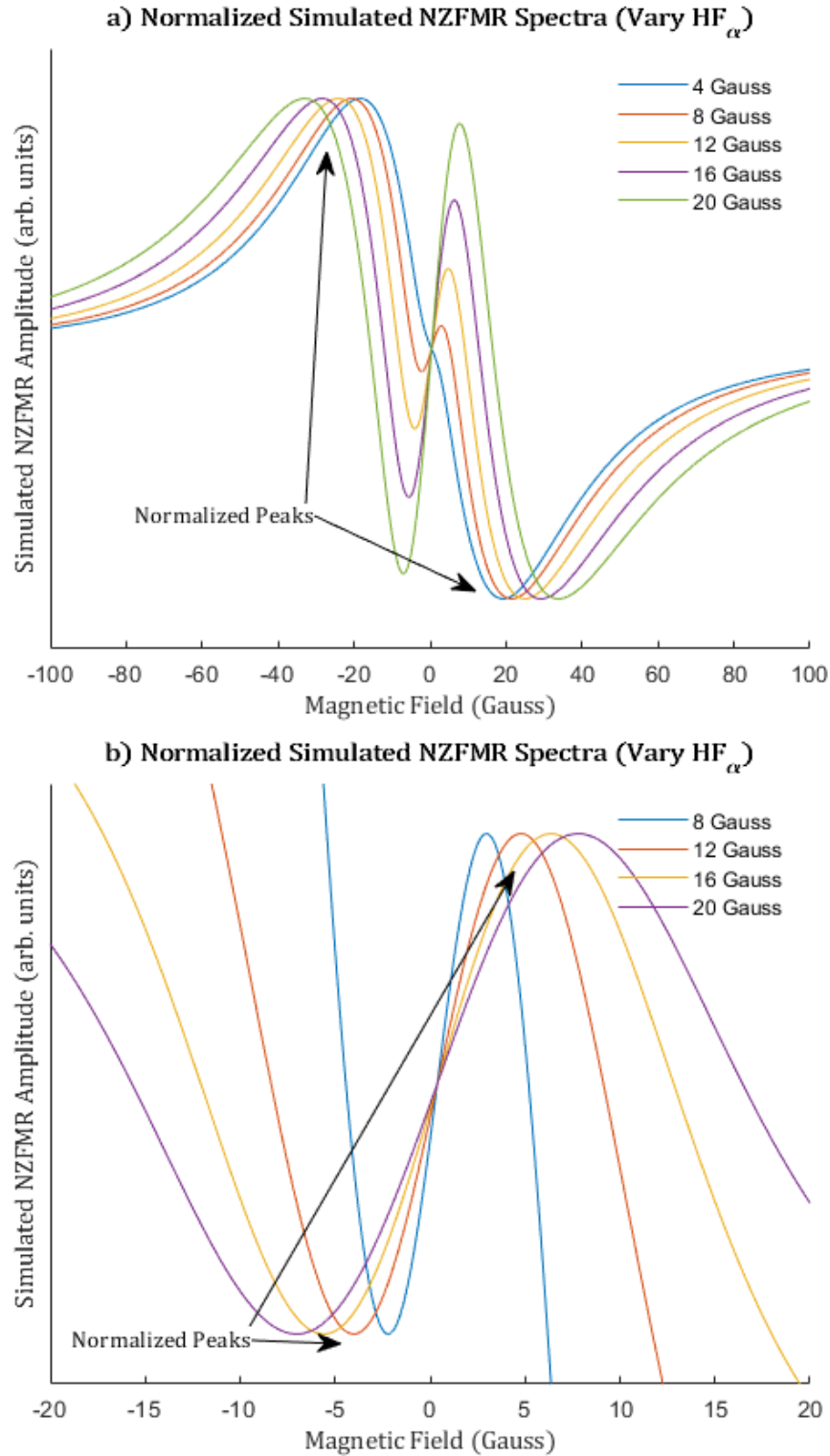


Figure 3.12: Normalized simulated spectra for different hyperfine values at site α with a non-magnetic nucleus at site β a) Spectra normalized to the broader line to show changes in broad line shape. b) Spectra normalized to inflection line to show changes in inflection line shape (one line is missing in the lower plot as the first simulation has no peaks in the inflection). Both the broad line and inflection line become wider for stronger hyperfine interactions.

Chapter 3.2.4.5 Simulated Effects of Two Magnetic Nuclei

This section will look at the effects of changing the hyperfine interactions between a charge carrier and a magnetic nucleus at site β with another magnetic nucleus at site α . The changes are similar to those seen in the previous section however not identical. The effect of changing the hyperfine interactions at site β are shown in Figure 3.13. For this case, the broad line increases in amplitude as the hyperfine interactions on site β become stronger, however the inflection line initially increases in amplitude, then decreases. These spectra were then normalized to the broad line and inflection line to see how the hyperfine interactions affect the shape and width of the lines. The normalized spectra are shown in Figure 3.14 with the spectra being normalized to the broad line in the top plot, and to the inflection line in the bottom plot. Both lines become broader as the hyperfine interactions are stronger however the effect is less pronounced than the previous case of only one site having a magnetic nucleus. Again, these values are unlikely to change during the course of a study unless a new defect is created (or an existing one is destroyed) due to stressing or charge injection.

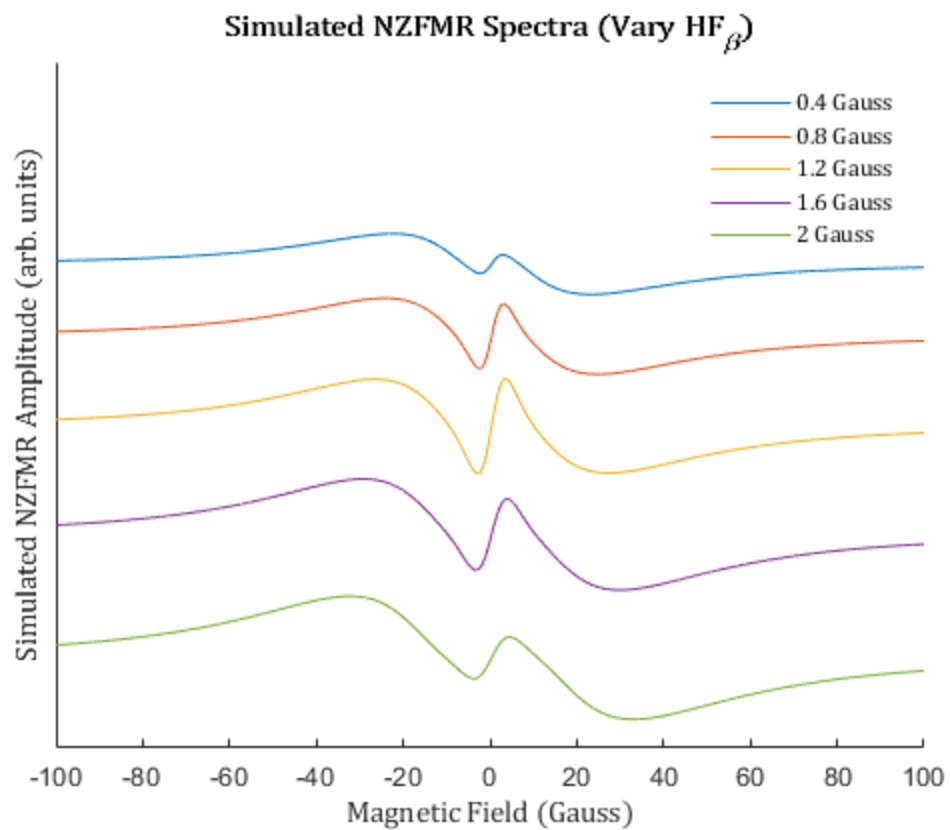


Figure 3.13: Raw simulated NZFMR spectra for different hyperfine values at site β with a magnetic nucleus at site α . The order shown in the legend matches the order of the plots.

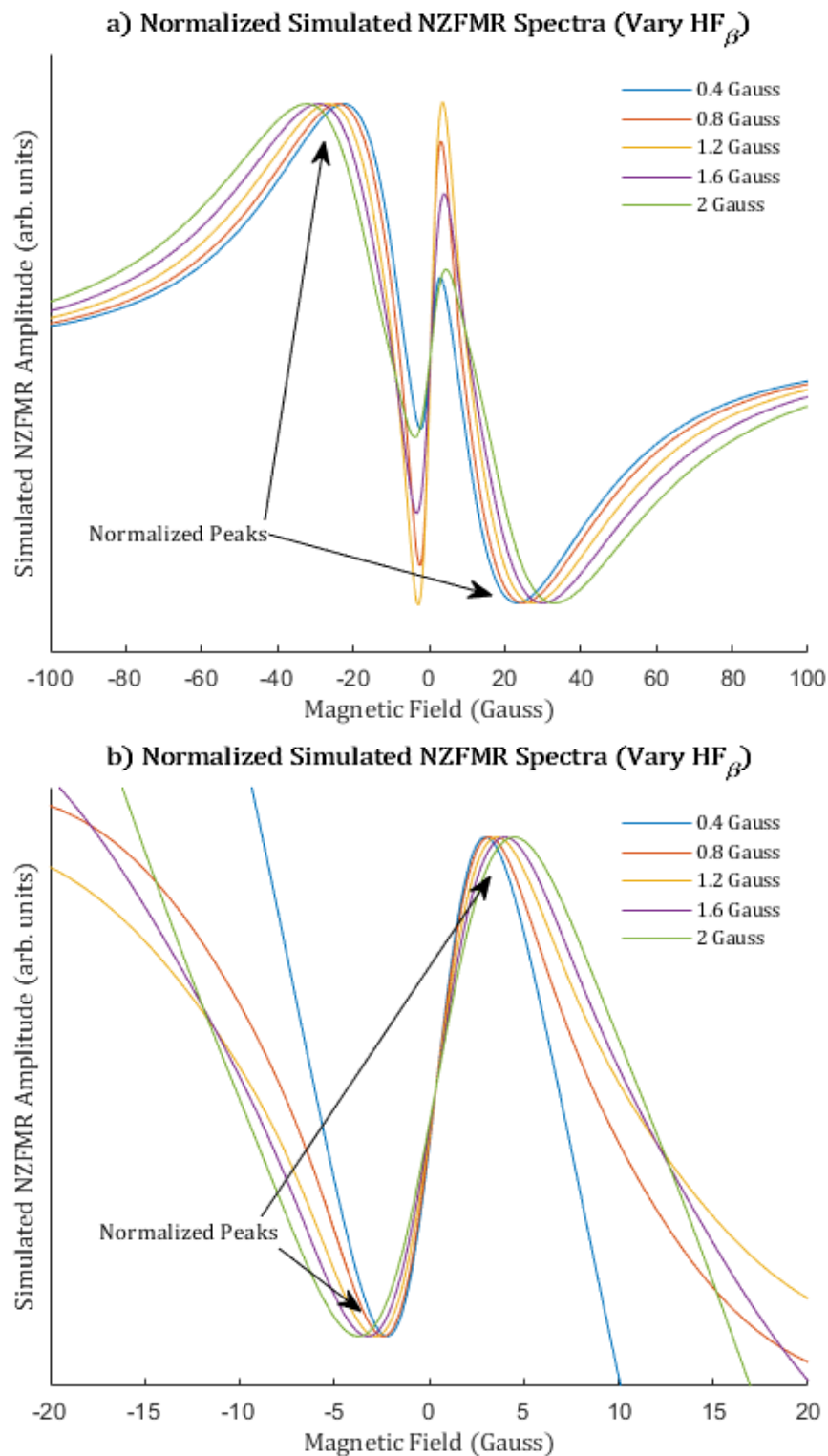


Figure 3.14: Normalized simulated spectra for different hyperfine values at site β with a magnetic nucleus at site α
a) Spectra normalized to the broader line to show changes in broad line shape. b) Spectra normalized to inflection line to show changes in inflection line shape. Both the broad line and inflection line become wider for stronger hyperfine interactions.

Chapter 4

Experimental Methods

The spectrometers utilized during the course of this study are described in this chapter. Each spectrometer will be given an identifying name which will be referred to when discussing experiments in this work. The breakdown of the spectrometers used includes two modified X-band EPR/EDMR spectrometers and two low-field EDMR/NZFMR spectrometers. One custom built wafer probing station spectrometer is utilized in appendix B however this will be discussed in greater length in that chapter.

Chapter 4.1 Bruker X-band EPR/EDMR Spectrometer

A modified commercial Bruker X-band EPR Spectrometer was utilized for some of the resonance measurements performed in this study. This spectrometer, which will be referred to as the “Bruker X-band spectrometer” consists of a commercial Bruker water cooled electromagnet and control system, Bruker 300 series X-band microwave bridge and plumbing, Bruker TE₁₀₄ X-band microwave cavity, Stanford Research Systems SR570 current to voltage pre-amplifier, Stanford Research Systems SR830 external lock-in amplifier, and computer for data acquisition with a National Instruments NI6259 data acquisition card and associated BNC 2110 DAQ connector block. Resonance measurements performed on this system utilized a modulation frequency of 12.5 kHz as it is the modulation frequency that provides the best signal to noise ratio in this spectrometer. The TE₁₀₄ cavity allows one to easily switch between resonance measurements on the sample and a known standard without removing anything from the cavity. A schematic diagram of this spectrometer is shown in Figure 4.1

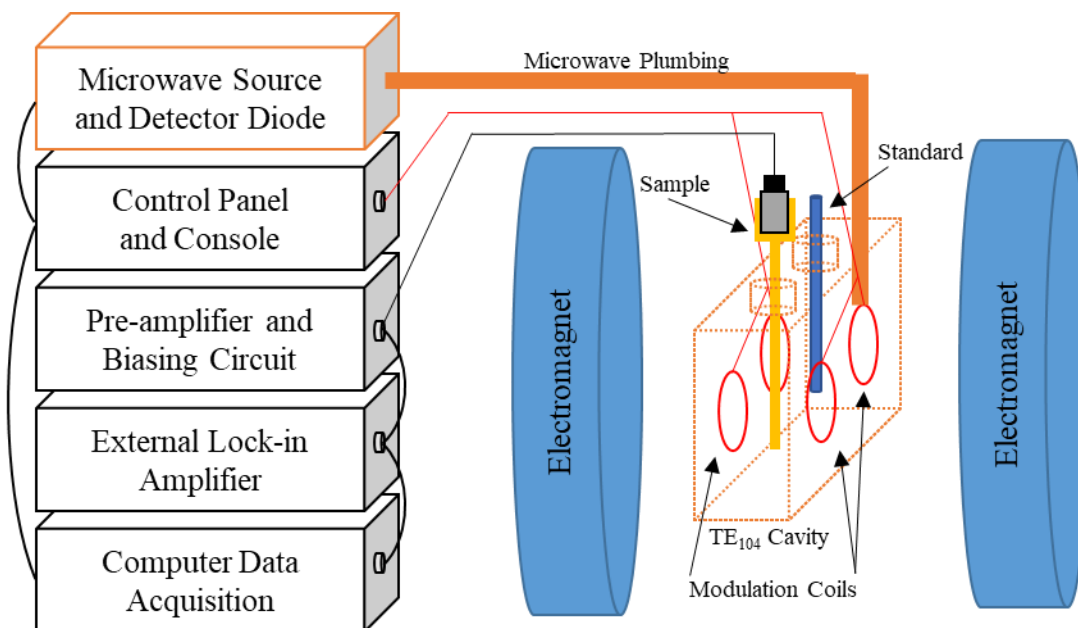


Figure 4.1: Cartoon schematic for Bruker X-band EDMR Spectrometer with TE_{104} microwave cavity.

Chapter 4.2 Lakeshore X-band EPR/EDMR Spectrometer

The second X-band spectrometer utilized in this study is a home-built system which shall be referred to as the “Lakeshore X-band spectrometer”. This spectrometer consisting of a Lakeshore water cooled electromagnet, a Lakeshore Cryotronics 450 temperature compensated Gaussmeter and Hall effect probe, a Micronow Systems microwave bridge with associated microwave plumbing and TE_{102} cavity, a Stanford Research Systems SR570 current to voltage pre-amplifier, and a computer for field control and data acquisition with a National Instruments PCIe-6374 data acquisition card and associated BNC 2110 DAQ connector block. Resonance measurements performed on this system could use any desired modulation frequency from near DC to approximately 20kHz. Modulation frequencies near 1 kHz were typically chosen. This system is capable of measuring the DC current in a sample during EDMR and NZFMR measurements. A cartoon schematic for the Lakeshore X-band is shown in Figure 4.2.

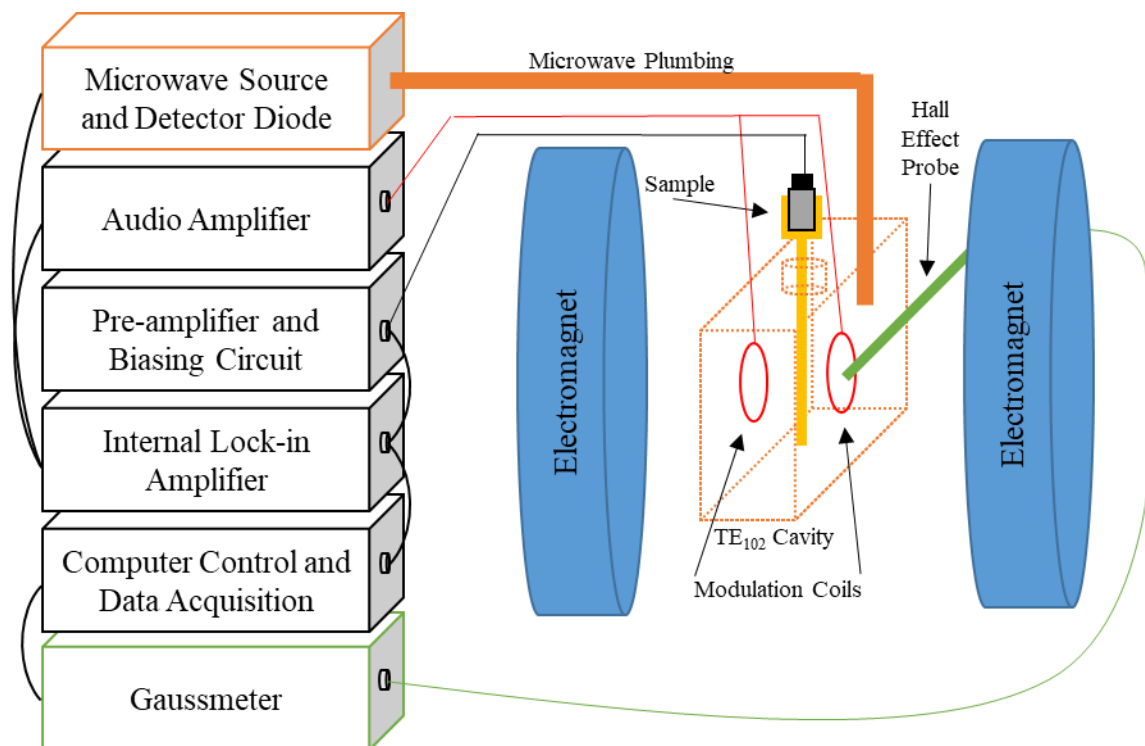


Figure 4.2: Schematic of Lakeshore X-band spectrometer with TE₁₀₂ microwave cavity.

Chapter 4.3 Home Built Kepco 100V 4A Low-Field EDMR/NZFMR Spectrometer

Some of the low-field EDMR and NZFMR measurements performed in this study were made on a home-built system which will be referred to as the “Kepco low-field/zero-field spectrometer” or “Kepco LF/ZF spectrometer”. This low-field EDMR and NZFMR spectrometer consists of a home-built nested Helmholtz coil electromagnet, a Kepco 400W 100V 4A Bipolar Power Supply, a Lakeshore Cryotronics 450 temperature compensated Gaussmeter and Hall effect probe, a Stanford Research Systems SR570 Current to Voltage Pre-amplifier, and a computer for field control and data acquisition with a National Instruments NI 6259 data acquisition card and associated BNC 2110 DAQ connector block. Resonance measurements performed on this system could use any desired modulation frequency from near DC to approximately 20kHz. Modulation frequencies near 1 kHz were typically chosen. The B_1 field

for the low field EDMR was provided by a DOTY 500MHz surface coil resonator with accompanying matching circuit. This system is capable of measuring the DC current in a sample during EDMR and NZFMR measurements. A schematic of the Kepco LF/ZF spectrometer is shown in Figure 4.3, adapted from Cochrane and Lenahan. [68]

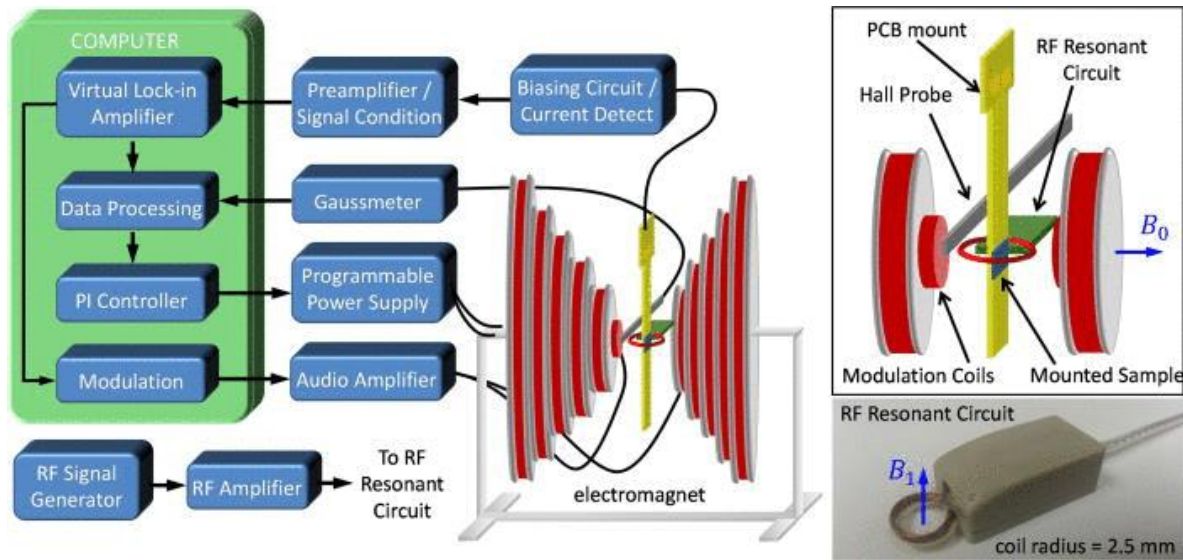


Figure 4.3: Schematic of Kepco LF/ZF spectrometer. Figure adapted from Cochrane and Lenahan. [68]

Chapter 4.4 Blue Spin, LLC Low Field EDMR/NZFMR Spectrometer

More low-field EDMR and NZFMR measurements performed in this study were made on a modified commercial Blue-Spin, LLC nested Helmholtz coil electromagnet and accompanying Blue-Spin electronics, computer, and data acquisition card, a Stanford Research Systems SR570 current to voltage preamplifier, SONY audio amplifier, and DOTY 350MHz surface coil resonator with accompanying matching circuit to provide the B_1 field for the low field EDMR. This spectrometer will be referred to as the “Blue Spin LF/ZF spectrometer”. The Blue Spin LF/ZF spectrometer is schematically very similar to the Kepco LF/ZF spectrometer in Figure 4.3.

Chapter 4.5 Other Equipment

Electrical measurements were made on other commercial testing equipment including an HP 4155 Semiconductor Parameter Analyzer and Boonton Electronics Model 72B capacitance meter. The Boonton capacitance meter will simply be referred to as the “capacitance meter”. A Kulicke and Soffa Model 4123 wire bonder was used to make electrical connections to the top contact of the samples. This process will be described as “wire bonding”.

Most samples were mounted onto custom made printed circuit board (PCB) “T’s” to make electrical contact while the sample is located inside the microwave cavities or low field electromagnets described previously. A USB connector is soldered to four contacts connected to parallel copper leads. Samples are bonded to a contact pad approximately four inches from the USB head using a conductive adhesive to create electrical contact between one lead and the back of the sample. The other three leads can then be wire bonded to bonding pads on the top of the sample to make electrical contact to the rest of the sample. A schematic of a “T” is shown in Figure 4.4. The sample and “T” are typically housed inside a quartz test tube to protect the sample from damage.

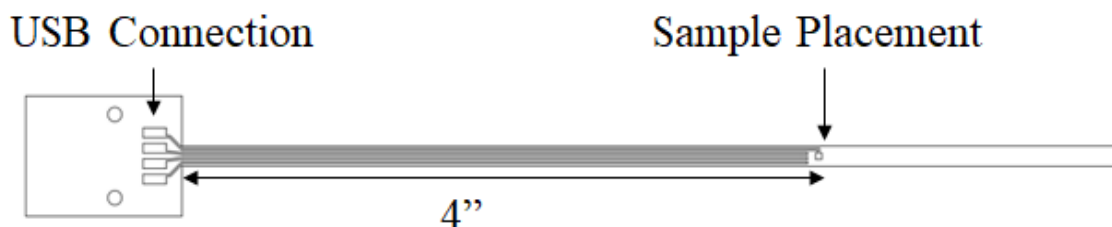


Figure 4.4: Schematic diagram of custom-made PCB "T" sample mount with copper leads. A USB connector is soldered to the 4 pads on the left side.

Chapter 5

Electrically Detected Magnetic Resonance and Near Zero Field Magneto-Resistance in Amorphous Carbon

SDTAT EDMR and NZFMR, along with other measurements, were performed on two types of amorphous carbon. The first of these samples were 500 nm amorphous hydrogenated carbon (a-C:H) hard mask film on p- silicon. The second was 1000 nm diamond-like (DLC/tetrahedral) carbon hard mask film on p- silicon. All amorphous carbon samples were grown using plasma enhanced chemical vapor deposition (PECVD) and have an aluminum/titanium top contact. The a-C:H samples are primarily sp^2 (triangularly) coordinated carbon with a hydrogen composition of 33% while the DLC samples are approximately half sp^2 coordinated and half sp^3 (tetrahedrally) coordinated carbon with a similar hydrogen content of 27%, there is only short and medium range order in these samples.

Atoms with sp^2 hybridized bonds have three nearest neighbors, with bond angles of 120 degrees in a planar triangle. Atoms with sp^3 hybridized bonds have four nearest neighbors, with bond angles of 109.5 degrees. Both of these bond angles can vary in materials due to strain and disorder. Illustrations of the sp^2 and sp^3 hybridized orbitals are shown in Figure 5.1.

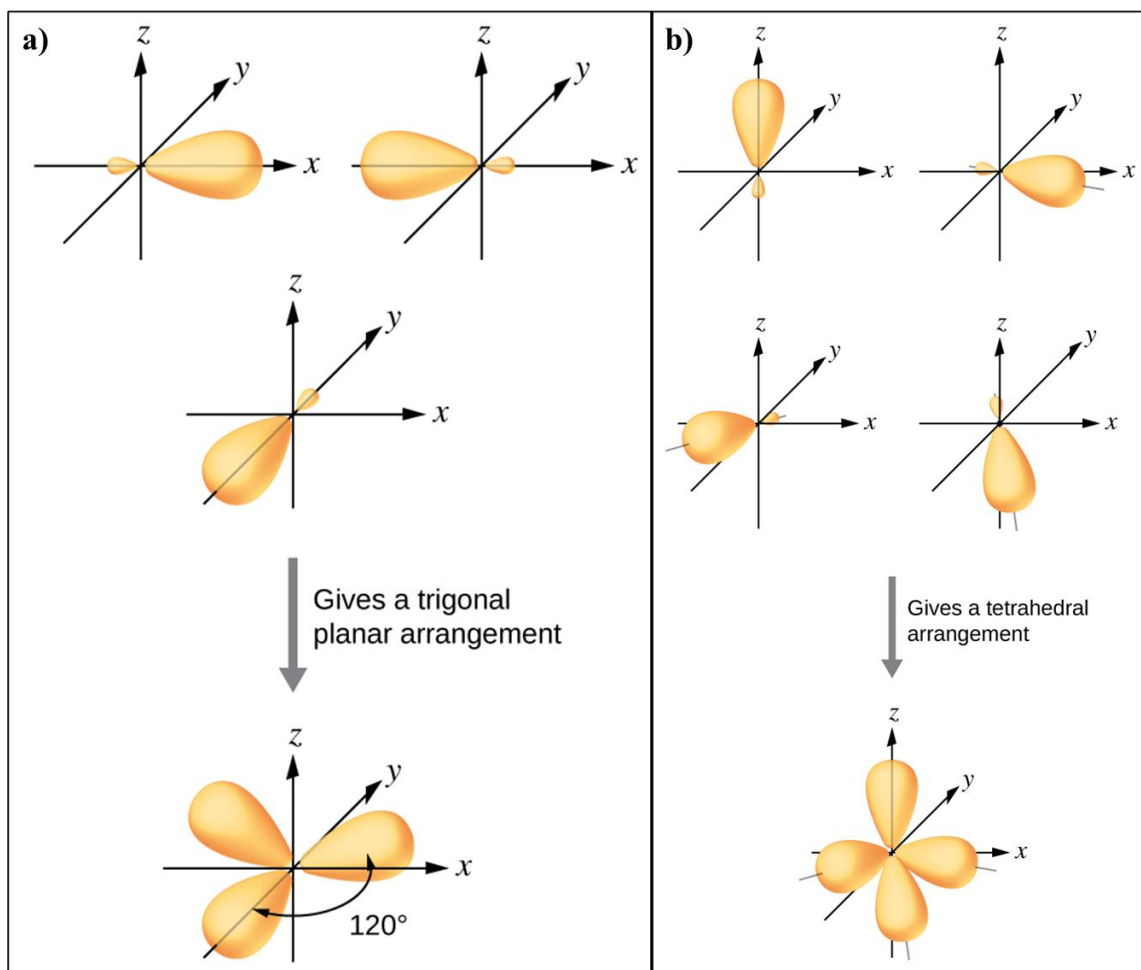


Figure 5.1: Diagrams of a) sp^2 hybridized orbitals with bond angles of 120 degrees and b) sp^3 hybridized orbitals with bond angles of 109.5 degrees. Adapted from [69].

Chapter 5.1 Defects and Electrical Conduction in Amorphous Carbon

The EDMR and NZFMR spectra measured in this work will be easier to understand with more knowledge of the known defects in these types of materials. Forms of amorphous carbon have been studied fairly extensively using EPR. [70]–[83] While a-C:H and DLC have different amounts of sp^2 and sp^3 coordination, the dangling bond defect in both materials occurs primarily on sp^2 sites because the sp^3 dangling bond tends to relax to sp^2 configuration. [70], [72], [84]

The small atomic size of carbon also leads to very little spin-orbit coupling, causing most defects

in amorphous carbon to have g values between 2.0024-2.0029 [71]–[82] although some diamond-like carbon samples have resonance spectra with g values as high as 2.0047. [71], [83] Paramagnetic impurities can also cause higher g values at approximately 2.0053. [71] We expect to observe spectra with zero-crossing g values in the range of 2.0024 to 2.0047. Defect states in amorphous carbon samples have been linked to π orbitals and found in a continuous distribution throughout the bandgap. [70], [84]

Chapter 5.2 EDMR and NZFMR in 500 nm Thick Amorphous Carbon:

SDTAT EDMR and NZFMR as well as other measurements were conducted on 500 nm amorphous hydrogenated carbon on p- silicon capacitor dots. The a-C:H was deposited via PECVD and the top contact is 2mm^2 , composed of approximately 100nm of titanium with approximately $1\mu\text{m}$ of aluminum on top. The composition of the carbon layer is 58% carbon, 33% hydrogen, 7% oxygen, and 1% nitrogen. The structure of the a-C:H is shown in Figure 5.2.

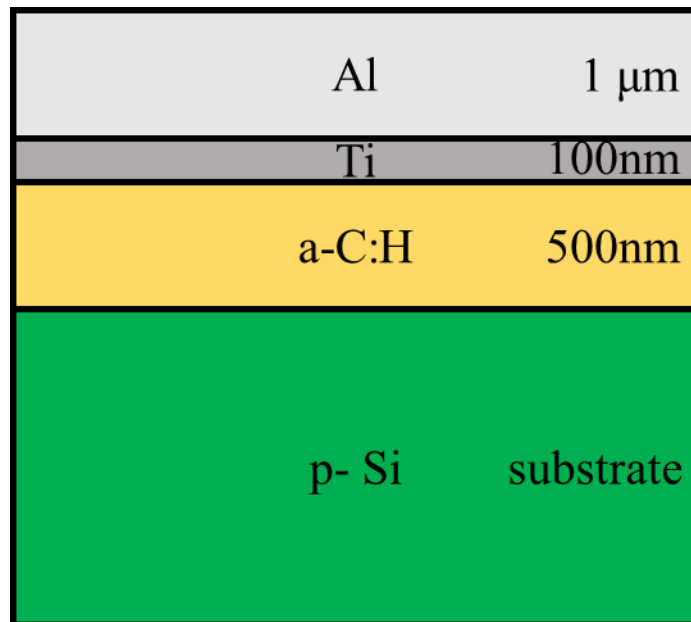


Figure 5.2: Schematic illustration of 500 nm a-C:H samples.

Samples were mounted on custom made USB connected PCB “Ts”, with the back contact connected to a bonding pad via electrically conductive paint. Electrical connection to the top contact was made via wire bonding. The samples were initially electrically characterized using capacitance vs voltage measurements conducted using the semiconductor parameter analyzer and capacitance meter. Some initial current vs voltage measurements were made using the semiconductor parameter analyzer, however, the current in these samples is not stable and decreases rapidly with time. The rate at which the trap assisted tunneling current drops in the samples is larger for larger magnitude biases, and a negative bias has a larger effect than a positive bias.

Chapter 5.2.1 X-band EDMR in 500 nm a-C:H

SDTAT EDMR measurements were made on 500 nm a-C:H in order to help identify the defects involved in the electronic transport in the material. The measurements were made on two X-band EDMR spectrometers. An X-band SDTAT EDMR spectrum from these samples, measured in the home-built Lakeshore X-band spectrometer, is shown in Figure 5.3. This measurement was performed with the sample biased at +4V on the top contact of the capacitor dot with a field modulation amplitude of 5 Gauss at 1150 Hz. The zero-crossing g value of the defect, 2.0022, was determined by comparing the measured spectra to a known standard.

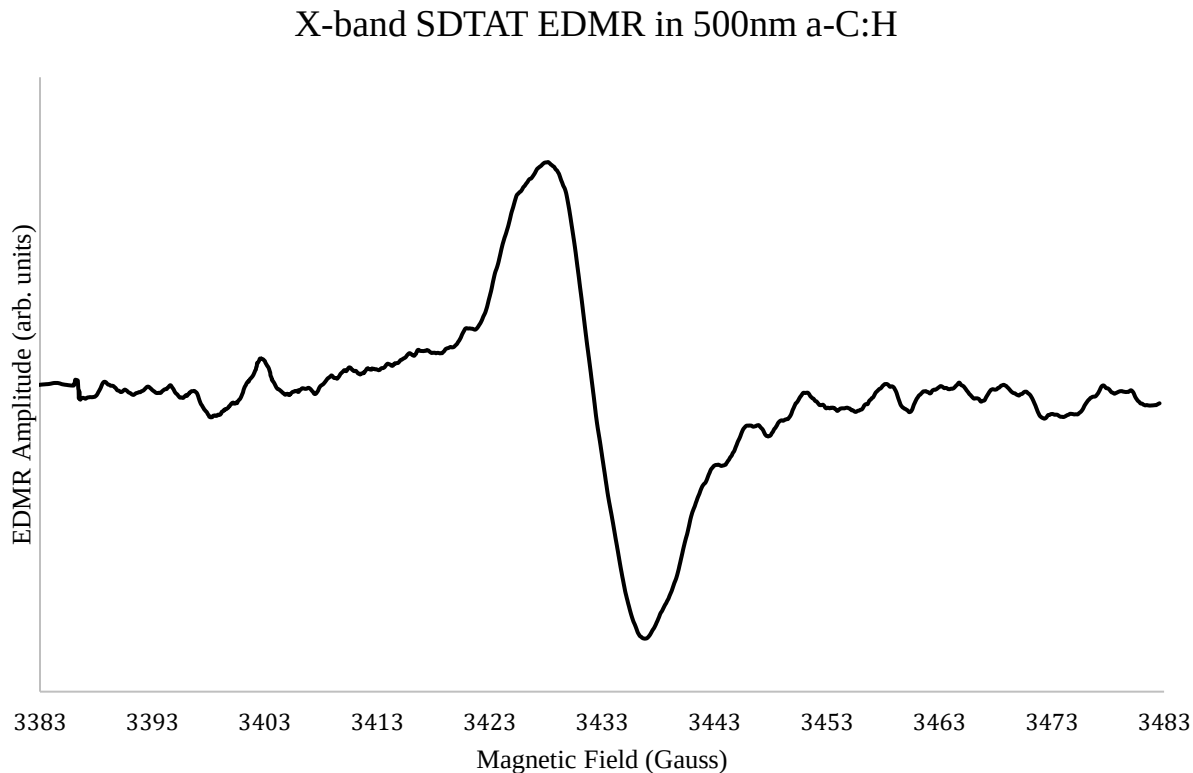


Figure 5.3: X-band SDTAT EDMR in 500nm amorphous carbon on Si. Applied bias of +4V. The signal width is 9.1 Gauss with a zero-crossing g value of 2.0022.

The measurement resulted in a 9.0 Gauss wide spectrum with a zero-crossing g value of 2.0022, which is very close to, and within experimental error of, the free electron g of 2.0023..., suggesting that there is very little spin-orbit coupling in the sample. This result makes sense when we consider the elements present in the sample. The sample is composed primarily of the small elements, carbon and hydrogen, which as discussed in chapter 2.1.3.1 will have very little spin orbit coupling. This defect is likely a carbon dangling bond, as there are no other atomic nuclei present in a high enough concentration that can have a dangling bond in the material, and the measured g -value aligns with the expected values discussed in chapter 5.1. Electrons likely hop from site to site across the material through these dangling bond defects.

X-band SDTAT EDMR was performed on a similar sample under negative bias using the Bruker X-band spectrometer. The sample was biased at a constant -3V and a magnetic field modulation of 5 Gauss at 12.5kHz was used. The increased modulation frequency was utilized to improve the signal to noise ratio on the spectrometer. The results are shown in Figure 5.4. The signal amplitude decreased rapidly over time and a representative set of spectra over approximately 30 minutes of measurements (1 minute per trace) are shown as individual scans without averaging. The zero-crossing g value of the spectra measured under negative bias is 2.0028, within the expected range of g values for sp^2 coordinated carbon dangling bonds.[71] The lines are all 9 Gauss wide, the same as the line width under positive bias. The defects giving rise to these spectra at both positive and negative bias are likely the same carbon dangling bond.

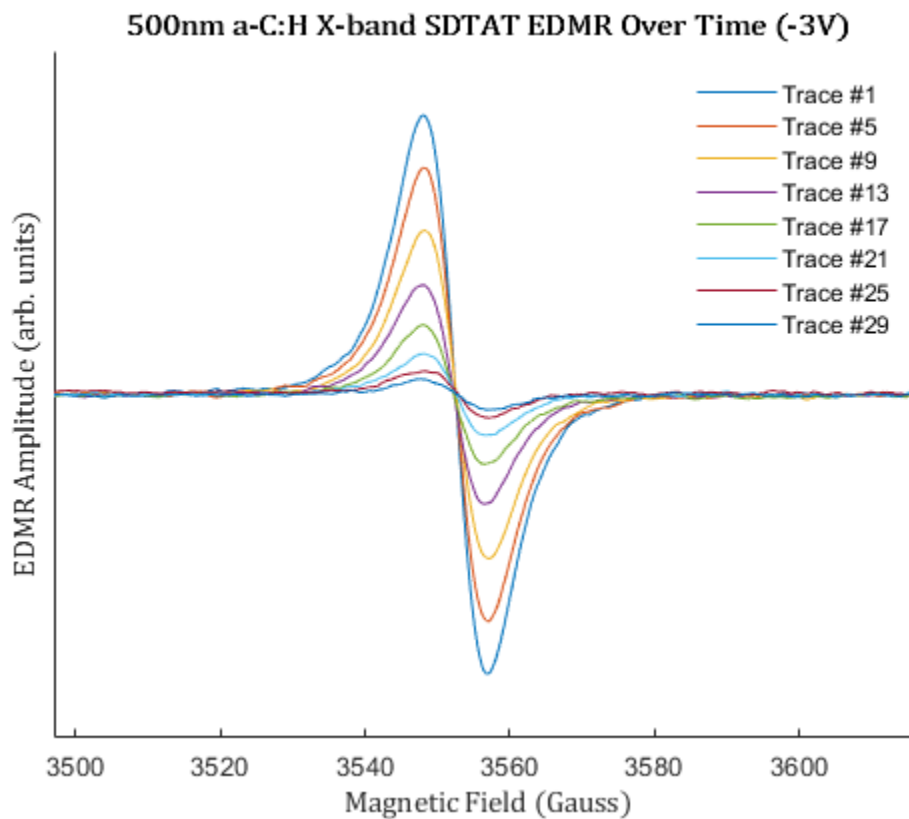


Figure 5.4: X-band SDTAT EDMR in 500nm a-C:H over time under constant applied bias of -3V. The signal amplitude decreases rapidly with time. Each trace takes approximately 1 minute to complete.

Typically, signal averaging is utilized in EDMR and NZFMR measurements in order to improve the signal to noise ratio of the measurement. The signal averaging technique requires that the DC current in the sample and the measured spectrum stay relatively consistent over the period of measurement, typically hours to days. The a-C:H samples do not exhibit such stable behavior as both the DC current through the sample and the amplitude of the measured spectra change rapidly with time. Fortunately, the signal to noise ratio of the measured spectra are large enough to not require signal averaging. The reduction in DC current over time is discussed in more detail in Chapter 5.2.2.

The amplitude of the measured spectra decreases rapidly over time until it levels off after prolonged applied bias. This effect occurs when the sample is under negative bias, and very little change is observed in samples in which only a positive bias is applied. The amplitude of the measured SDTAT EDMR spectra over time is shown in Figure 5.5.

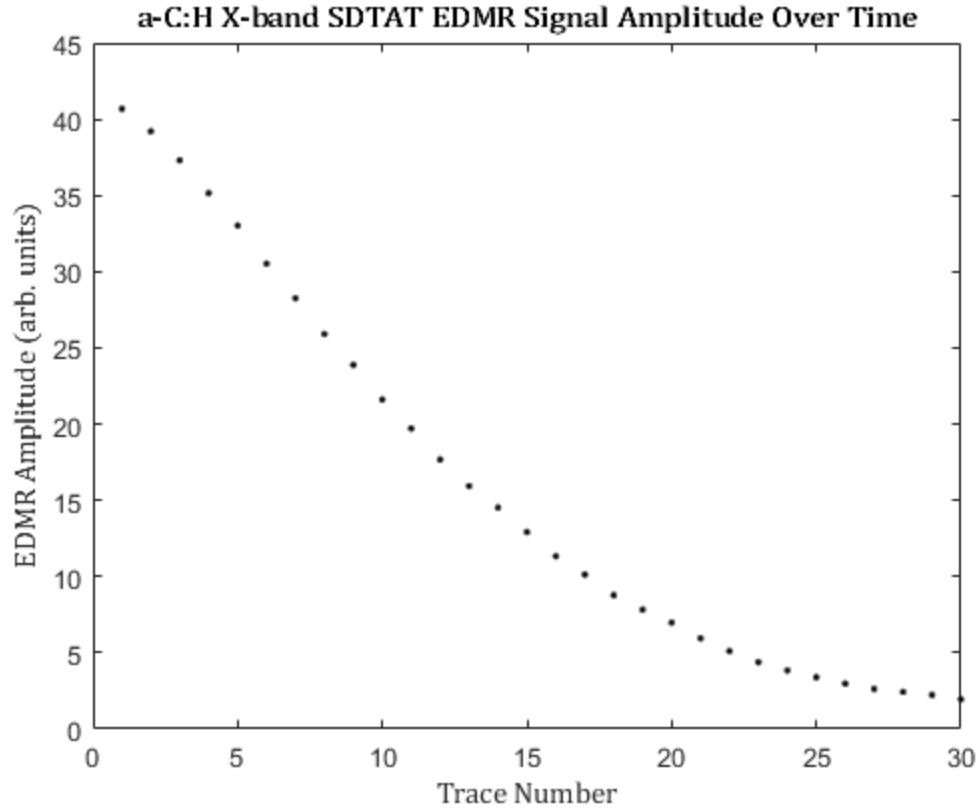


Figure 5.5: Amplitude of measured SDTAT EDMR spectra in 500nm a-C:H over time. The amplitude of the measured spectra decreases rapidly at first, then levels off after a prolonged period of time.

Although the signal amplitude decreases over time at negative bias, the shape and linewidth of the signal do not change. Normalized spectra are shown in Figure 5.6 in which the shape of the line is unchanged, indicating that the same defect is responsible for conduction through the material and there is little or now change in the types of defects involved in leakage current during the stressing.

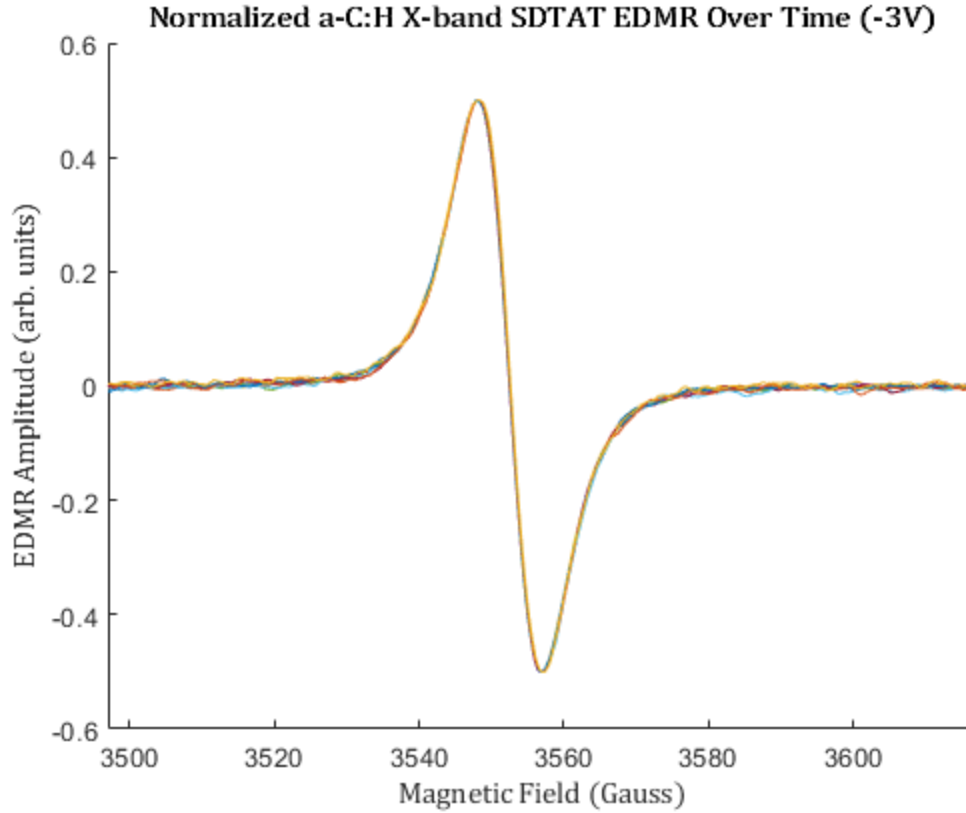


Figure 5.6: Normalized SDTAT EDMR in 500nm a-C:H over time. The signal shape and linewidth do not change over approximately 30 minutes.

Chapter 5.2.2 NZFMR and Low Field EDMR in 500 nm a-C:H

After performing X-band SDTAT EDMR on these samples, SDTAT NZFMR and low field EDMR measurements were performed on fresh samples. These measurements were made on the Kepco LF/ZF spectrometer because DC current can be measured while performing EDMR and NZFMR. Some of the NZFMR results from these samples are shown in Figure 5.7. These measurements were made with a magnetic field modulation of 1 Gauss at a frequency of 1000 Hz. The device was biased at -5V on the top contact. The signal amplitude decreases in the same way that the X-band EDMR signal did.

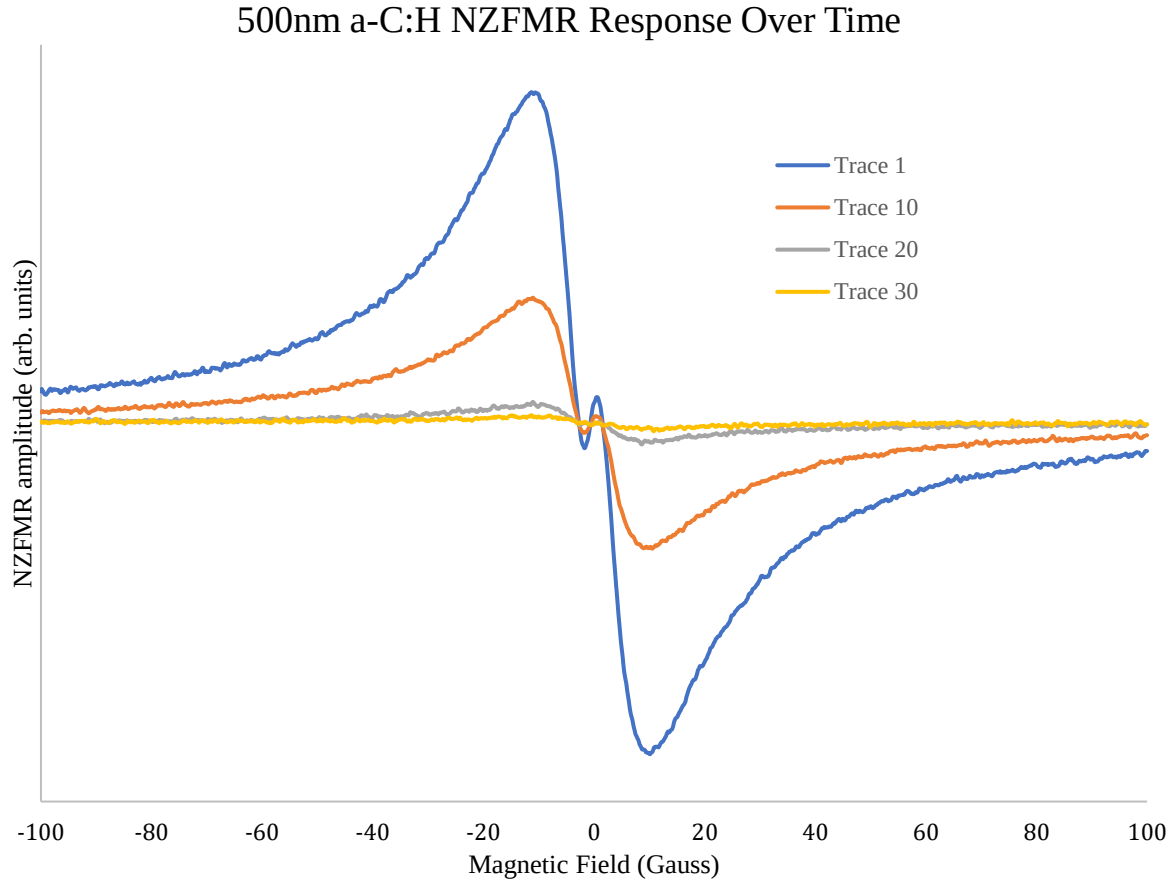


Figure 5.7: SDTAT NZFMR response in 500 nm amorphous carbon over time. Each trace takes approximately 10 minutes to measure. The signal amplitude decreases over time but remains 21 Gauss wide. Signal amplitude is not normalized. Measurements were made using a 1 Gauss modulation amplitude at 1kHz. Sample biased at -5V.

The high signal to noise ratio and accurate line-shape of these signals make them good candidates to analyze in terms of the stochastic Liouville equation described in chapter 3. As stated previously, the defects involved in conduction are likely carbon dangling bonds. The hyperfine interactions in this particular material likely arises from nearby hydrogen nuclei, as they are the only magnetic species in the sample in high enough concentration to be relevant. Only 1.1% of the carbon nuclei possess a magnetic nucleus, whereas nearly 100% of the hydrogen nuclei are magnetic. Recall that the hydrogen content in these films is very high, about

33%. A simulated spectrum was fitted to the 21 Gauss wide experimental spectrum. The simulated and measured spectra are shown in Figure 5.8.

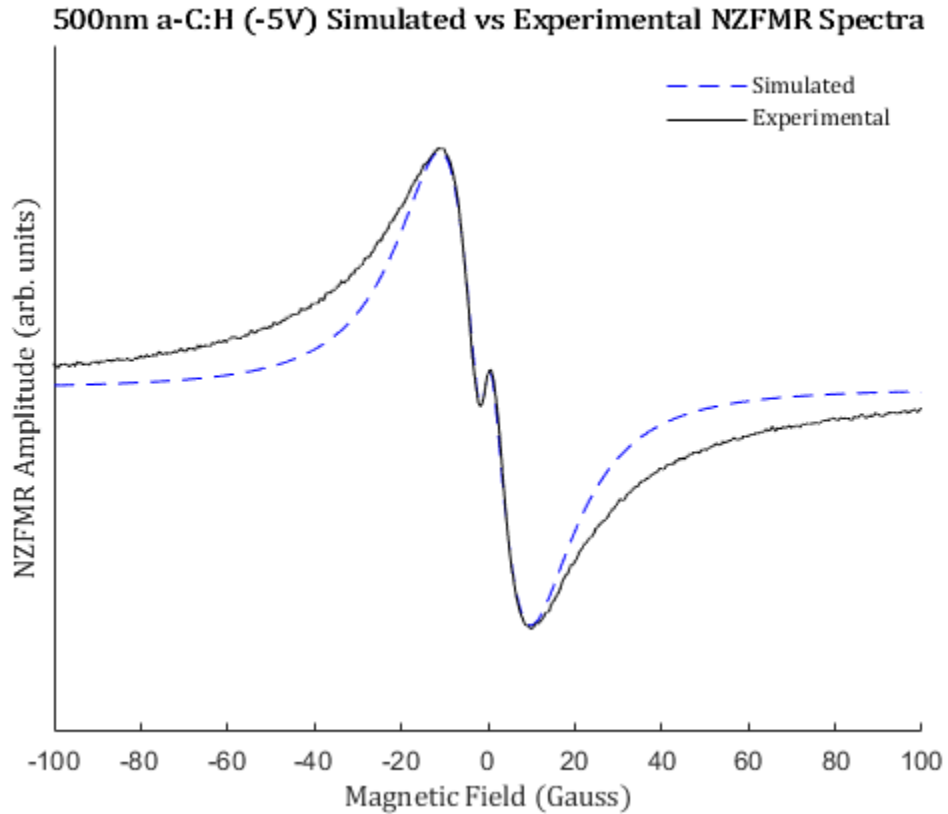


Figure 5.8: Simulated and measured NZFMR spectra in 500nm a-C:H. The measured spectrum is the largest of the spectra shown in the previous data set. The simulated spectrum utilizes values of $r_{e\alpha} = 530 \pm 25 \text{ GHz}$, $r_{ae} = 97 \pm 10 \text{ MHz}$, $r_{\alpha\beta} = 340 \pm 50 \text{ MHz}$, and hyperfine coupling constants on both sites of $3.4 \pm 0.5 \text{ Gauss}$.

The simulated spectrum was fit using a least squares method outlined by Frantz et al. [64]. The resulting hyperfine coupling constants for sites α and β were both found to be approximately $3.4 \pm 0.5 \text{ Gauss}$, as reasonable fits could be made using coupling constants within approximately 0.5 Gauss . This is considered a hop from a site with a nearby hydrogen to another site with a nearby hydrogen. The resulting generation rate from the fit, $r_{e\alpha}$, was approximately $530 \pm 25 \text{ GHz}$, while the resulting dissociation rate from the fit, r_{ae} , was approximately $97 \pm 10 \text{ MHz}$. The

resulting singlet probability of hopping rate, $r_{\alpha\beta}$, was approximately 340 ± 50 MHz. This fitting algorithm was constructed to consider the contribution from a single species, assumed to be hydrogen because its concentration of magnetic nuclei was 2 orders of magnitude higher than that of carbon in this film.

These values appear to be reasonable for this sample, indicating that the computational model, though not perfect, is a useful analytical tool that can be used to extract certain parameters from NZFMR spectra. The value of the hyperfine coupling constant is physically plausible. It must be a value between the magnetic field due to a nearby hydrogen nucleus at a distance of a few angstroms (approximately 1 Gauss) and the low-field EDMR linewidth (approximately 7 Gauss). The linewidth in low-field EDMR measurements is dominated by hyperfine interactions and a hyperfine coupling constant greater than 7 Gauss in this sample would result in a wider low-field EDMR line. A hopping rate larger than the dissociation rate is also plausible as the current is dominated by variable range hopping events.

Current through the sample dropped significantly during the measurement along with the signal amplitude. The drop in the steady state DC current over time is shown in Figure 5.9.

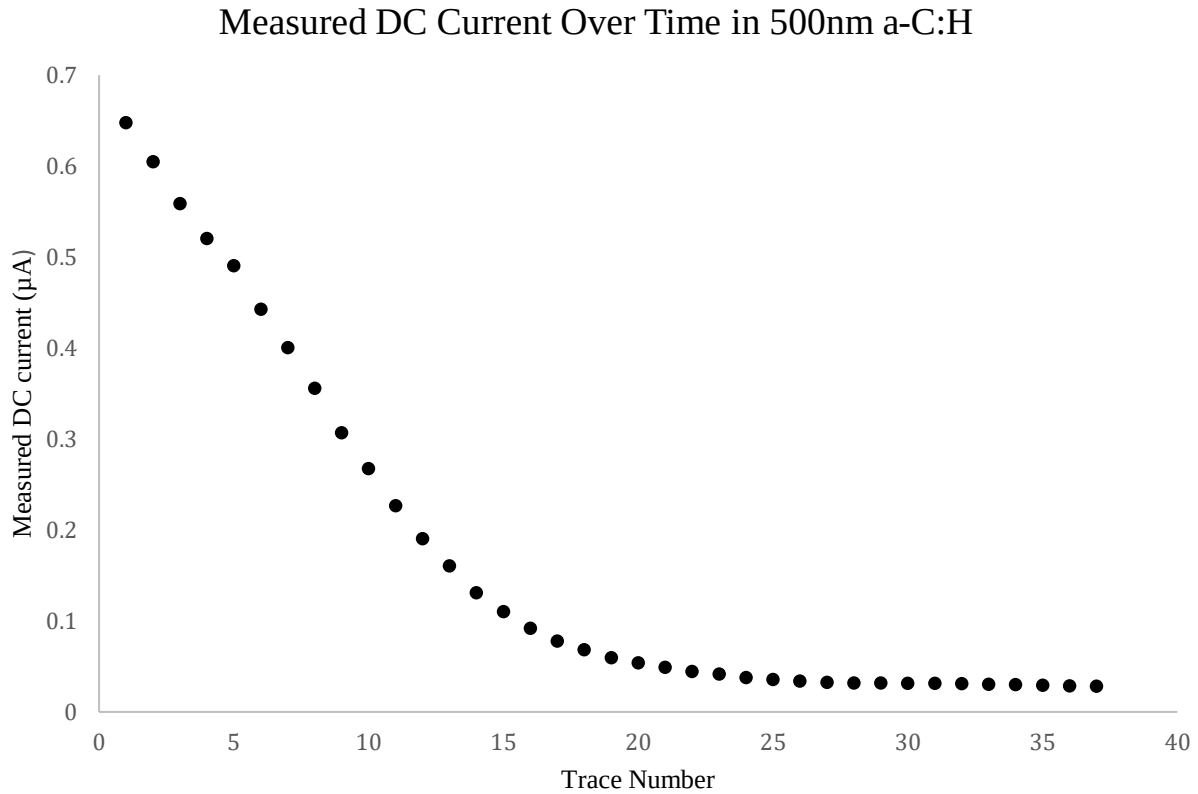


Figure 5.9: DC current over time in 500nm a-C:H under constant -5V bias during NZFMR measurement.

While both the DC current and the NZFMR signal amplitude both decrease over time, the NZFMR signal amplitude decreases faster, with the percent change of the current due to NZFMR decreasing over time under a constant bias. This is shown in Figure 5.10.

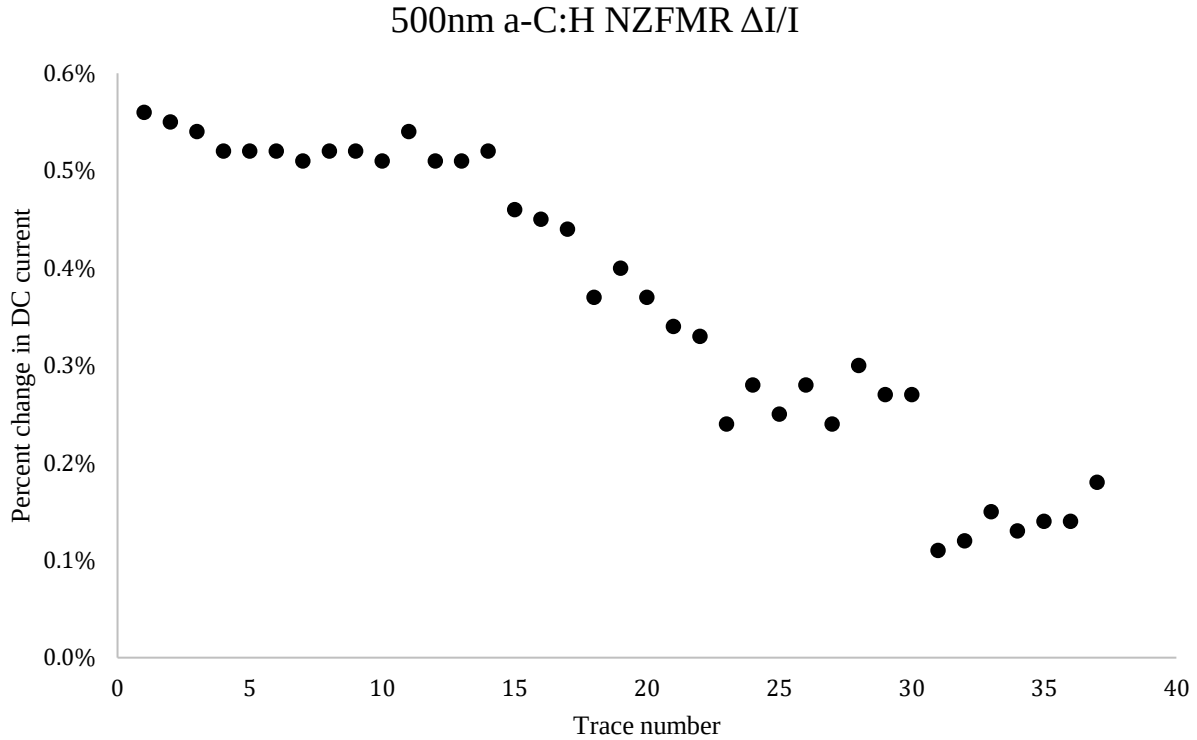
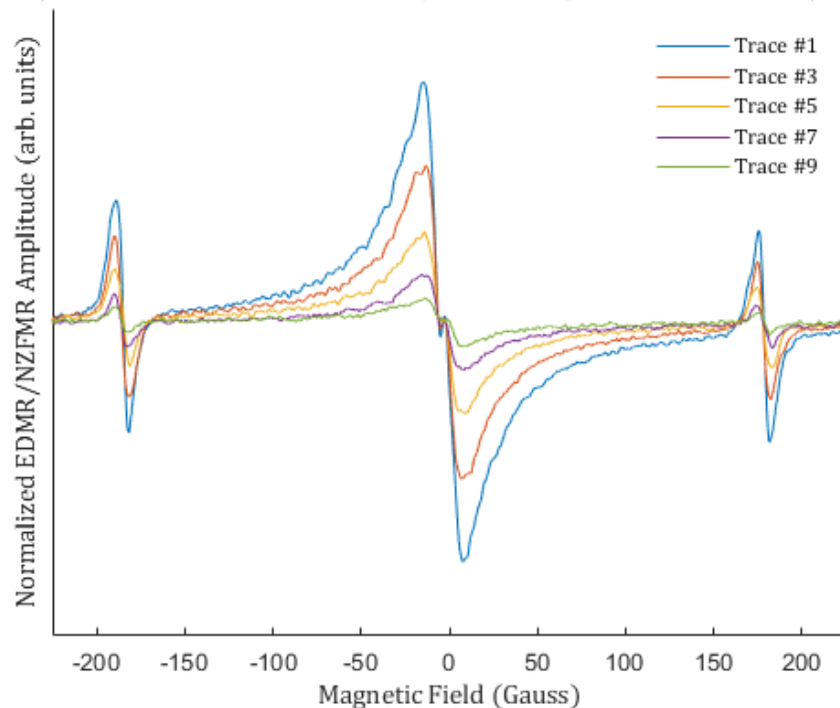


Figure 5.10: Percent change in current due to NZFMR response during measurement under constant -5V bias.

Further studies on these samples showed that the current through the device would decrease monotonically when under negative bias until it stabilized at a lower value which could be two orders of magnitude smaller than the initial current. During this change in current and signal amplitude under a constant bias, there is no discernable change to the line shape or line width of the signals. The reduction in amplitude is shown in Figure 5.11 a) which depicts both the NZFMR and low-field EDMR response over time under a constant -4V bias. The constant shape is shown in Figure 5.11 b) with the normalized spectra. The current in this measurement decreases like in Figure 5.9, with a starting current of approximately 10 μA at the moment the bias was applied. This current dropped to approximately 500 nA after 20 minutes under constant bias. For these measurements, each trace took approximately 2 minutes to complete.

a) 500nm a-C:H SDTAT NZFMR/EDMR Amplitude over Time (-4V)



b) a-C:H Normalized SDTAT NZFMR/EDMR Amplitude over Time (-4V)

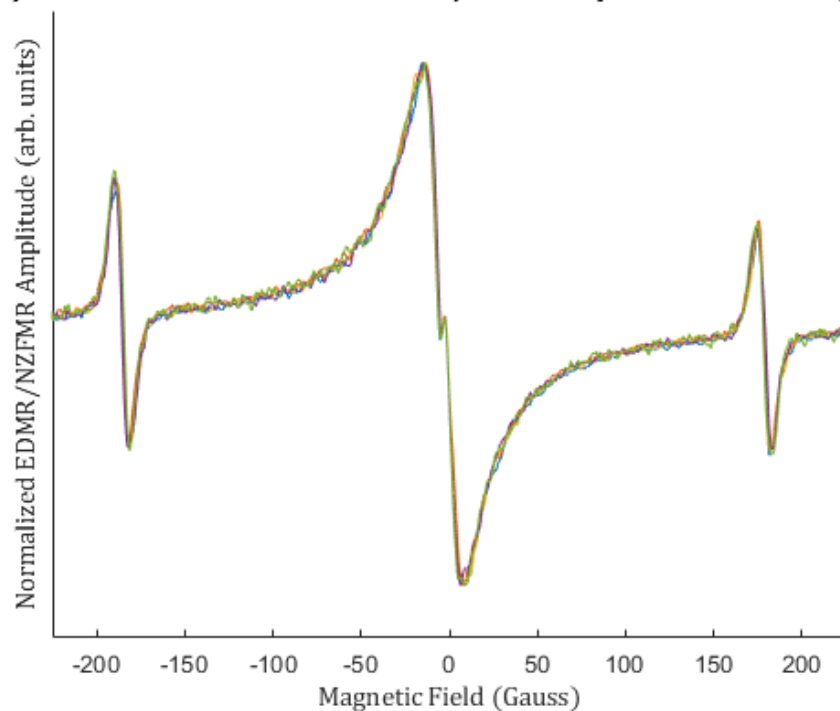


Figure 5.11: a) Change in NZFMR and EDMR amplitude over time under constant -4V bias during measurement. b) Normalized traces from a) showing no discernable change in the shape or linewidth over time, the smaller signals have reduced SNR. The legend applies to both parts of the figure.

The line shape in these measurements, with both low-field EDMR and NZFMR, does not change over time. This would suggest that, like in the pure NZFMR measurement, only the generation rate at which charge carriers enters the material, r_{ea} , is changing. The low-field EDMR linewidth is 7.5 Gauss wide, which is somewhat smaller than the 9 Gauss wide X-band EDMR linewidth.

The DC current through the sample initially recovers slightly when subjected to positive bias until dropping again after a short period of time. The measured current through the sample is shown in Figure 5.12. The percent change in current due to NZFMR also increases during this recovery period, doubling over approximately 10 minutes then somewhat stabilizing, when subjected to a constant +4V bias. This is shown in Figure 5.13.

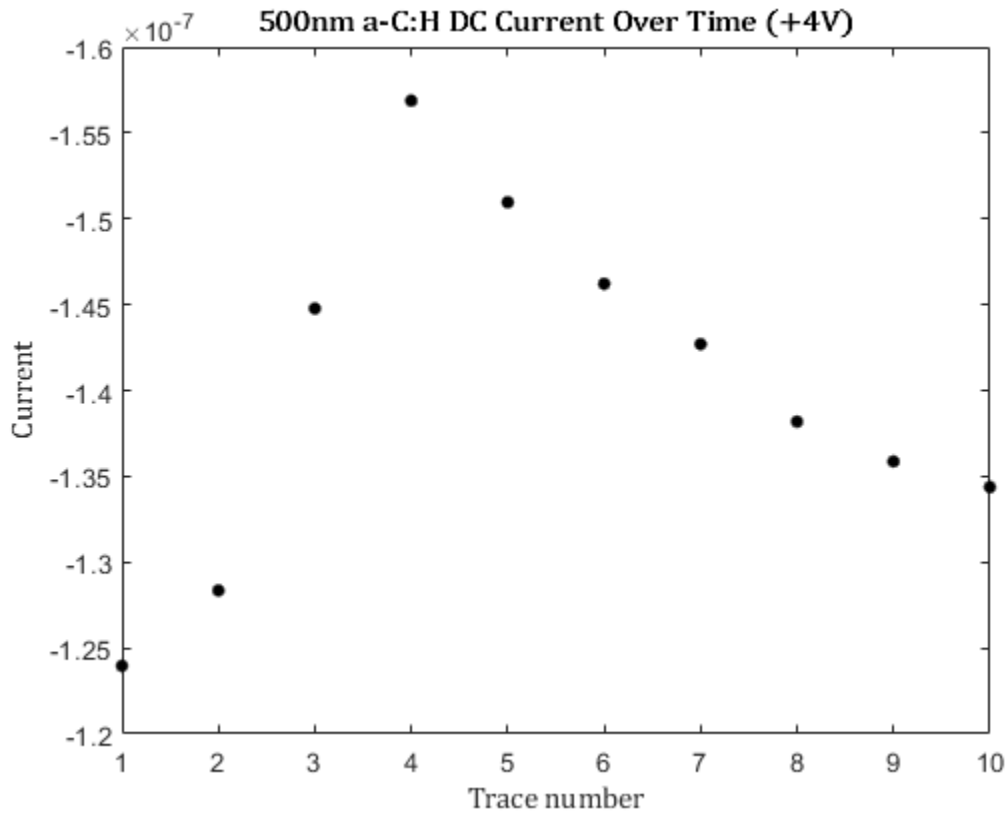


Figure 5.12: Measured DC current in 500nm a-C:H sample over time under constant +4V bias. This measurement was taken after the measurement at a -4V bias shown in Figure 5.11.

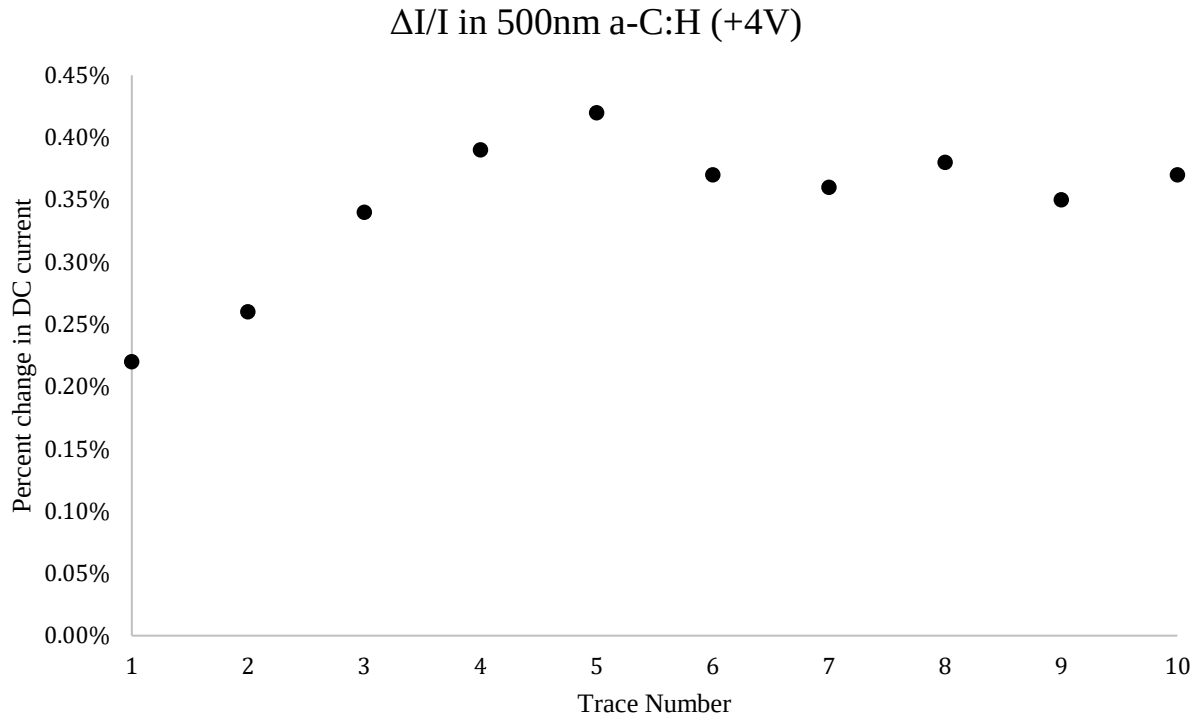


Figure 5.13: Percent change in current due to NZFMR over time under constant bias of +4V.

After a series of biases over a prolonged period of time, the applied bias must be increased in order to measure an NZFMR/EDMR spectrum. After applying a bias of -8V, a DC current of 280 nA is initially measured, much less than the 10 μ A measured as soon as the initial -4V was applied for the previous measurements. This current dropped to approximately 118 nA indicating that the relative decrease in current is reduced the longer the device is under an applied bias.

Capacitance vs voltage measurements were made on the sample between some NZFMR measurements and are shown in Figure 5.14. These measurements show a leftward shift after negative bias, and a small rightward shift after positive bias. This indicates that there is positive space charge accumulating in the insulating material when under negative bias, and positive

space charge leaves the insulating material under positive bias. It is also clear to see the effect of time on the amount of space charge in carbon film. Each of the measurements in Figure 5.14 was made sequentially and stressing occurred during NZFMR/EDMR measurements. There is a large leftward shift (approximately 3V) between the measurement made on the fresh sample and the measurement made after the initial -4V bias over 10 minutes. An additional 8 hours biased at a constant -4V shows a much smaller leftward shift. A small rightward shift in the plot is shown after the same sample is biased at +4V for a period of 20 minutes. Finally, there is a large leftward shift in the plot after being biased at -8V for approximately 20 minutes.

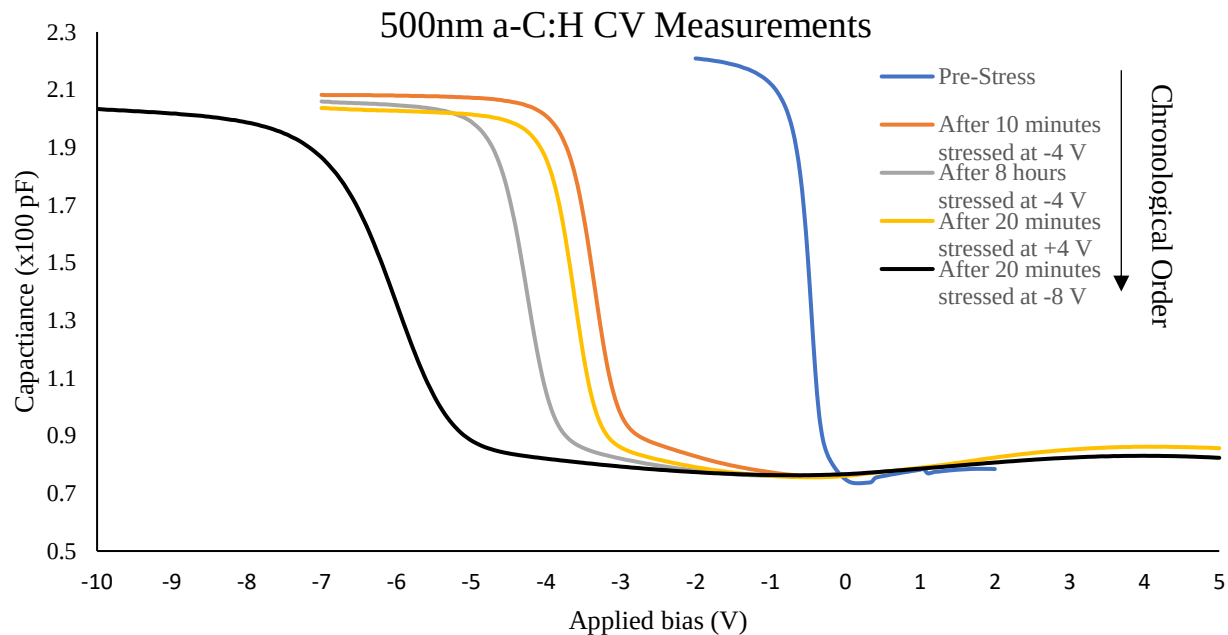


Figure 5.14: Capacitance vs voltage measurements made on 500nm amorphous carbon on Si MOS capacitors after EDMR and NZFMR measurements. The measurements were made in the order shown. The sample was first biased at -4V for approximately 10 minutes, then at -4V for approximately 8 hours, then at +4V for approximately 20 minutes, then at -8V for approximately 20 minutes. Capacitance vs voltage measurements were made between EDMR/NZFMR experiments. The shift in the curve to the left indicates positive space charge in the insulating layer. A 1-volt shift left corresponds to approximately 6×10^{10} charges cm^{-2} .

The leftward shift in the capacitance vs voltage plots in Figure 5.14 corresponds to approximately 6×10^{10} charges cm^{-2} per volt. The slight downward shift in the accumulation capacitance between the pre-stress and post-stressed measurements is likely due to the rapidly changing electrical conductivity of the sample and relatively large current through the sample during the initial measurement. Both of these effects have been observed to cause the capacitance meter to measure slightly different values in other samples. There does not appear to be a way to make a more accurate initial measurement as the biasing used in the measurement causes the change in the sample, therefore a second, more stable measurement would yield different results. The accumulation and inversion capacitances do not change once the sample is biased long enough to become more stable throughout the rest of the measurements made.

Approximate band diagrams for the a-C:H samples with different levels of fixed space charge in the carbon layer are shown in Figure 5.15. Band diagrams were modeled using the Boise State University College of Engineering Multi-Dielectric Energy Band Diagram Program version 3.1.6. The space charge in the a-C:H is modeled using 2-D sheet charge as the program's fixed charge modeling capabilities are limited to 2-D sheet charge. Band alignment and energy level offsets were determined using data from Robertson [72], [85] and Schäfer et al. [86]. It is clear in the modeled band diagrams how positive space charge in the carbon would raise the tunneling barrier that holes experience, thereby reducing the tunneling current. These models show how the reduction in current seen in the a-C:H samples can be related to the larger barrier due to positive space charge observed in the capacitance vs voltage measurements.

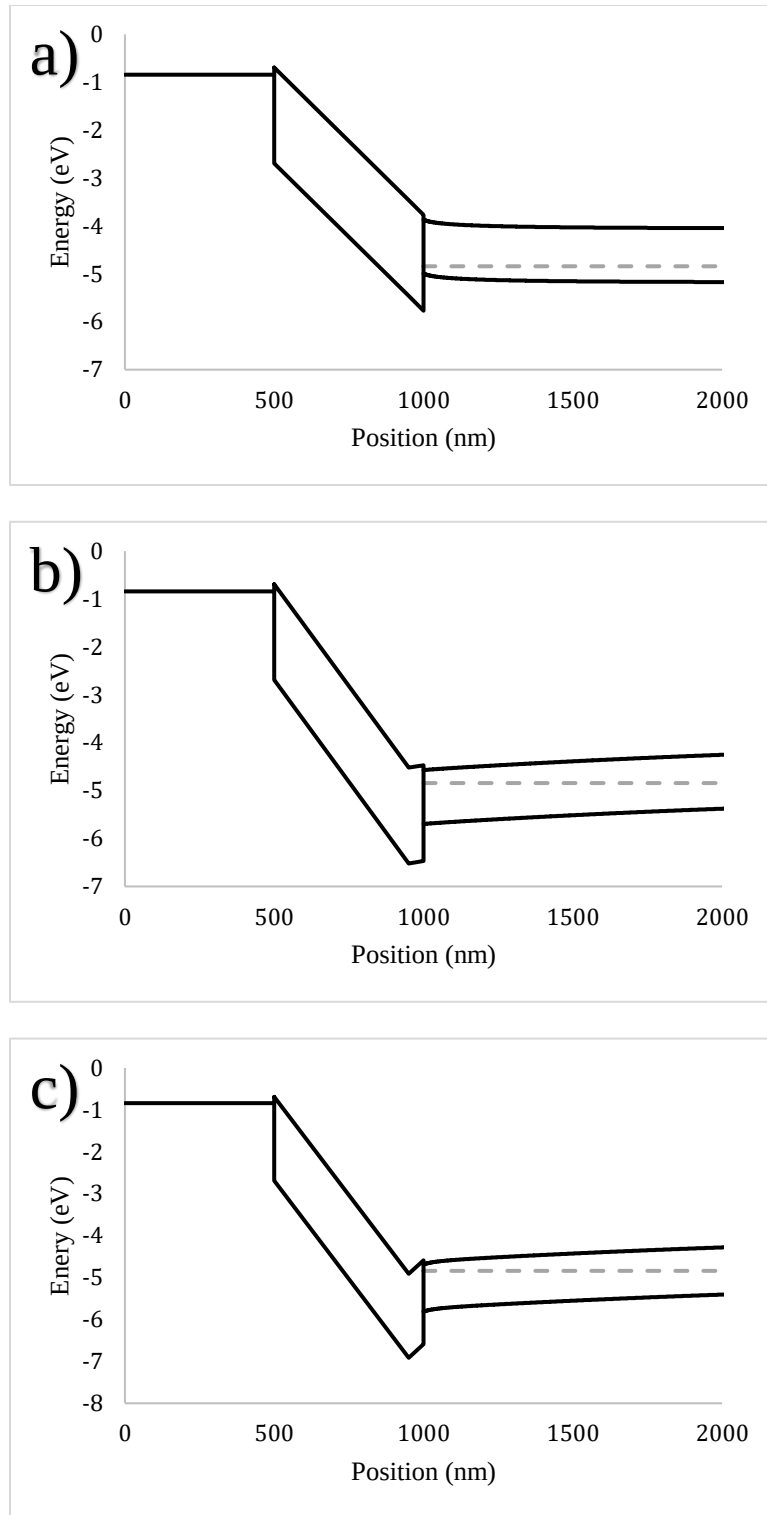


Figure 5.15: Band diagrams of the Ti/a-C:H/p-Si samples under bias of -4 V for a) no space charge in the a-C:H, b) 2×10^{11} positive charges cm^{-2} positioned 50 nm from the C/Si interface, and c) 4×10^{11} positive charges cm^{-2} positioned 50 nm from the C/Si interface.

The largest bias applied to these samples was -60V after a series of smaller biases. The sample showed the same monotonic decrease in DC current and NZFMR/EDMR amplitude over time during this measurement. The DC current started at 1 μA , once again smaller than the 10 μA measured immediately after applying the initial -4V bias. The DC current dropped to 200 nA under the constant -60V bias. No bias larger than 60V was used in order to prevent damage to the electronic components of the spectrometer. The averaged spectra measured under a bias of -60V can be seen in Figure 5.16.

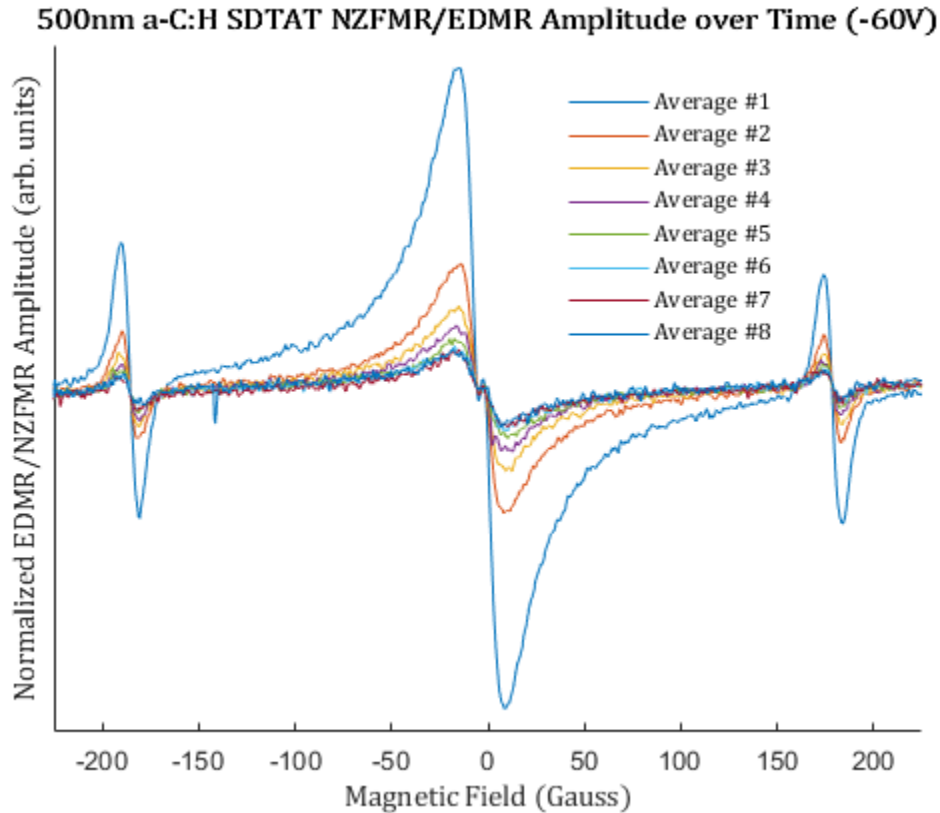


Figure 5.16: SDTAT NZFMR and EDMR in 500nm a-C:H under -60V bias over time.

There was no discernable change in the shape of either the EDMR or NZFMR lines even at this very large bias of -60V. Comparing the shape of the both the EDMR and NZFMR lines

for multiple biases shows a similar trend. Even for different biases, the line shapes do not change, only the signal amplitude drops as the DC current through the sample also drops.

Normalized spectra for different biasing conditions is shown in Figure 5.17.

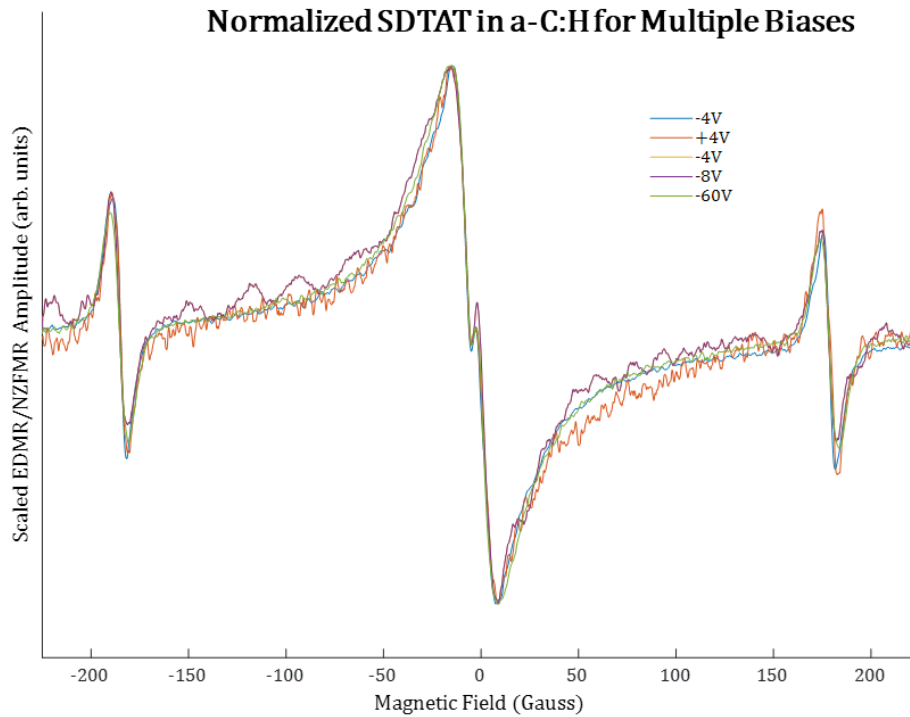


Figure 5.17: Normalized SDTAT spectra for multiple biases during the course of the study. No appreciable change to the shape of either the EDMR or NZFMR spectra can be seen.

We can compare the changes in the measured NZFMR spectra in the a-C:H to the simulated changes from chapter 3.2.4. In this case, only the amplitude of the traces is changing over time and there are no changes to line shape or width. This indicates that only the parameter r_{ea} , the generation rate, is changing during the measurement. This is the rate at which charge carriers enter the material from the environment. This suggests that fewer charge carriers are entering the material which aligns well with the decrease in the measured DC current through the

sample. It would make sense that this reduction in $r_{e\alpha}$ contributes to both the reduction in DC current and the size of the NZFMR response under a constant bias. The likely physical cause of this reduction is trapped charge in the carbon layer increasing the tunneling barrier height. Several samples were annealed at 400K under small positive bias after stressing with no lasting effects on the electronic properties of the samples.

The low field EDMR spectra measured in the a-C:H follow the same trends as the NZFMR spectra. The low field EDMR spectra were narrower than the NZFMR spectra however the line shape and width did not change as the current or bias changed.

The X-band EDMR (9.5 GHz) spectra measured were also broader than the measured low field EDMR (500 MHz) spectra. This broadening is likely due to the disorder in the amorphous system, as no two paramagnetic defects will have identical surroundings. Small strains in the lattice can change the zero-crossing g-value of each individual defect which causes the resonance condition of each defect to be slightly different as well. This range in g-values causes the signal to be spread over a wider range of magnetic field values. This effect is small at low magnetic fields and frequencies; however, it becomes more prominent at higher fields and frequencies. This can be easily understood by looking at the case of only two different g values, 2.002 and 2.003, at two different resonance frequencies, 500 MHz and 9.5 GHz. The resonance condition from chapter 2 is met for the g value of 2.003 at 500 MHz at a magnetic field of 178.35 Gauss. The resonance condition is met for the g value of 2.002 at 500 MHz at a magnetic field of 178.44 Gauss. These two lines would therefore be separated by 0.089 Gauss and would essentially add together to form a single line. The resonance condition is met for the g value of 2.003 at 9.5 GHz at a magnetic field of 3388.7 Gauss. The resonance condition is met for the g value of 2.002 at 9.5 GHz at a magnetic field of 3390.3 Gauss. These two lines would therefore

be separated by 1.7 Gauss. If the lines are broad enough, they may overlap and add to form a single line. This separation is 19 times larger for the 9.5 GHz measurements and would result in a broader line. It is not a coincidence that this factor is exactly the ratio of 9.5 GHz to 500 MHz as the resonance field will increase linearly with resonance frequency. A wider spectrum of g values in an amorphous system would therefore produce a single broad line as all of the different lines for each g -value overlap. The amount that the signal broadens at high resonance fields will correlate to the range of g values in the system, with samples with wider ranges becoming broader as the resonance field and frequency are increased.

Chapter 5.3 EDMR and NZFMR in 1000 nm Thick Amorphous Diamond-like Carbon

SDTAT EDMR and NZFMR as well as other measurements were conducted on 1000nm diamond like (tetrahedral) carbon on p- silicon capacitor dots. The DLC films were deposited using PECVD. The top metal contact is 2mm^2 , composed of approximately 100nm of titanium with approximately $1\mu\text{m}$ of aluminum on top. The composition of the DLC layer is 69% carbon, 27% hydrogen, and 3% nitrogen. The structure of the DLC is shown in Figure 5.18.

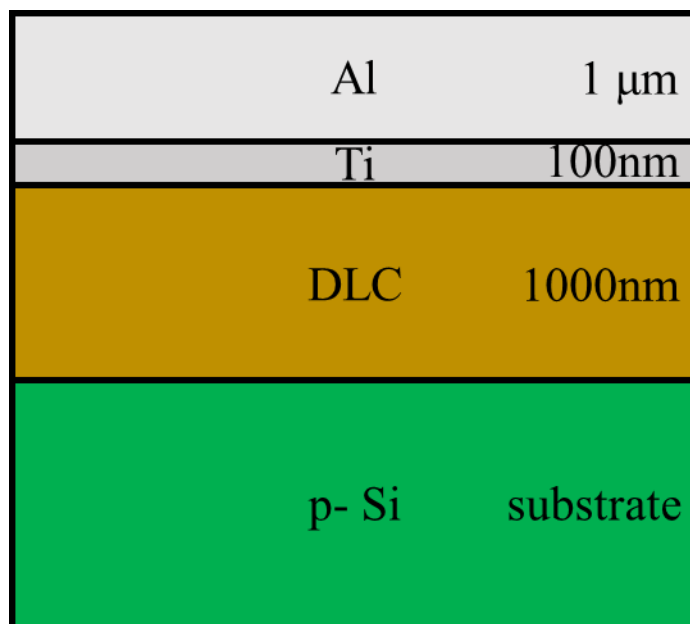


Figure 5.18: Schematic illustration of 1000 nm DLC samples.

Samples were mounted on custom made USB connected PCB “Ts”, with the back contact connected to a bonding pad via electrically conductive paint. Electrical connection to the top contact was made via wire bonding. Under low applied bias, these samples were very stable over long periods of time, however there appear to be changes in the sample at higher applied bias. EDMR and NZFMR signals were only observed when samples were negatively biased.

Chapter 5.3.1 X-band SDTAT EDMR in 1000nm DLC

SDTAT EDMR measurements were made on 1000 nm DLC in order to help identify the defects involved in the electronic transport in the material. The measurements were made on the Bruker X-band spectrometer. A SDTAT EDMR spectrum was measured with the sample biased at -1.5V. This spectrum is shown in Figure 5.19. The magnetic field modulation had an amplitude of 5 Gauss at 12.5 kHz. This modulation frequency is higher than most of the other measurements in this study because this frequency provides the largest signal to noise ratio on

this particular spectrometer. This measured spectrum has a zero-crossing g-value of 2.0041. The line width of the spectrum was approximately 14 Gauss wide.

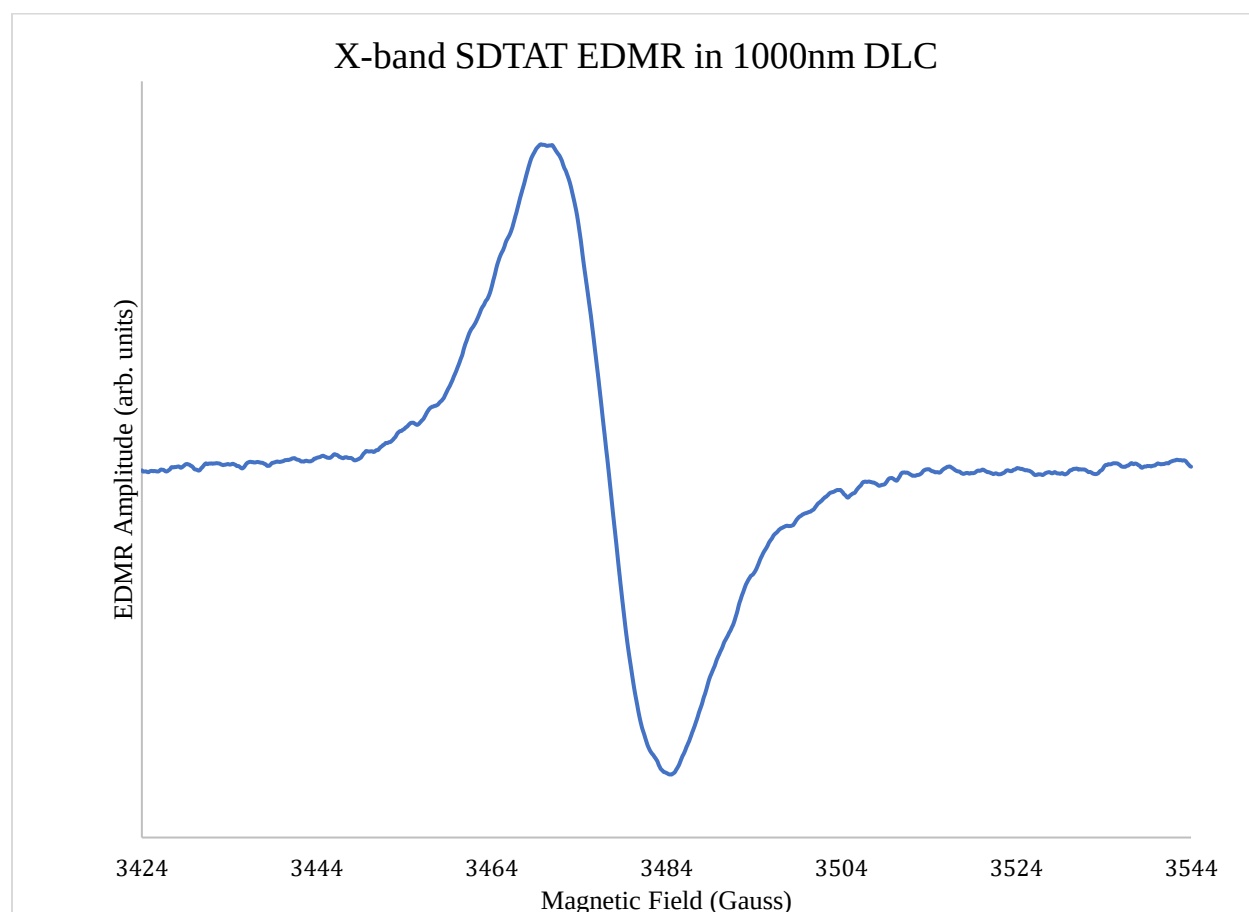


Figure 5.19: X-band SDTAT EDMR spectra in 1000nm DLC. The signal is 14 Gauss wide with a g value of 2.0041.

The defect responsible for the SDTAT EDMR in the 1000nm DLC is likely a dangling bond on sp^2 coordinated sites despite being comprised of half sp^3 coordinated carbons, as discussed in Chapter 5.1. The measured g-value is higher than originally expected but has been seen before in DLC films [83] It could be caused by impurities in the material, [71] however this seems unlikely, as the sample is composed almost entirely of carbon and hydrogen, with only 2% nitrogen.

Chapter 5.3.2 SDTAT NZFMR and Low Field EDMR in 1000nm DLC

After performing X-band SDTAT EDMR on these samples, SDTAT NZFMR and low-field EDMR were performed on fresh samples. These measurements were made on both the Kepco LF/ZF spectrometer and the Blue Spin LF/ZF spectrometer. The Kepco LF/ZF spectrometer was primarily used when the DC current in the samples had to be measured during the low-field EDMR/NZFMR measurements.

SDTAT NZFMR measurements on the DLC at multiple applied biases is illustrated in Figure 5.20. The NZFMR amplitude increases with bias until approximately -1.5V. Higher magnitude bias (more negative) causes the NZFMR signal to become slightly smaller, then hold relatively constant at biases greater than -3V. The wider signal appears to maintain its shape and line width however the center inflection appears to change shape and increases in amplitude with increasing bias. Under a large constant bias, the inflection continues to change over time. The low bias measurements were made on the Blue-Spin LF/ZF spectrometer.

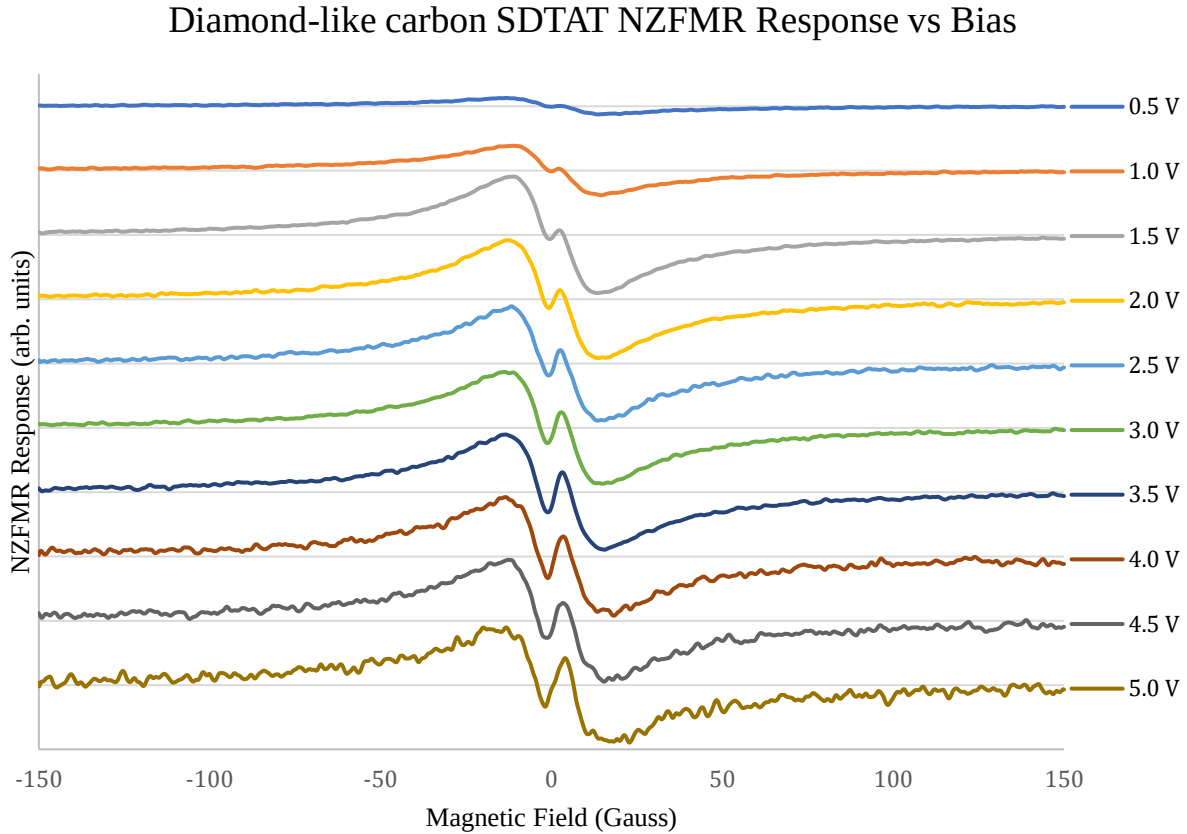


Figure 5.20: SDTAT NZFMR response vs bias in 1000nm diamond-like carbon. Amplitude of signals is not normalized. Measurements made using 3 Gauss modulation amplitude.

Figure 5.20 shows that, unlike in the a-C:H samples, the shape of the line changes with bias. In this case with low applied biases, there are different variables changing from Figure 3.4. Here the amplitude of the spectra increases as the bias is increased from -0.5V to -1.5V, indicating that $r_{e\alpha}$ changes. The change in the amplitude of the inflection also indicates that more than one variable is changing. The changes seen in these measured spectra are similar to the changes seen in the simulated spectra for a decrease in $r_{\alpha e}$ from Figure 3.4.

Spectra were simulated and fit utilizing the least squares fitting outlined by Frantz et al. [64] for the cases of -1V, -2V, and -3V applied to the samples. These cases were chosen because

of the high signal to noise ratio of the measured spectra, and the stability of the samples during the measurements. There were no large changes in the DC current or measured spectra at the low biases. The fitting algorithm was again constructed to only consider the hyperfine contributions of a single species which was assumed to be hydrogen, the most abundant magnetic nucleus in the sample.

The simulated spectra are shown in Figure 5.21. The resulting hyperfine coupling constants for sites α and β were both found to be 3.8 ± 0.5 Gauss. This value was constant for all three simulations as the defect responsible for conduction does not appear to be changing, with all measured spectra being repeatable at low biasing conditions. For this reason, the singlet probability of hopping rate was also constant for all three simulations. The fitting produced a hopping rate, $r_{\alpha\beta}$, of 490 ± 50 MHz, while the generation rate, $r_{e\alpha}$, and dissociation rate, $r_{\alpha e}$, were allowed to vary for the different cases. The fitting for the case of -1V applied to the sample produced values for a generation rate, $r_{e\alpha}$, of 3.8 ± 0.25 GHz and a dissociation rate, $r_{\alpha e}$, of 110 ± 10 MHz. The fitting for the case of -2V applied to the sample produced values for a generation rate, $r_{e\alpha}$, of 6.4 ± 0.25 GHz and a dissociation rate, $r_{\alpha e}$, of 100 ± 10 MHz. The fitting for the case of -3V applied to the sample produced values for a generation rate, $r_{e\alpha}$, of 4.3 ± 0.25 GHz and a dissociation rate, $r_{\alpha e}$, of 90 ± 10 MHz.

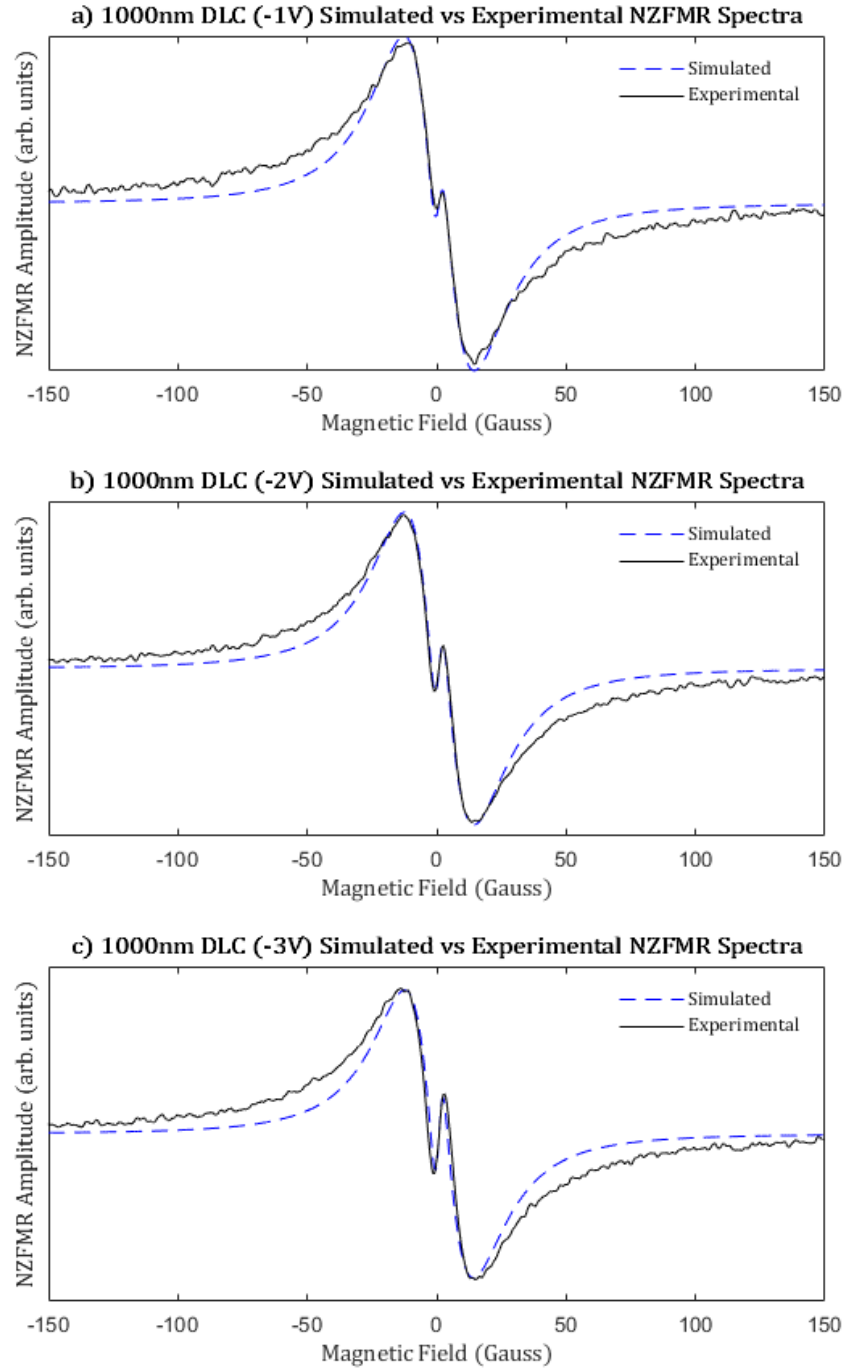


Figure 5.21: Simulated and measured NZFMR spectra in DLC for multiple biasing conditions. a) -1V applied to sample, simulation parameters $r_{ea} = 3.8 \pm 0.25 \text{ GHz}$, $r_{ae} = 110 \pm 10 \text{ MHz}$, $r_{\alpha\beta} = 490 \pm 50 \text{ MHz}$, and hyperfine coupling constants on both sites of $3.8 \pm 0.5 \text{ Gauss}$. b) -2V applied to sample, simulation parameters $r_{ea} = 6.4 \pm 0.25 \text{ GHz}$, $r_{ae} = 100 \pm 10 \text{ MHz}$, $r_{\alpha\beta} = 490 \pm 50 \text{ MHz}$, and hyperfine coupling constants on both sites of $3.8 \pm 0.5 \text{ Gauss}$. c) -3V applied to sample, simulation parameters $r_{ea} = 4.3 \pm 0.25 \text{ GHz}$, $r_{ae} = 90 \pm 10 \text{ MHz}$, $r_{\alpha\beta} = 490 \pm 50 \text{ MHz}$, and hyperfine coupling constants on both sites of $3.8 \pm 0.5 \text{ Gauss}$.

The fittings from these simulations indicate that the dissociation rate, r_{ae} , is the primary variable that changes in the DLC as the bias is changed. The generation rate, $r_{e\alpha}$, changes as well, but is mostly used as a scaling factor in the simulations. The dissociation rate decreases in this sample as the bias is increased, from a value of 113 MHz at -1V to 87MHz at -3V, causing the center inflection line to become larger.

These values appear to be plausible as the hyperfine coupling constant is a value between the approximately 1 Gauss lower limit and the 9 Gauss low-field EDMR line width. A hopping rate larger than the dissociation rate is also plausible as the current is dominated by hopping events and larger biases lead to increased hopping rates relative to dissociation rates. The hyperfine coupling constant is larger for the DLC than for the a-C:H even though it has a slightly lower hydrogen content. This may be due to the higher coordination of the carbon atoms in the DLC as about half of them are sp^3 coordinated unlike the a-C:H in which all of the carbon atoms are sp^2 coordinated. This would provide a higher probability of one or more hydrogen atoms being near the sp^3 coordinated carbon. This higher hyperfine coupling constant is also consistent with the broader low-field EDMR line width of 9 Gauss in the DLC compared to 7 Gauss in the a-C:H. The low-field EDMR linewidth is dominated by hyperfine interactions so a larger hyperfine coupling constant would cause the low-field linewidth to be larger.

One might expect a larger dissociation rate at higher bias as the tunneling barrier shifts from trapezoidal towards a triangular shape and transport shifts from spin dependent site to site hopping to more of a spin-independent direct tunneling, however, this does not appear to be the case. This may be due the thickness of the samples, approximately 1 micron thick, resulting in very few direct tunneling events at the biasing conditions used in this study. Further studies at higher bias may provide some insight into this somewhat unexpected behavior.

The DC current does not change in the DLC samples at low applied biases and these signals did not change over time, however, the change in center inflection over time at higher bias (greater than -5V) suggested that it would be useful to increase the bias further and monitor both the DC current in the sample and the changes to the SDTAT NZFMR signal over time. This meant using the Kepco LF/ZF spectrometer to perform NZFMR on the DLC under a larger bias of -7.5V. There were significant changes to both the DC current and the measured spectra over time. Unlike the 500nm a-C:H, the DC current in the DLC increased over time and the spectrum changed shape, with the narrow center inflection becoming larger in amplitude than the wider signal. This change is shown in Figure 5.22. The signal to noise ratio is severely reduced at higher biases and currents in these samples. This measurement was made in conjunction with other students in the laboratory group, namely Elias Frantz and Ryan Waskiewicz.

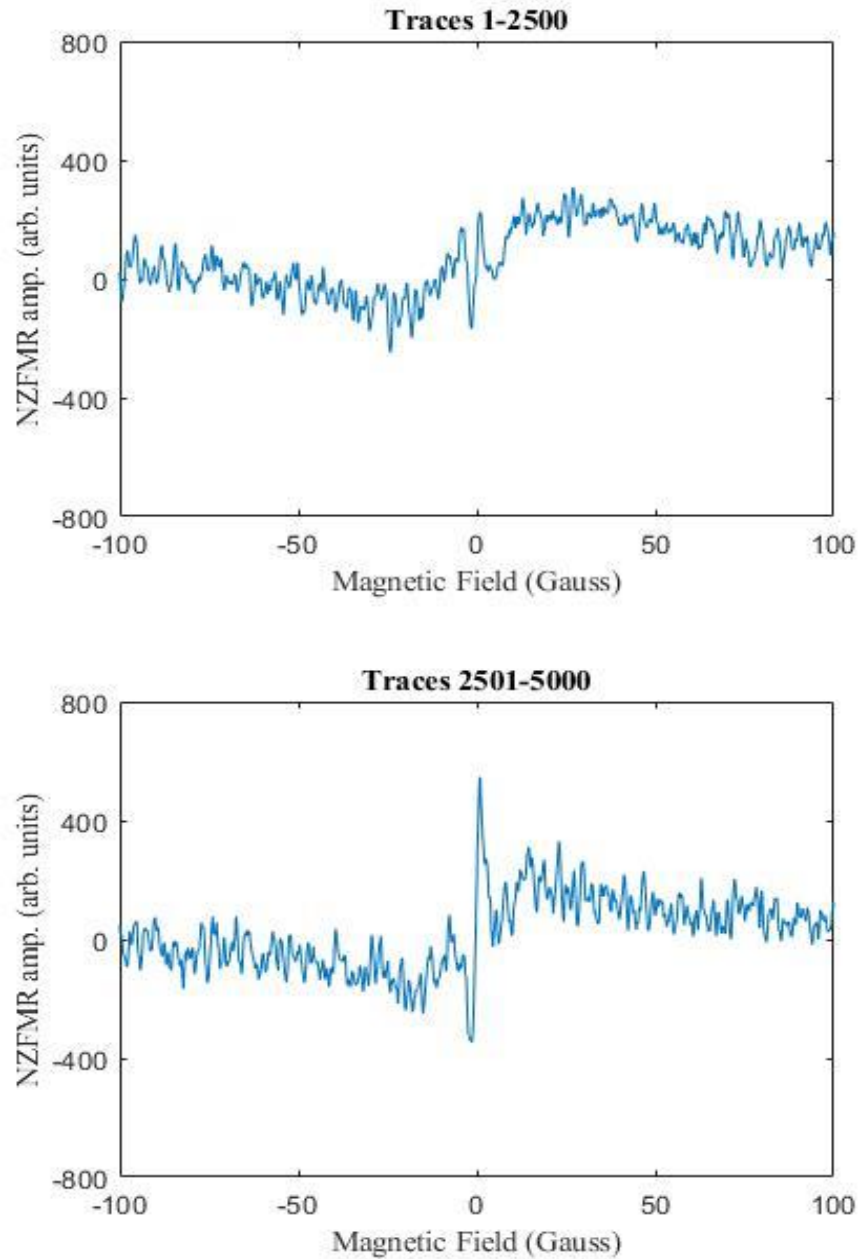


Figure 5.22: Top) Averaged SDTAT NZFMR spectrum of 1000nm DLC biased at -7.5V (first 2500 traces)
 Bottom) Averaged SDTAT NZFMR spectrum of 1000nm DLC biased at -7.5V (last 2500 traces)
 Note the change in the size of the center inflection over time while the wider signal remains mostly unchanged.
 Both plots use the same scale for the vertical axis.

The current in the sample was monitored during the SDTAT NZFMR measurement and is shown in Figure 5.23. The current increases over time corresponding to the increase in the size of the center inflection signal.

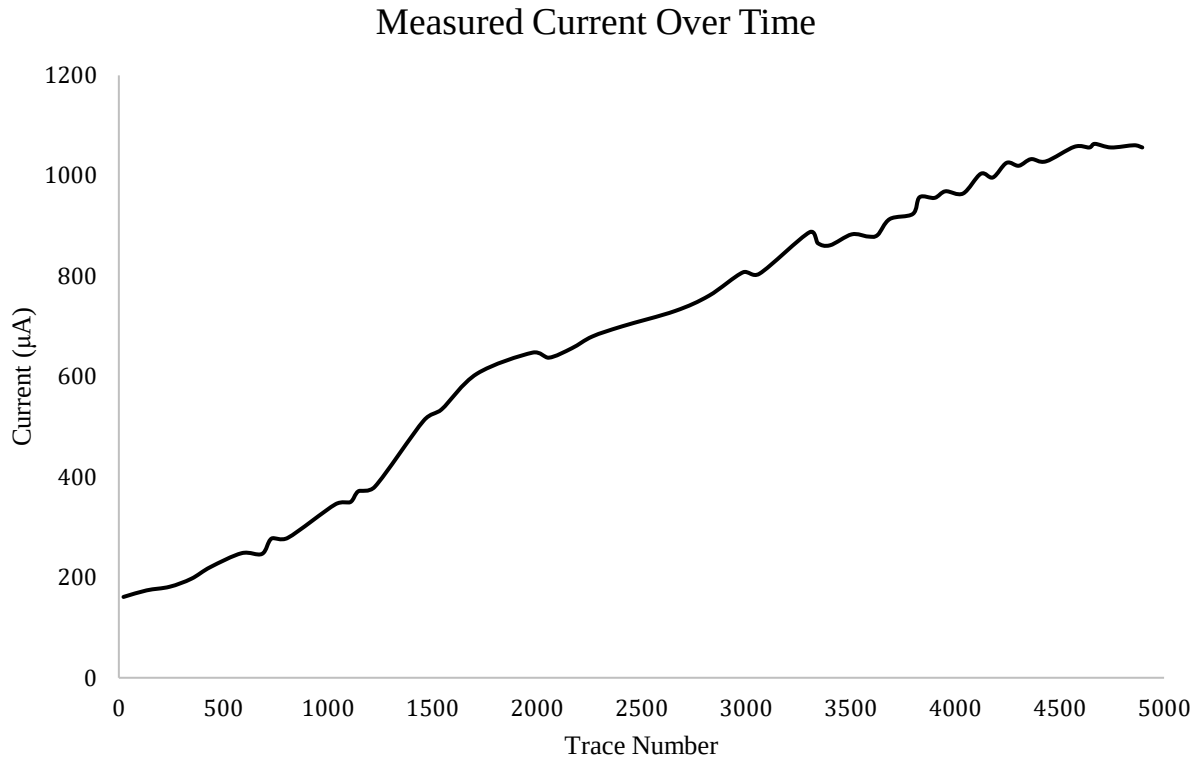


Figure 5.23: DC current in 1000nm DLC under constant -7.5V bias during SDTAT NZFMR.

The change in line shape over time, and not just signal amplitude, suggests that an additional defect forms and is likely also responsible for the spin dependent conduction through the material. The low signal to noise of the spectra measured at higher biasing conditions prevents accurate simulations from being calculated, although the introduction of a new defect in the system could be a symptom of stress induced leakage current or a time dependent dielectric breakdown-like process, if not both.

The low field SDTAT EDMR response at low bias in the 1000nm DLC follows a pattern similar to the NZFMR response at low bias, with the signal amplitude increasing until the applied bias is approximately -1.5V. After -1.5V the signal decreases slightly but stays fairly constant at higher magnitude bias. This can be seen in Figure 5.24 in which the spectra are not normalized. The EDMR response signal to noise ratio also suffers at higher bias and increased DC current. These measurements were made on the Blue-Spin LF-EDMR/NZFMR spectrometer.

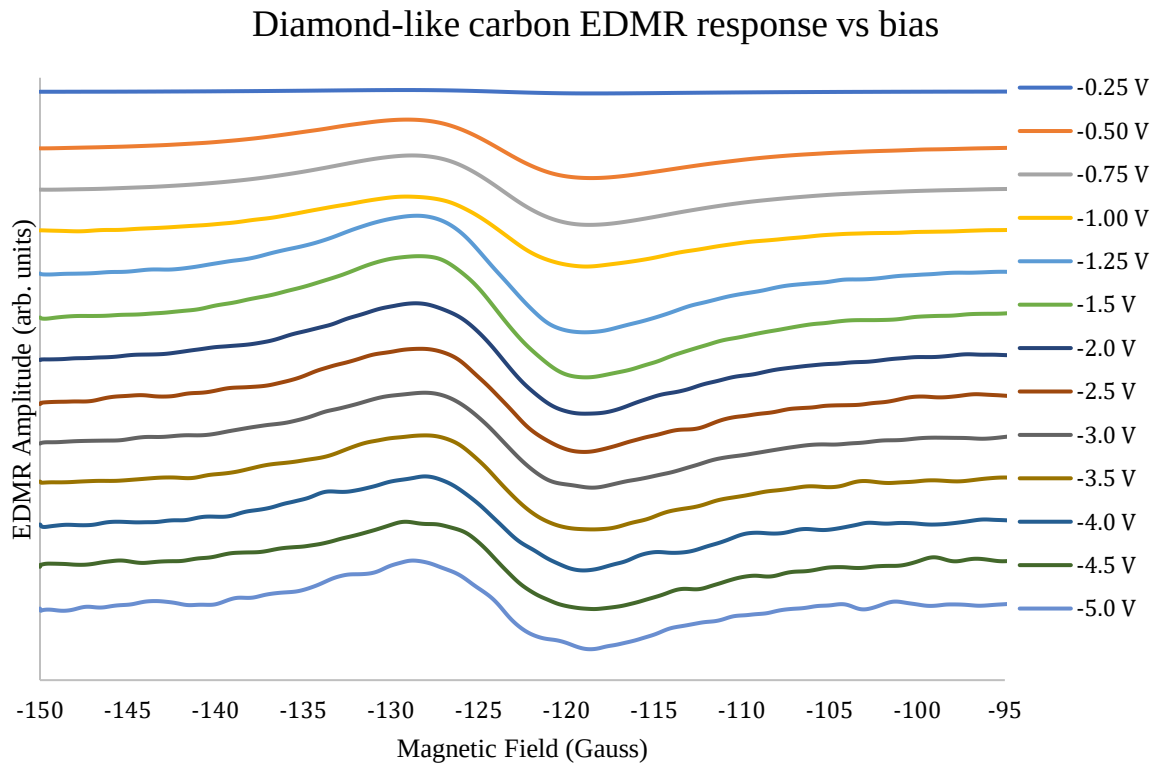


Figure 5.24: SDTAT EDMR response vs bias in 1 μm diamond-like carbon. Amplitude of signals is not normalized. Measurements were made using a 5 Gauss modulation amplitude.

The change in the low-field EDMR signal at low bias, with the amplitude reaching a maximum around -1.5V then staying roughly constant, matches the NZFMR measurements,

suggesting that the defects responsible for conduction in these cases are likely the same. Low field EDMR is difficult to perform at higher biases due to the reduced SNR, however, these measurements will likely be part of any future work on these samples. The line-width of this sample also changes from 9 Gauss wide at 370 MHz to 14 Gauss wide at 9.5 GHz, suggesting a wider range of g values in the DLC than in the a-C:H, as described earlier.

Chapter 5.4 Discussions about a-C, Contributions to Theory, and Future Work

The changes in the amorphous carbon samples over time under constant bias provide an opportunity to improve our models of both EDMR and NZFMR and also, to some extent, develop an understanding of SILC and TDDB in these model systems. The changes observed in the 500nm a-C:H, the diminishing DC current and SDTAT signals over time, might be linked to a TDDB process in which hopping sites close to the edge of the material are occupied by charge carriers. Occupation by a charge carrier may increase the barrier that another charge carrier must overcome in order to tunnel through the material as seen in Figure 5.15. The larger barrier makes it more difficult for a charge carrier to tunnel to another site which could potentially explain the reduction in current and the drop in signal amplitude if the critical hopping distance is getting longer as sites are blocked due to being filled and the tunneling probability is decreased due to the potential well getting deeper as the barrier height increases.

The general shape of the spectra measured in the a-C:H remains the same across a wide range of biases and charge states and only the amplitude of the signal appears to change as measurements are performed on the samples. Comparing the changes in the SDTAT NZFMR response in the a-C:H to the simulations in Chapter 3.2.4 indicates that the only parameter that is changing in these samples is the generation rate at which carriers are entering the initial site from

the environment. That is, only r_{ea} is changing while the other variables from equations 3.2.5 and 3.2.6, r_{ab} , r_{ae} , a_α , and a_β , are remaining constant. This would indicate that the trapped charges are likely localized to a thin region near the interface as the density of defects through which the carrier's travel does not appear to change because the hopping rate, r_{ab} , remains constant.

The changes seen in the 1000nm DLC provide evidence that the center inflection signal likely arises from a different pathway than the broader, usually larger signal. Some attempts have been made by collaborators to model this change using the adapted two site model for SDTAT in inorganic materials. The most promising attempt, though not fully developed and discussed in a private communication, appears to use two separate spin mixing pathways in order to achieve the measured spectra, with the relative contributions of each to the total spectrum changing over time. [87]

The variables from equations 3.2.5 and 3.2.6 that appear to be changing in the DLC samples are r_{ae} , and r_{ea} , as variations in the first value affects the size of the inflection response relative to the size of broader signal, while the increased current is likely due to an increase in the initial generation rate. It appears that there may be a reduced dissociation rate, r_{ae} , the rate at which carriers hop from a trapping site is not spin dependent when hopping to an environment with more possible states to occupy. These changes would indicate that the path the carrier is taking through the material is changing as the voltage, current, and time stressed are increased.

The simulated spectra fitted in this section agree well with the peaks of the measured spectra at lower magnetic field. There is, however, some discrepancies between the tails of the measured and simulated spectra. This could be caused by another interaction that is not accounted for in the models. The model only accounts for the effects of a single species,

assumed to be nearby hydrogen nuclei as they are the most abundant magnetic species in the samples. Further development of the computational model may be able to account for more interactions.

Both material systems have broader EDMR linewidths at X-band (9.5 GHz) than at low-field (370 MHz). The spectra measured in the DLC samples appear to broaden more than the spectra measured in the a-C:H. The EDMR linewidth of the a-C:H changes from 7.5 Gauss at 500 MHz to 9 Gauss at 9.5 GHz, a difference of 1.5 Gauss. The EDMR linewidth of the DLC changes from 9 Gauss at 370 MHz to 14 Gauss at 9.5 GHz, a difference of 5 Gauss. This would suggest that there is wider range in g values in the DLC. This result seems to make sense as most of the electron orbitals around the carbon atoms in the a-C:H are sp^2 hybridized while the electron orbitals around carbon atoms in the DLC are a mix of sp^2 and sp^3 hybridized, leading to more disorder and a wider range of g values for the individual defects that contribute to the spectrum. The broader low-field linewidth in the DLC suggests that the hyperfine interactions are stronger in these samples, which is consistent with the larger extracted hyperfine coupling constant from the simulated spectra.

Future work on these samples will likely involve finding ways to safely stress the 500nm a-C:H until hard breakdown occurs to observe the change in the EDMR and NZFMR signals as the material approaches failure. The signal to noise ratio of the 1000nm DLC samples will also have to be increased at the higher, -7.5V, bias to be able to fit the data with simulated spectra. This can likely be achieved by increasing the amount of time the samples are signal averaged for while monitoring the changes in the DC sample current and NZFMR spectrum over time. The changes in the DLC samples at higher bias might then be described quantitatively by fitted simulated spectra.

Chapter 6

Electrically Detected Magnetic Resonance and Near Zero Field Magneto-Resistance in Amorphous Silicon Carbide

SDTAT EDMR and NZFMR measurements were made on 5nm amorphous hydrogenated silicon carbide (a-SiC:H) on p⁺⁺ silicon MOS capacitors. These thin film samples contain approximately 32% hydrogen. The a-SiC:H was deposited via PECVD. The top metal contact is 2mm², composed of approximately 100nm of titanium with approximately 1μm of aluminum on top. The structure of the a-SiC:H is shown in Figure 6.1.

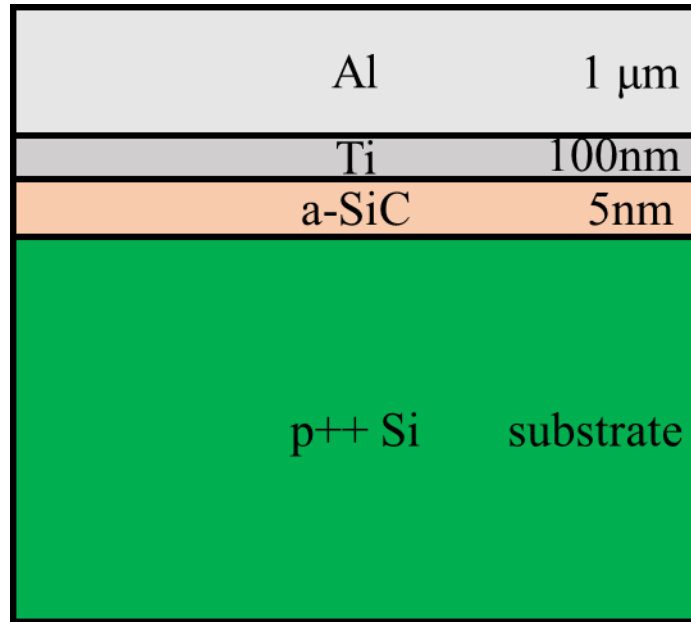


Figure 6.1: Schematic illustration of 5nm a-SiC:H samples.

These a-SiC:H samples showed no instabilities over a wide range of biases. The very high signal to noise ratio of these samples could be potentially useful in the development of spin-based magnetometers like the one proposed by Cochrane et al. [43] as materials with large signals and narrow line widths allow for a higher sensitivity in this type of magnetometer.

Most samples were mounted on custom made USB connected PCB “Ts”, with the back contact connected to a bonding pad via electrically conductive paint. Electrical connection to the top contact was made via wire bonding. These samples were characterized using the HP model 4155 semiconductor parameter analyzer and Boonton model 72B capacitance meter; however, no changes were seen over time.

Some samples were not attached to any external mount for NZFMR measurements. These samples were placed onto a wafer probing station in a small home-built wire wound electromagnet with enclosed modulation coils. Electrical contact to the back of these samples was made by placing the sample on a sheet of copper foil which was electrically isolated from the chuck of the probing station. The probes used to make contact to the sample were placed on the top contact of the MOS capacitor and the copper foil the sample was resting on. It should be noted that unlike the other samples in this work which have to be bonded to the PCB “T” and connected via wire bonds, no sample preparation was required in order to make NZFMR measurements on these samples. This is discussed further in appendix B.

Chapter 6.1 X-Band SDTAT EDMR in 5nm a-SiC:H

SDTAT EDMR measurements were made on 5nm a-SiC:H using the home-built Lakeshore X-band spectrometer. The lock-in detection for these measurements utilized a field modulation amplitude of 5 Gauss at approximately 1000 Hz. An X-band SDTAT EDMR spectrum for the a-SiC:H is shown in Figure 6.2. This measurement was made with the sample top contact biased at +2V and a modulation frequency of 1150 Hz.

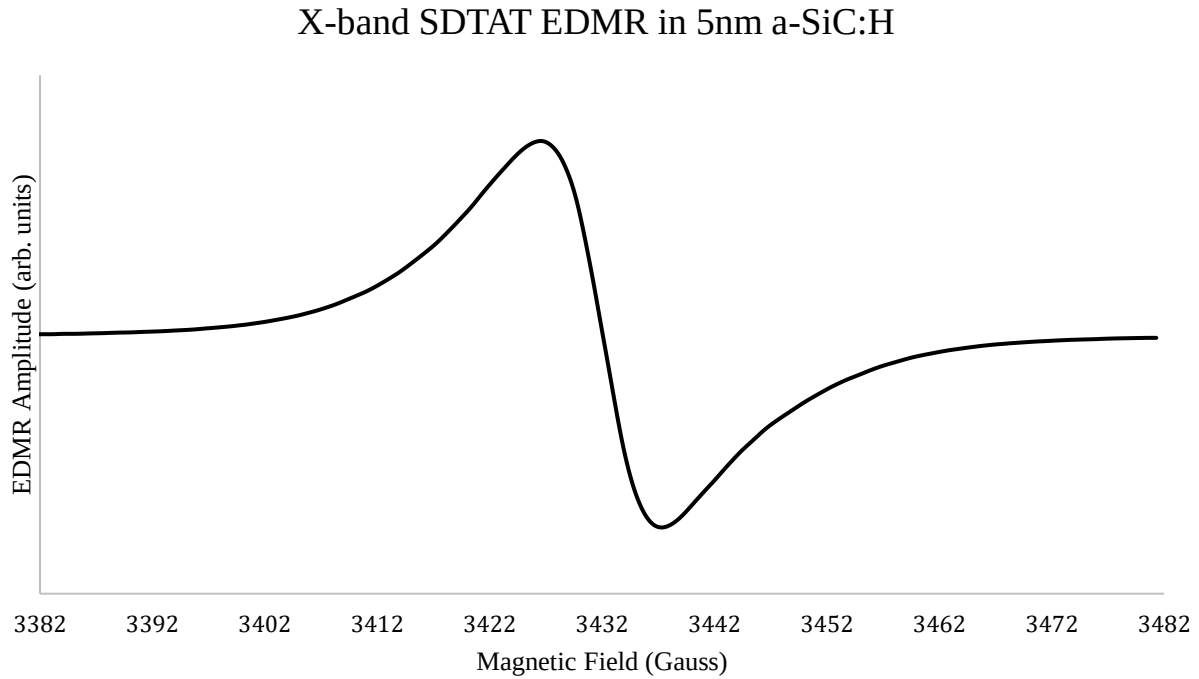


Figure 6.2: X-band SDTAT EDMR in 5nm a-SiC:H. Sample biased at +2V. The signal is 10.8 Gauss wide. This defect has a zero-crossing g value of 2.0032.

The measured spectrum has a linewidth of 10.8 Gauss and a zero-crossing g value of 2.0032. A series of measurements at different biases was made yielding spectra of different amplitudes but with similar line shapes. The line shapes of the normalized measured spectra are compared in Figure 6.3.

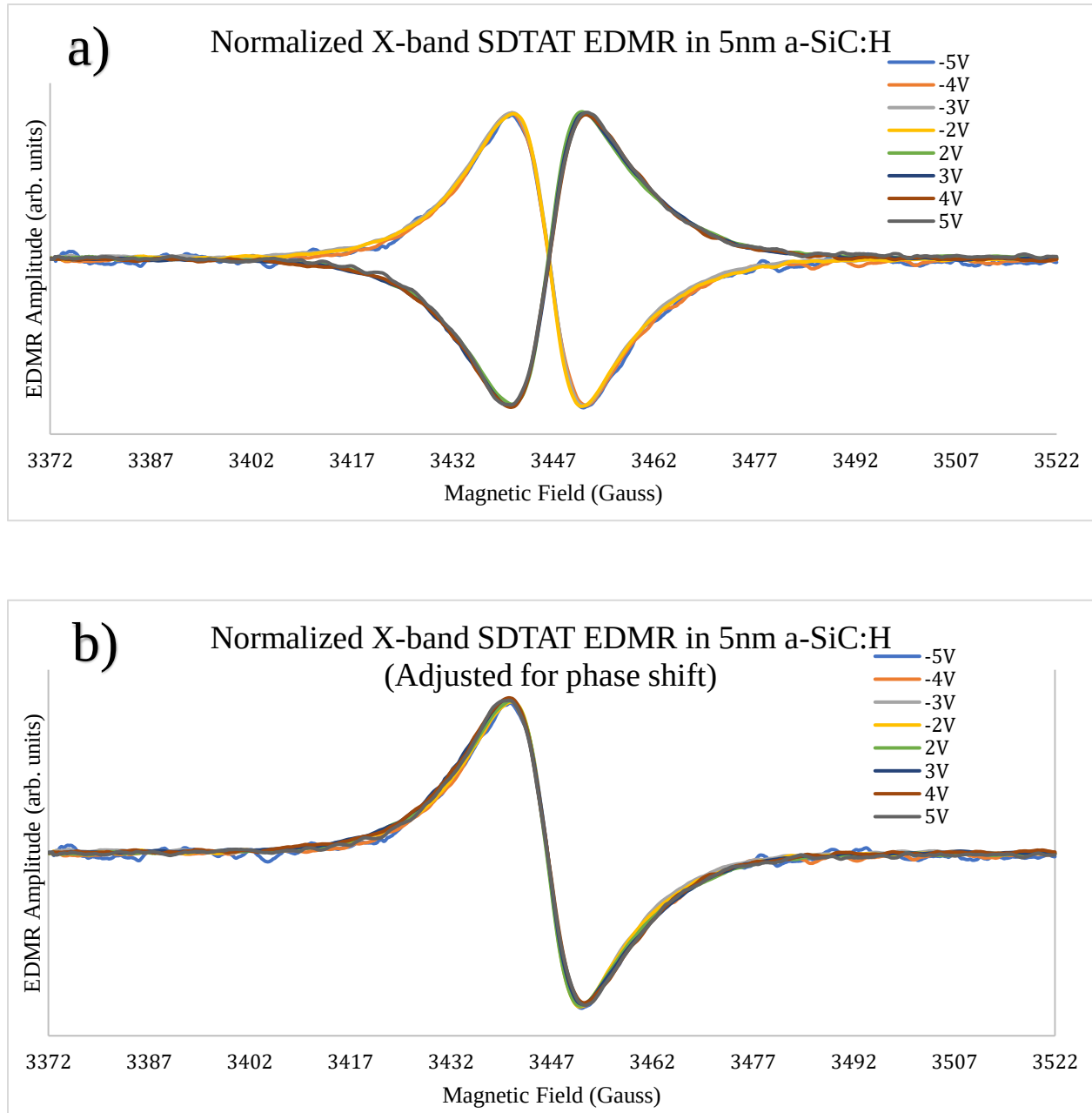


Figure 6.3: Normalized SDTAT EDMR spectra in 5nm a-SiC:H. a) Spectra showing phase of signal changes due to inverting polarity of the applied bias. b) All spectra with same phase.

There is no change in the linewidth or general line shape of the signals with the same polarity of applied bias. Inverting the normalized signals measured under a positive applied bias shows that the line shape of all signals is essentially identical across all measured applied biases.

The only difference between the X-band SDTAT EDMR spectra at different biases is the amplitude of the response. The amplitude of the EDMR response at different biases is shown in Figure 6.4, showing an exponential growth in the signal amplitude with bias, although the signal amplitude is larger at positive applied bias.

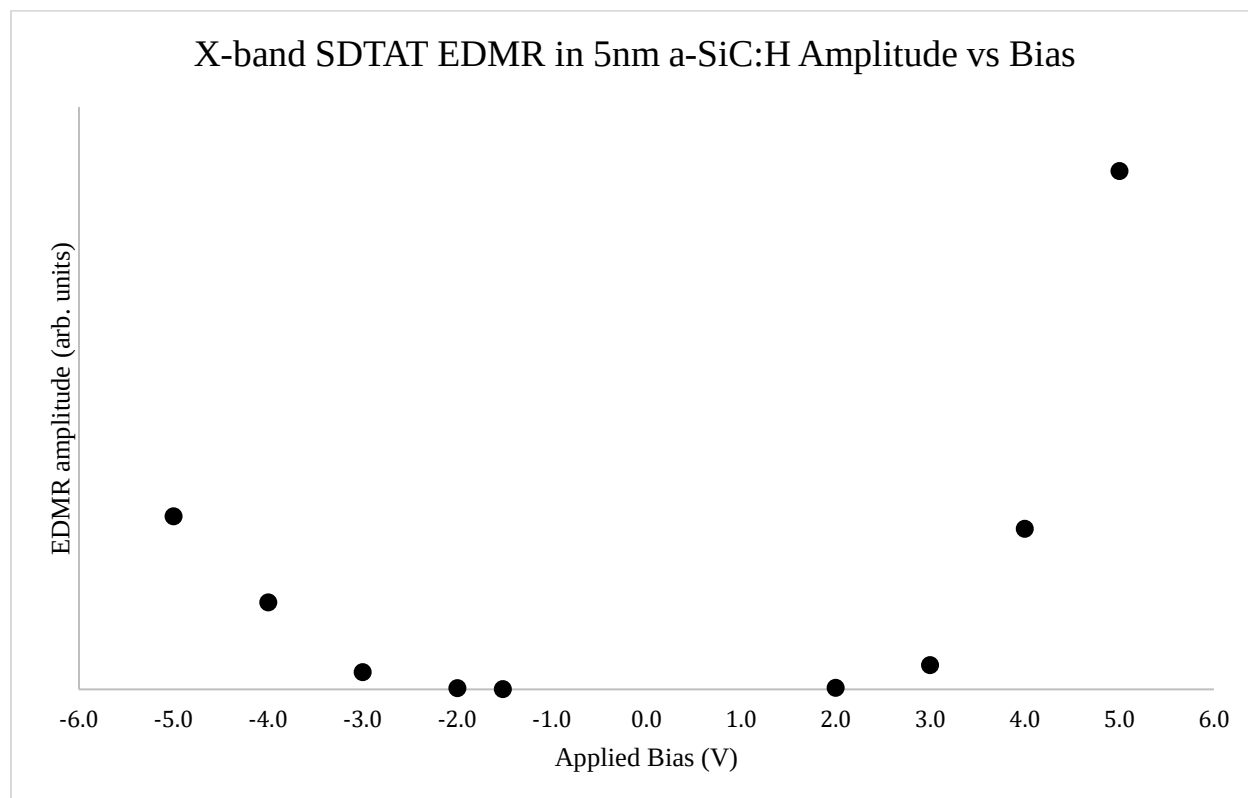


Figure 6.4: X-band SDTAT EDMR amplitude vs bias in a-SiC:H. Larger signals are measured at positive bias.

Chapter 6.2 NZFMR and Low Field EDMR in 5nm a-SiC:H

Due to the stability of the 5nm a-SiC:H, the same samples studied at X-band were studied using NZFMR and low field EDMR. These measurements were made on the Blue-Spin LF/ZF spectrometer. EDMR and NZFMR measurements were made at biases between -5V and 5V. Normalized spectra at several applied biases are shown in Figure 6.5. Unlike the X-band and low-field SDTAT EDMR response, the NZFMR linewidth changes with applied bias, becoming

slightly wider with increasing bias magnitude (both positive and negative). The relative size of the EDMR to the NZFMR response also changes with bias, initially increasing with increased bias, reaching the maximum relative size of the EDMR at an applied bias of +3V, then becoming smaller with increasing bias. This is a curious effect and likely arises from the different mechanisms that give rise to each signal in this material, with EDMR being caused by spin flipping, and NZFMR caused by spin mixing.

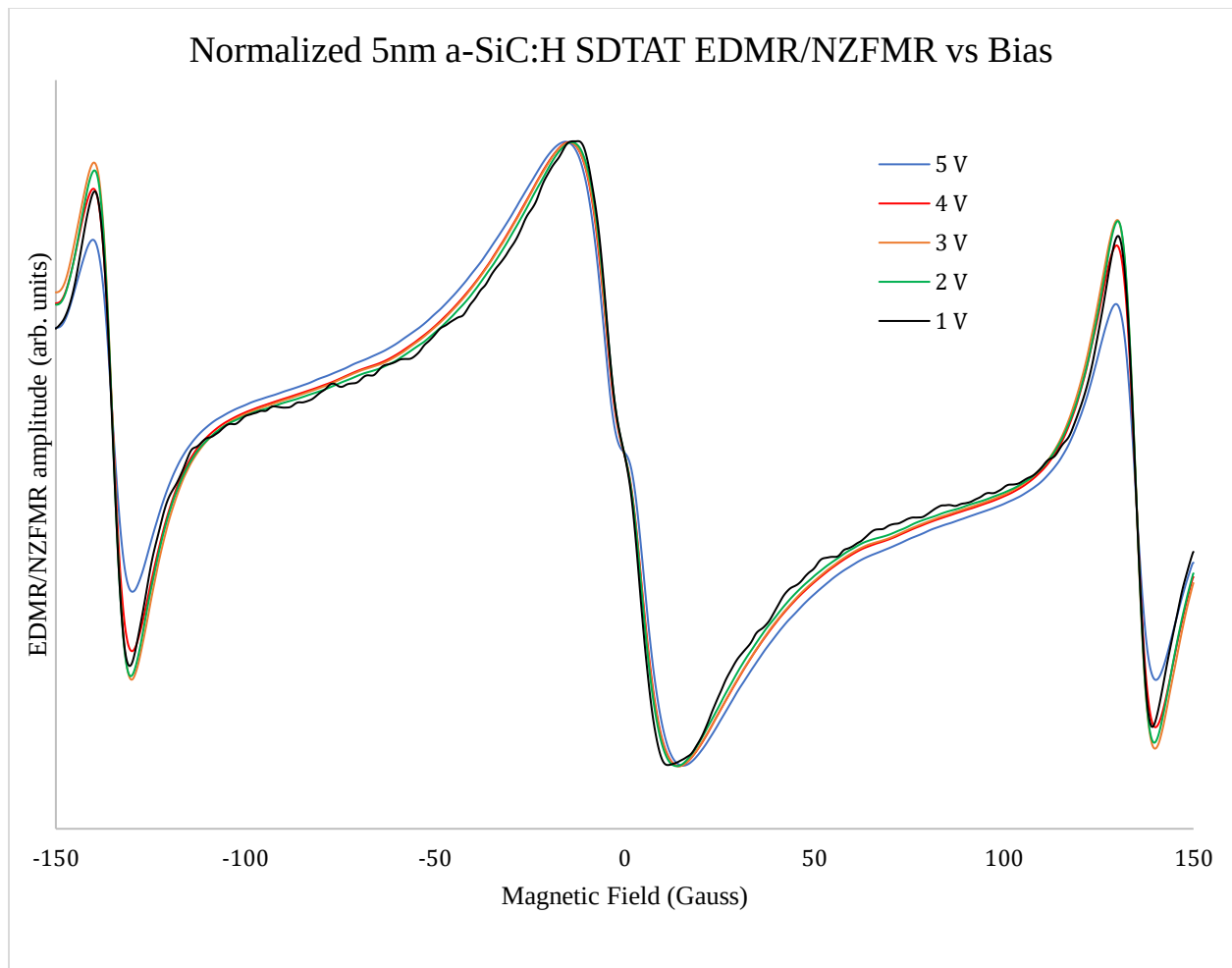


Figure 6.5: Normalized SDTAT EDMR/NZFMR response in 5nm a-SiC:H. Relative to the NZFMR response, the amplitude of the 8 Gauss wide EDMR signal proportionally increases until it reaches 3 V, then decreases as the bias is increased. The NZFMR response becomes wider with increased bias from 25 Gauss at 1V to 31 Gauss at 5V. Measurements were made using a 5 Gauss modulation amplitude.

The magnitude of the SDTAT NZFMR response, shown in Figure 6.6, follows a similar pattern to the X-band SDTAT EDMR, increasing at higher bias and larger for positive applied bias than for negative.

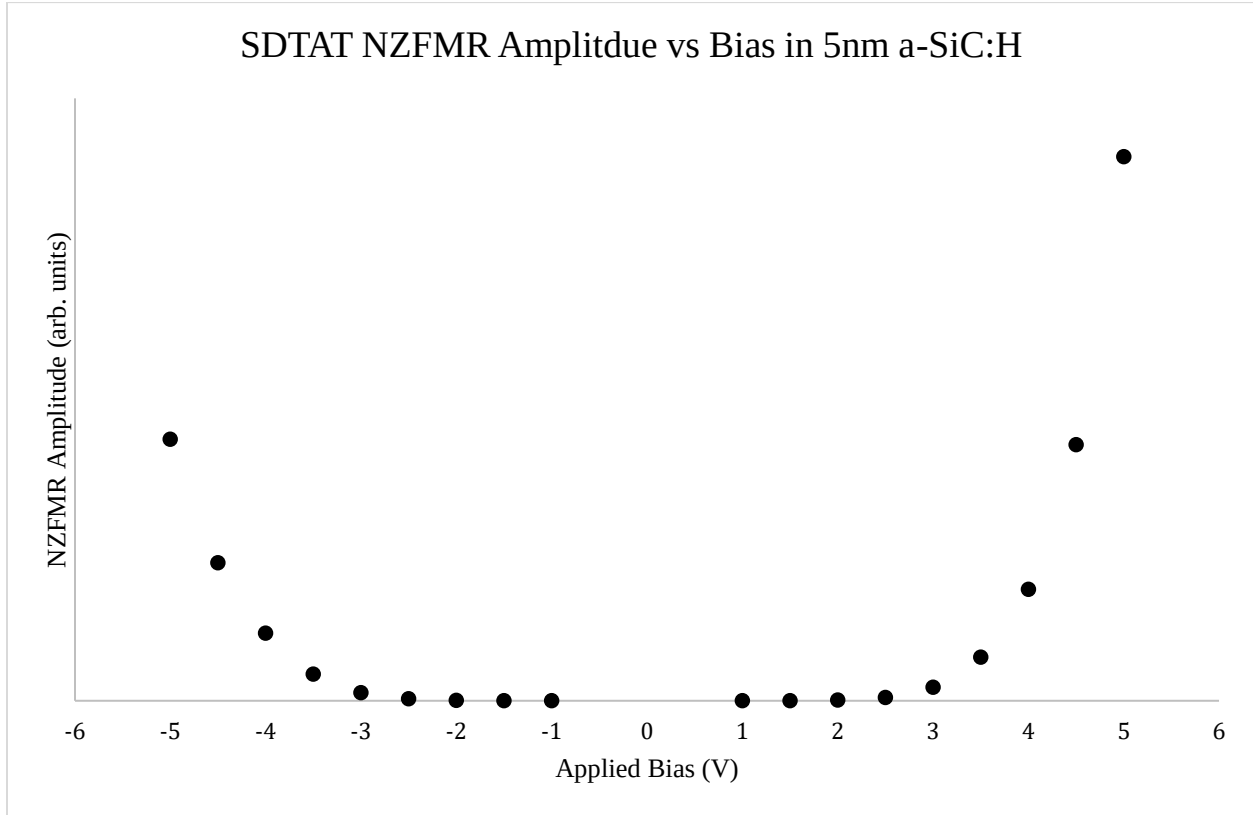


Figure 6.6: SDTAT NZFMR amplitude vs bias in 5nm a-SiC:H.

The defect responsible for conduction could either be related to silicon or carbon atoms as both are present in the material. One method to determine which species is likely involved is discussed by Mutch. [88] An approximate range of g values, Δg , can be calculated for a sample by using the difference in the linewidths of spectra measured at very different resonance frequencies. The linewidth measured at X-band was 10.8 Gauss while the low-field EDMR signal was 8 Gauss wide; a difference of 2.8 Gauss. This corresponds to a $\Delta g = 0.0016$. The

Δg value for a silicon dangling bond in a-Si:H is approximately 0.0044 while the Δg value for a carbon dangling bond in a-C:H is approximately 0.0011. [88] The Δg value for the defect in the a-SiC:H in this study is very close to the Δg value for carbon dangling bonds in a-C:H and therefore these spectra are attributed to carbon dangling bonds.

SDTAT NZFMR measurements without the RF oscillating B_1 were made over a series of applied biases using a smaller magnetic field modulation amplitude of 1 Gauss to resolve the center inflection signal. These measurements are shown in Figure 6.7. The NZFMR measurements made without the RF B_1 oscillating magnetic field appear to be essentially identical to the NZFMR response when the oscillating B_1 field is present and the line width of the NZFMR response increases with larger magnitude applied bias.

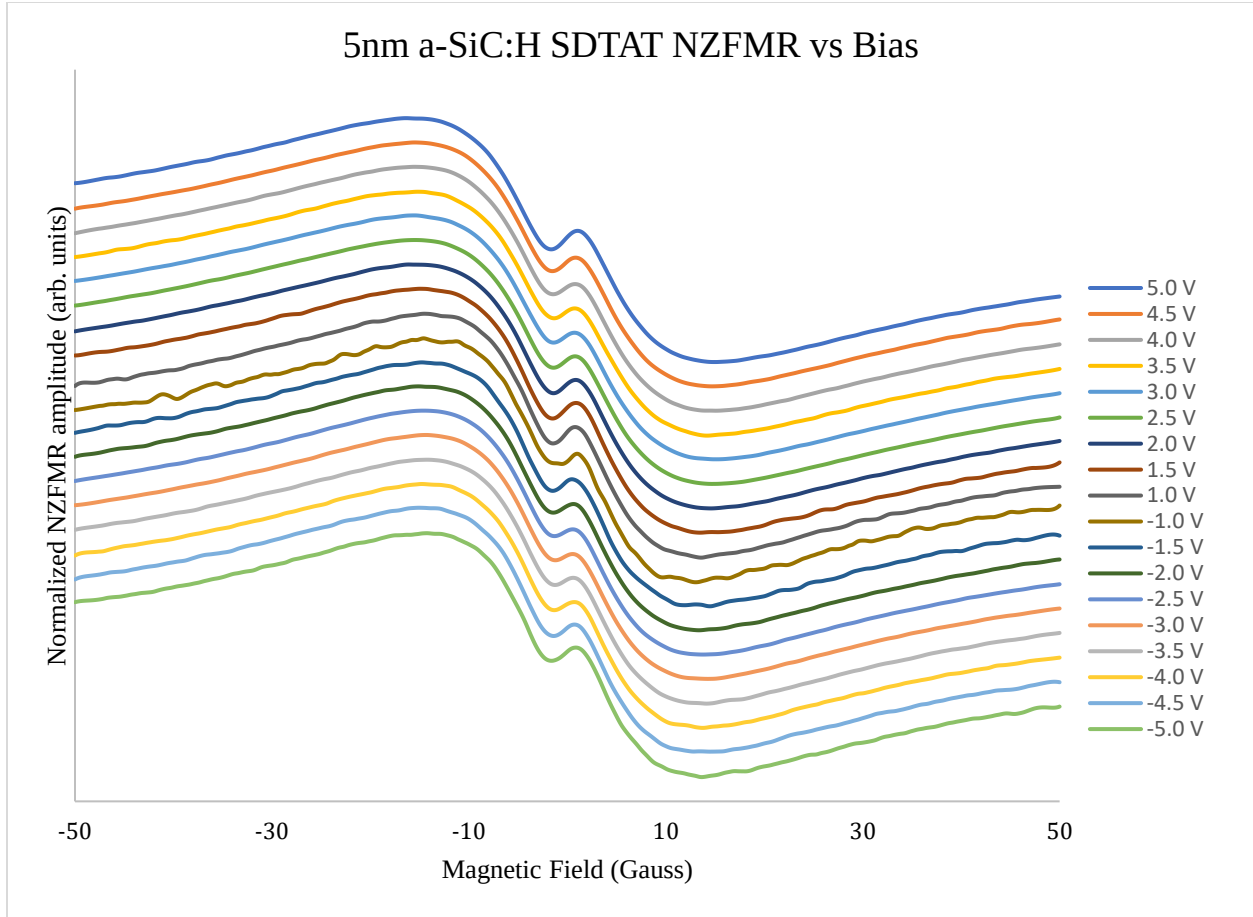


Figure 6.7: Normalized SDTAT NZFMR in a-SiC:H for multiple applied biases without oscillating B_1 magnetic field. The NZFMR response becomes wider with increased bias from 25 Gauss at 1V to 31 Gauss at 5V. The inflection linewidth stays at a roughly constant value of 2.3 Gauss.

This response suggests that the parameters from Figure 3.4 that are changing include $r_{e\alpha}$ due to the increased signal amplitude, and $r_{\alpha\beta}$ and $r_{\alpha e}$ which can both affect the width of the NZFMR response. The spin system for a-SiC:H is probably too complicated [89]–[91] for the current models to simulate, and as such, no simulations have been made regarding the a-SiC:H. Qualitative comparisons to the plots from chapter 3.4.2, however, strongly suggest that the dissociation rate, $r_{\alpha e}$, is decreasing and the hopping rate, $r_{\alpha\beta}$, is increasing as both of these changes to these parameters would cause the broad line to become wider. Each change would

affect the amplitude of the inflection line in an opposite way so a combination of these two changes could keep the center inflection line constant, essentially canceling each other out. Future work on this material system will likely involve improving modeling capabilities to account for the more complex nature of the SiC spin system.

Chapter 6.3 Lock-in Phase Dependent Responses

The large signal to noise ratio of the a-SiC:H samples allows for a curious phenomenon to be seen in which the spectrum line shape is different depending on the phase detection angle of the lock-in amplifier. This is most easily seen during an EDMR and NZFMR measurement where the phase of the EDMR response can be flipped with respect to the phase of the NZFMR response by careful selection of a lock-in detection phase. This is shown in Figure 6.8 where the two spectra have the same phase orientation for the NZFMR response (down-up-down) but have opposite phase orientation for the EDMR response (up-down-up vs down-up-down). These two spectra were measured simultaneously under identical experimental conditions, with the only difference between them being the detection phase used by the lock-in amplifier. These signals are interpreted by the lock-in amplifier 90 degrees apart. It should be noted that the top spectrum is much smaller than the bottom spectrum. There is also a discrepancy between the size of the center inflection of the NZFMR signals, with the relative size of the inflection in the top signal being significantly larger than that of the bottom. This measurement was made using the Blue Spin LF/ZF spectrometer. The change in the relative amplitude of the center inflection suggests that the center inflection and broad signal arise from different physical mechanisms which newer models will have to account for in order to extract information about what those mechanisms may be.

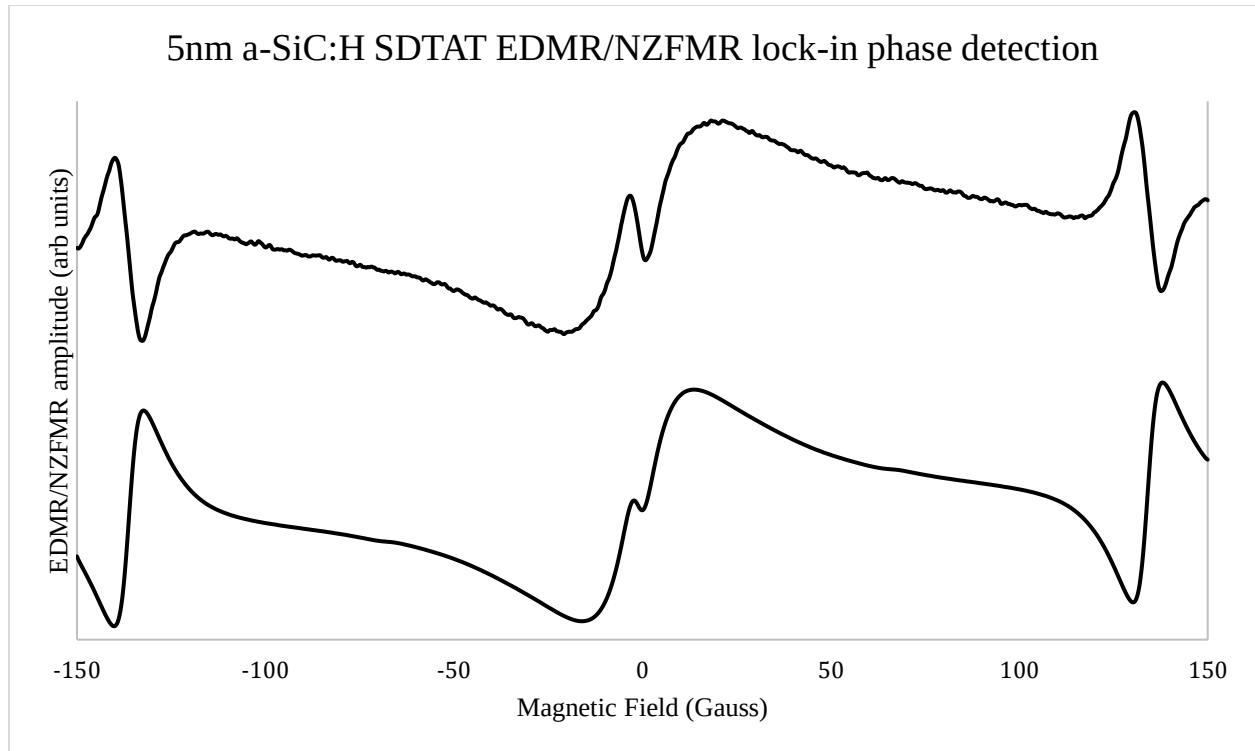


Figure 6.8: Phase shift in EDMR response in 5nm a-SiC:H relative to NZFMR response. Also note the relative size difference between the center inflection in the top and bottom trace. Measurements were made simultaneously at identical bias. Signal amplitudes are normalized to show different line shapes.

A similar response appears at X-band with two differently shaped spectra measured simultaneously under identical experimental conditions. This may be due to a less dominant conduction mechanism that has a slightly different phase response to the magnetic field modulation that is obscured by the larger signal and can be seen once the primary conduction mechanism is removed by using the correct lock-in phase. This difference is illustrated in Figure 6.9. The Lakeshore X-band spectrometer was used to perform this measurement.

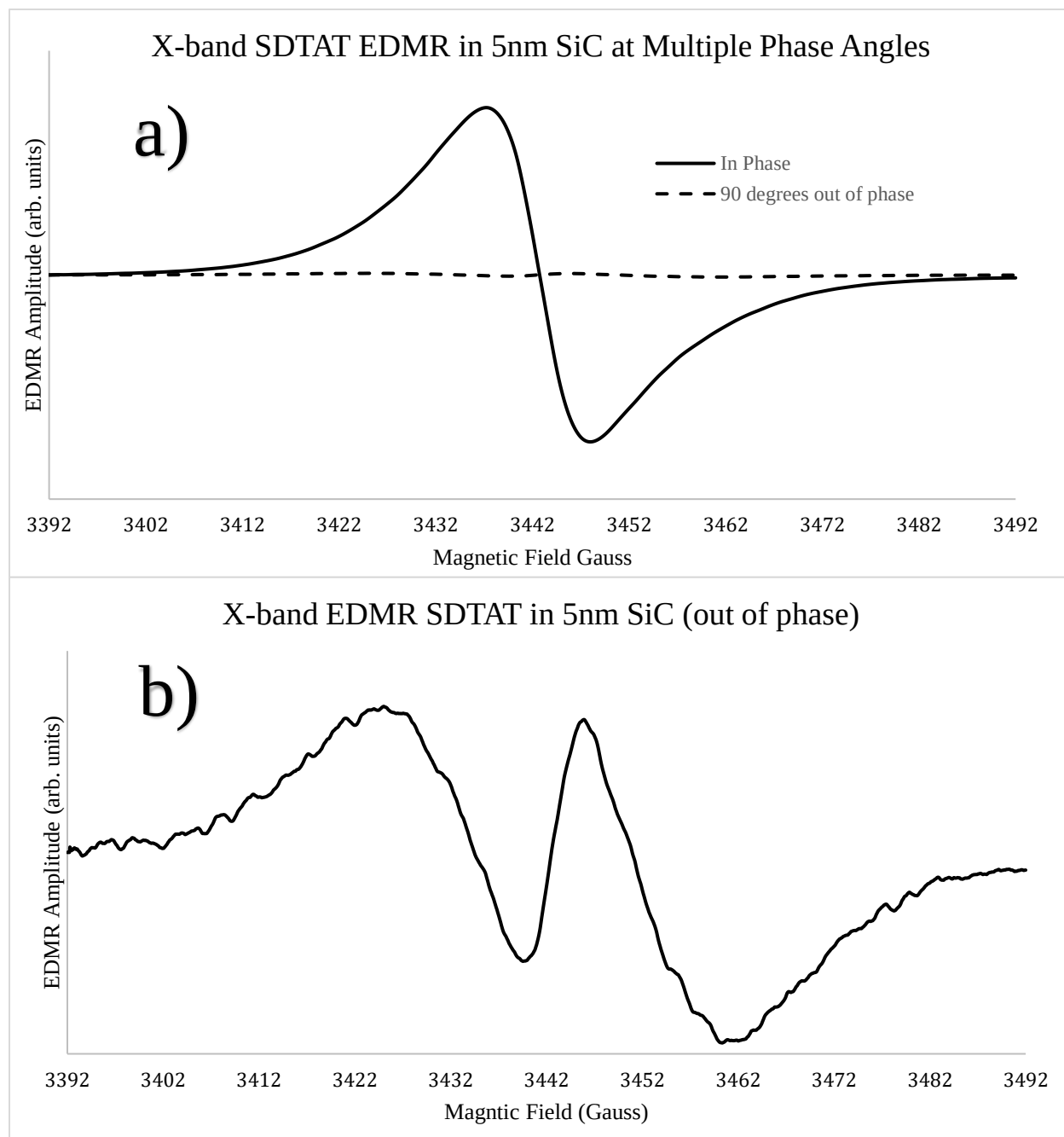


Figure 6.9: a) Relative amplitude of “in phase” vs “out of phase” SDTAT EDMR response. b) Zoomed in “out of phase” measurement in 5nm a-SiC:H, note the different shape of the EDMR response compared to the larger “in phase” response.

Chapter 6.4 Discussions about SiC, Contribution to Theory, and Future Work

The computational models for this system are still being developed, as such no direct comparisons to simulated spectra are shown. The data provided on this material in this work will, however, aid in the development of a computational model for this material system.

The SDTAT response as well as the DC current in the a-SiC:H is very stable over a wide range of biases and stressing time, unlike the amorphous carbon samples. The 5nm a-SiC:H exhibits different SDTAT NZFMR responses under different applied biases, becoming wider at higher positive bias. This would indicate that the parameters from equations 3.2.5 and 3.2.6 that change are likely r_{ea} , r_{ab} , and r_{ae} . The increased current and larger signal at higher biasing conditions would be a result of larger initial hopping rate, r_{ea} . The slight broadening of the NZFMR response with increased bias is consistent with an increase in r_{ab} . A combination of an increase in r_{ab} , and a decrease r_{ae} could also account for the approximately constant ratio the amplitude of the broad signal to that of inflection line, as the two rates changing with bias would essentially work against each other in this aspect, while allowing the broad signal to become wider.

The difference in the relative size of the center inflection in the SDTAT NZFMR response due to the lock-in detection phase may also indicate that the center inflections observed in almost all NZFMR measurements may be due to a different conduction pathway than the wider and larger signal which occurs at a slightly different phase relative to the magnetic field modulation. When combined with the changes seen in the NZFMR response in the DLC, this seems to be fairly strong evidence of each portion of the signal (the broader signal and the center inflection) arising from different pathways and spin mixing.

The SiC spin systems, for both crystalline and amorphous materials, are difficult to model as there are several spin states depending on whether a site is occupied by an electron and only certain spin transitions are allowed. [89]–[91] The spectra measured in this work will be useful for our collaborators who are focused on modelling the NZFMR phenomenon in a wider range of material systems, including different forms of SiC such as the most common and technologically relevant hexagonal 4H-SiC.

These samples offer incredibly good signal to noise ratios compared to other samples studied by the laboratory group. The simple nature of the capacitor dot samples, the simplicity of biasing scheme utilized in the measurements, and the good signal to noise ratio could allow these samples to be utilized as test samples when developing new spin-based tools and spectroscopic techniques such as the wafer probing station spectrometer outlined in appendix B.

Chapter 7

Conclusions

This study sought to develop an understanding of the atomic scale defects involved in electronic transport in amorphous carbon and amorphous silicon carbide using NZFMR and EDMR, as well as to determine the analytical power of new computational models for NZFMR.

Chapter 5 presented a variety of measurements performed on amorphous hydrogenated carbon and amorphous diamond-like carbon metal-oxide-semiconductor capacitor dots. The defect responsible for conduction in both is likely a carbon dangling bond. The a-C:H samples exhibited drastic changes to the electrical properties over time under constant biasing conditions, with the current decreasing by up two orders of magnitude over the course of several minutes. These changes in current were accompanied by trapped charge in the carbon layer. The trapped charges raise the tunneling barrier experienced by charge carriers, explaining the reduction in current. The shape of the EDMR and NZFMR spectra never changed, indicating that the electrically active defects are not changing. The line shape is also constant for different biasing conditions up to 60V. It is plausible that the trapped charges and reduced current could be symptoms of a time dependent dielectric breakdown process.

The measured NZFMR spectra were analyzed using the computational model outlined by Frantz et al. [64] to determine a number of physical parameters of the measurement. Simulated spectra were fit to the experimental data to extract hyperfine coupling constants, hopping rates, and dissociation rates. The values produced by the fitted data are physically plausible and likely correct to at least the order of magnitude. The hyperfine coupling constant could not be larger than the low field EDMR linewidth which is dominated by hyperfine interactions. The extracted hyperfine coupling constant was about 3.4 Gauss, below the 7 Gauss maximum of the low-field

EDMR linewidth and a reasonable value for nearby hydrogen. All of the parameters are of the same order of magnitude as those extracted from a recent study involving simulated spectra fitted to a spectrum for a P_b center at the Si/SiO₂ interface. [64] The hyperfine coupling constant attributed to hydrogen near a P_b center was found to be approximately 2 Gauss. Our larger value of approximately 3.4 Gauss appears to make sense given that nearby hydrogen nuclei are likely to be closer to the carbon atoms in a-C:H than the silicon atoms at the Si/SiO₂ interface, given the size difference of the respective atoms. No changes in the shape of the NZFMR spectra for different biasing conditions suggests that only the generation rate is changing, while the rest of the parameters are constant. The change in the generation rate can be attributed to the lower current due to the increased tunneling barrier caused by the trapped charge in the carbon layer. The computational model appears to be an effective tool for analyzing the a-C:H samples.

The other amorphous carbon sample, diamond-like carbon, also exhibited unstable electrical characteristics, however the changes seen in the DLC were much slower and required a larger bias to achieve. At low biasing conditions, the samples were fairly stable and simulated spectra could be fit to the experimental data to extract some physical parameters. Unlike the a-C:H, the shape of the NZFMR spectra changed with bias in the DLC samples. Simulated spectra were fitted to three experimental spectra corresponding to the lower biasing conditions of 1V, 2V, and 3V. The value for hyperfine coupling constant, 3.8 Gauss, was found to be similar to the value for the other carbon samples and less than the 9 Gauss wide low-field EDMR line width. The larger value for the DLC make sense as it has a broader low field EDMR line-width, which is likely dominated by hyperfine interactions. The hopping rate in the DLC is similar to that in the a-C:H although the changes in the line shape for increasing bias have been attributed

to a reduction in the dissociation rate, likely because more hopping can occur at higher bias. The values for this rate are also similar to the dissociation rate in the a-C:H samples.

The changes in the DLC samples over time under constant bias are different than those seen in the a-C:H samples, with the leakage current increasing over time. This change in current was accompanied by a change in the shape of the NZFMR spectra, although the signal to noise ratio of these measurements was not sufficient for fitting simulated spectra. The drastic change in the line shape indicates that another defect is likely forming in the sample and is contributing to conduction. Some progress has been made on determining the hyperfine coupling constants that could result in the changed spectra, and the best attempt so far indicates that two pathways with different hyperfine coupling constants are likely contributing to the measured spectra. The change in line shape can be achieved by adjusting the contribution of each of the pathways over time. [87] The computational model appears to be an effective tool for analyzing the DLC at low bias and will likely become more effective at higher biasing conditions as the model is developed.

Chapter 6 presented EMDR and NZFMR measurements performed on amorphous hydrogenated silicon carbide. The a-SiC:H samples were very stable over time and no changes were observed in the electrical characteristics or the EDMR/NZFMR spectra. X-band EDMR data was not enough to determine the defect responsible for conduction, however, comparing the X-band spectra with those measured at low magnetic field allowed for a value to be calculated for $\Delta g = 0.0016$. This indicates that the defect responsible for the spectra is likely a carbon dangling bond.

Computational models for SiC systems are still being developed, however, due to the potential complexity of the SiC structures [89]–[91], no simulated spectra were fitted. A

qualitative understanding of the changes to the NZFMR spectra due to bias indicates that the increased signal amplitude likely due to a larger generation rate, due to a higher overall current. The generation and dissociation rates change due to bias in manner that causes the center inflection shape to remain constant and the broad line to become wider. This is likely caused by an increase in the hopping rate and a decrease in the dissociation rate as the bias is increased. This may seem counterintuitive as one might expect the spin-independent dissociation rate to become larger at higher bias as tunneling through the thin samples becomes easier. The data collected on these model samples will be utilized to further develop the computational model in order to study and analyze other devices and material systems.

A curious phenomenon was observed in these a-SiC:H samples. A different line shape was observed for both NZFMR and EDMR measurements for different lock-in detection phases. This may be due to a separate, and much less prominent signal, that is usually obscured by the large signal. Careful selection of the lock-in detection phase allows one to essentially cut out the large signal and see the smaller one. The different pathway may also be responsible for the inflection line in the NZFMR signal as the relative amplitude of the inflection to the broad line also changes with detection phase. The large signal to noise ratio of these samples also allows them to be used when developing new spin dependent spectroscopic and measurement tools such as the spin-based magnetometer proposed by Cochrane [43].

APPENDIX A

NZFMR AND LOW FIELD EDMR IN SIGE P-MOSFETS

Silicon germanium ($\text{Si}_{1-x}\text{Ge}_x$) alloys are potentially useful materials with tunable bandgaps, depending on the concentration of each species. The bandgap can be tuned between the values for pure germanium and pure silicon, 0.66 eV and 1.1 eV respectively. The material can be easily implemented into silicon manufacturing lines and the tunability of the bandgap allows for faster heterojunction bipolar transistors and thin film CMOS transistors to be manufactured. [92]–[95] This makes SiGe alloys good candidates to be studied using spin-based spectroscopy techniques to identify the performance limiting defects in the material.

EDMR and NZFMR measurements were made on SiGe alloy p-MOSFETs with varying levels of success. Low field EDMR and NZFMR were successfully performed using the SDGP and DCIV biasing schemes, however the same measurements could not be successfully performed at X-band. Several attempts were made to boost the signal to noise ratio of the measurements at X-band including increasing the power of the microwaves in the microwave cavity and performing the measurements at low temperatures ranging from 77K to room temperature (300K). The low temperature measurements were made using liquid nitrogen in a Janis continuous flow cryostat installed into the modified commercial Bruker EDMR spectrometer used in other portions of this work. These noise reduction methods did not produce usable results at X-band and as such only low-field EDMR and NZFMR spectra are reported for the SiGe p-MOSFETs.

It is possible that the reason no spectra were observed in X-band EDMR measurements is a large spin-orbit coupling from germanium atoms, which have much higher nuclear charge (32 protons) than the silicon (14 protons) and carbon (6 protons) atoms from other measurements in

this work. As stated before, large nuclear charges increase spin-orbit coupling and can broaden EDMR signals at higher frequencies. The increased broadening of the signal due to the spin orbit coupling from germanium nuclei could widen the signal, thereby reducing the signal amplitude below the detection level of the spectrometer. Several attempts were made to signal average for more than seven days without producing recognizable spectra. One of these attempts is shown in Figure A.1.

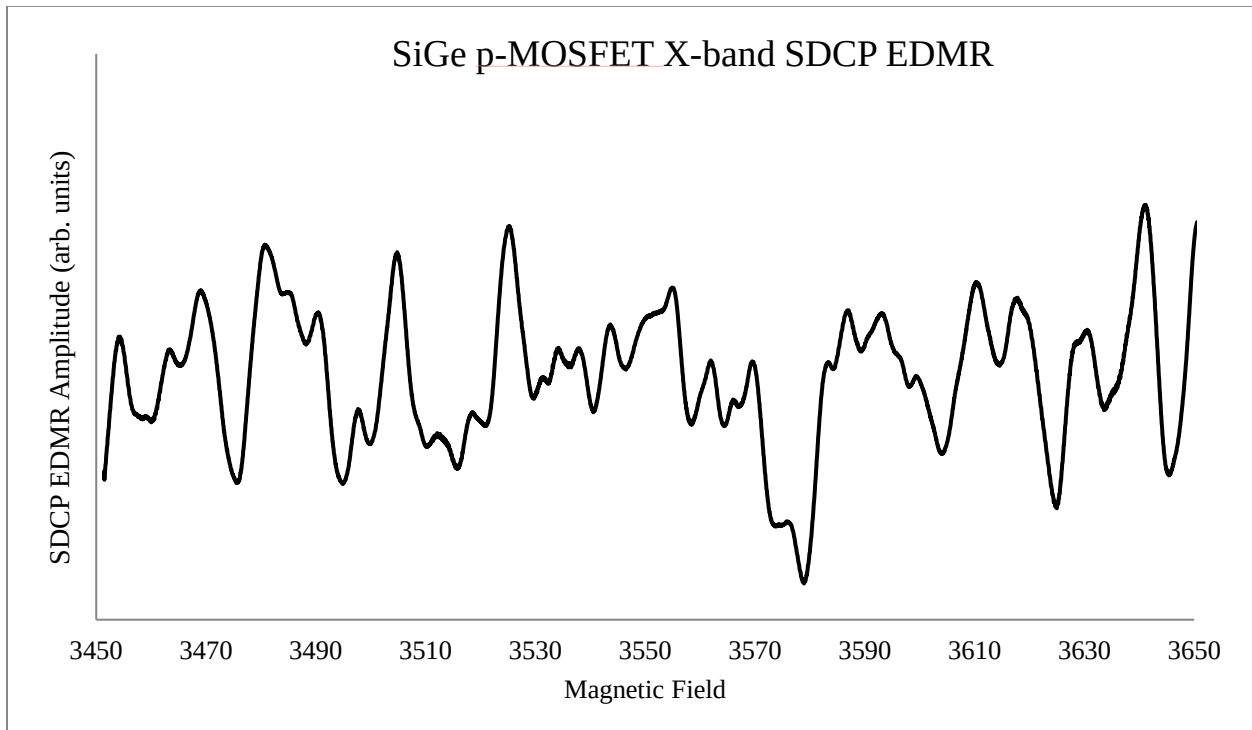


Figure A.1: X-band SDCP EDMR measurement in $\text{Si}_{0.6}\text{Ge}_{0.4}$ pMOSFET, no spectrum is measured even though extensive signal averaging was utilized. The trace was averaged 5000 times over the period of one week.

The SiGe p-MOSFET samples are epitaxial $\text{Si}_{0.6}\text{Ge}_{0.4}$ grown on silicon with effective an effective gate oxide thickness of 1.2nm. The gate area was approximately $100\mu\text{m}$ wide by $1\mu\text{m}$ long, making them ideal for charge pumping measurements. The first measurement performed on the SiGe MOSFET samples was near zero field spin dependent charge pumping (NZFSDCP).

The gate pulse used for this measurement was a 1 MHz, $1.6V_{pk-pk}$ (-.4V to 1.2V), trapezoidal waveform with rise and fall times equal to 100ns ($1/10^{th}$ of the total waveform period). This waveform was chosen to bring the interface into accumulation and inversion and the rise and fall times were chosen to provide enough time to allow free carriers at the interface to evacuate before the opposite charge carriers were brought to the interface for recombination to occur. The measured SDCP EDMR and NZFSDCP response is shown in Figure A.2.

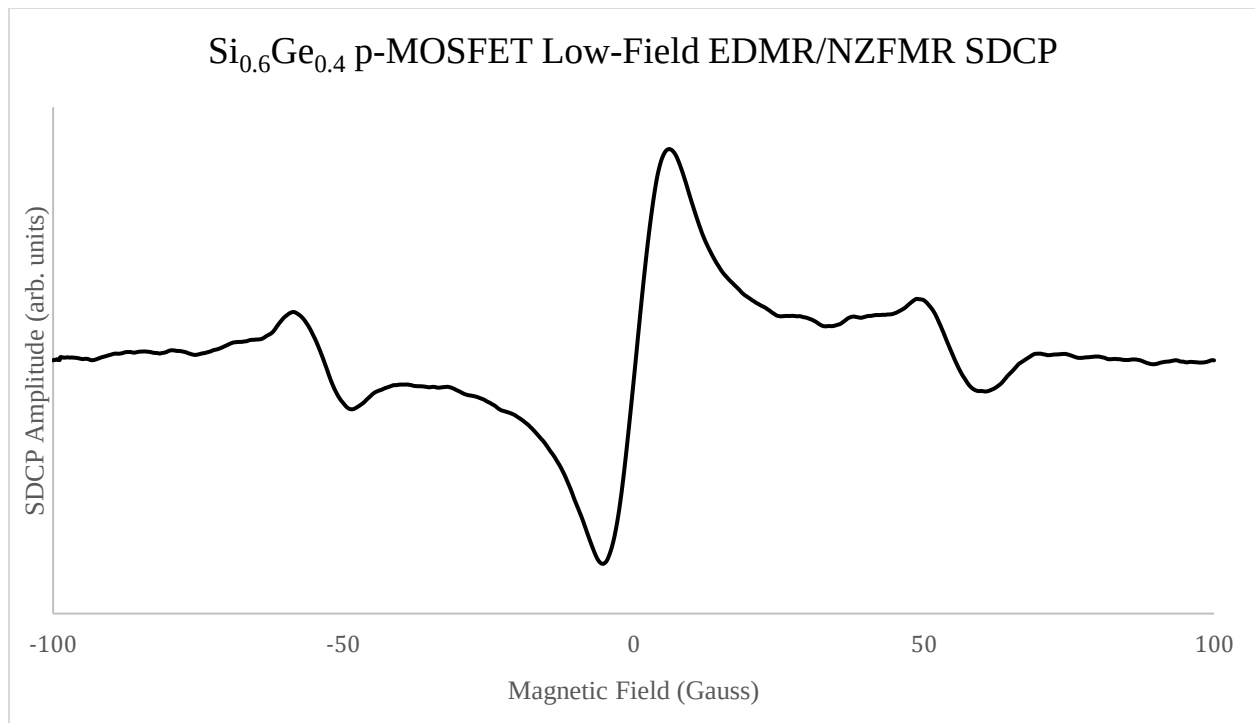


Figure A.2: Near zero field and low field EDMR spin dependent charge pumping in $Si_{0.6}Ge_{0.4}$ pMOSFET.

NZFMR was also performed on a similar sample that was biased in the DCIV scheme with the gate held at 0V and the source/drain to body bias of +0.3V. This scheme was chosen to ensure the interface was held in depletion to facilitate recombination at the interface. The signal to noise ratio of this measurement was much lower than that for the SDCP EDMR and NZFSDCP. The measured spectrum is shown in Figure A.3.

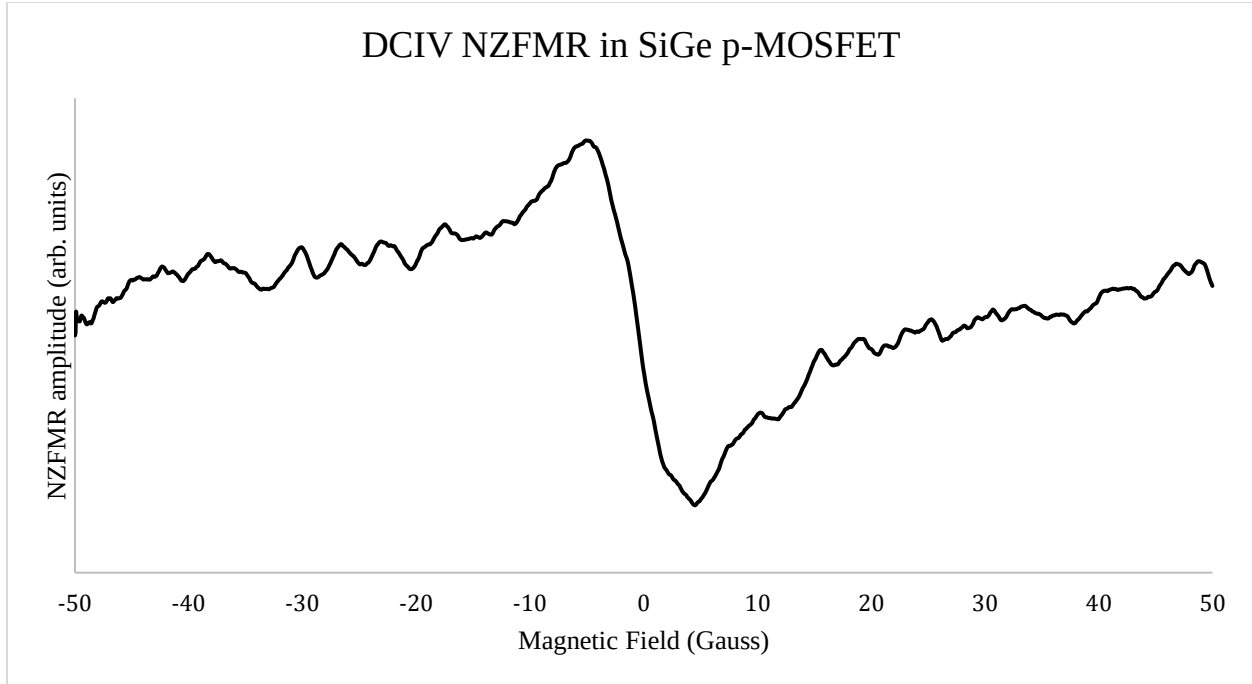


Figure A.3: DCIV NZFMR spectrum in $\text{Si}_{0.6}\text{Ge}_{0.4}$ pMOSFET.

The phase shift phenomenon seen in the a-SiC:H is also seen in the SDCP measurements in the $\text{Si}_{0.6}\text{Ge}_{0.4}$ MOSFETs where the EDMR (up-down-up) is out of phase with the NZFMR (down-up-down) as seen in Figure A.2. This phase shift makes sense as the two signals arise from very different mechanisms. Both signals occur because of a change in the singlet population of the electron spins allowing for recombination to occur; however, the EDMR response is due to electron spins flipping at resonance while the NZFMR response is due to spin mixing with nearby magnetic nuclei. It is therefore understandable why these two different mechanisms could occur at different phases relative to the modulation field phase.

APPENDIX B

SDTAT NZFMR SPECTROSCOPY IN A WAFER PROBING STATION

B.1 Construction of NZFMR Spectrometer in Wafer Probing Station

NZFMR is a simpler spectroscopic technique than EPR and EDMR; however, it typically involves quite a lot of sample preparation. Samples must be diced from their wafers and mounted onto special PCBs used to hold the sample in the electromagnet and to make electrical contact to the sample. The ability to bypass sample preparation and perform spin-based spectroscopy on samples at the wafer level would add to the speed and throughput of the technique, allowing for more samples to be studied in a shorter time. EDMR spectroscopy has been performed on a wafer probing station [96]–[98], and some forms of NZFMR, specifically SDCP, have been performed at the wafer level [99].

An apparatus similar to that proposed by Anders et al. [99] was designed and built to allow for wafer level SDTAT NZFMR on the samples studied in this work. This undertaking was initially proposed in order to better study the 500nm a-C:H due to a problem with the metal top contact having poor adhesion to the carbon film. The metal layer could be accidentally removed from the sample during wire bonding, rendering the sample essentially useless. It was decided that this problem would be addressed in two ways, the first of which was to determine a method by which the samples could be wire bonded without causing damage to the samples or the metallic contact. The second method used to address this problem was to incorporate NZFMR spectroscopy into a wafer probing station, removing the need to bond wires to the sample entirely. It was decided that the first method, determining the proper way to wire bond to the samples without causing damage to them, would be the most effective as it would allow for the samples to be studied using X-band and low field EDMR as well as NZFMR, however,

NZFMR would also be incorporated into a wafer probing station in order to determine how the technique could be improved upon.

A small electromagnet consisting of 400 turns of enamel coated magnet wire with a modulation coil consisting of 200 turns of finer gauge enamel coated magnet wire was built. The electromagnet and modulation coils were powered using components from the Blue Spin low field EDMR/NZFMR spectrometer used in other parts of this work. The applied magnetic field was measured directly using a Gaussmeter and Hall effect probe built into the Blue Spin spectrometer.

B.2 NZFMR Measurements in a Wafer Probing Station

The first NZFMR measurement on the wafer probing station was made on SiC pn-diodes that were mounted onto the PCB Ts. These samples were chosen as a good test platform for the new apparatus as they have a known and well understood spectrum and provide a very good signal to noise ratio, making them an ideal choice to test new systems. The measured SDTAT NZFMR spectra are shown in Figure B.1.

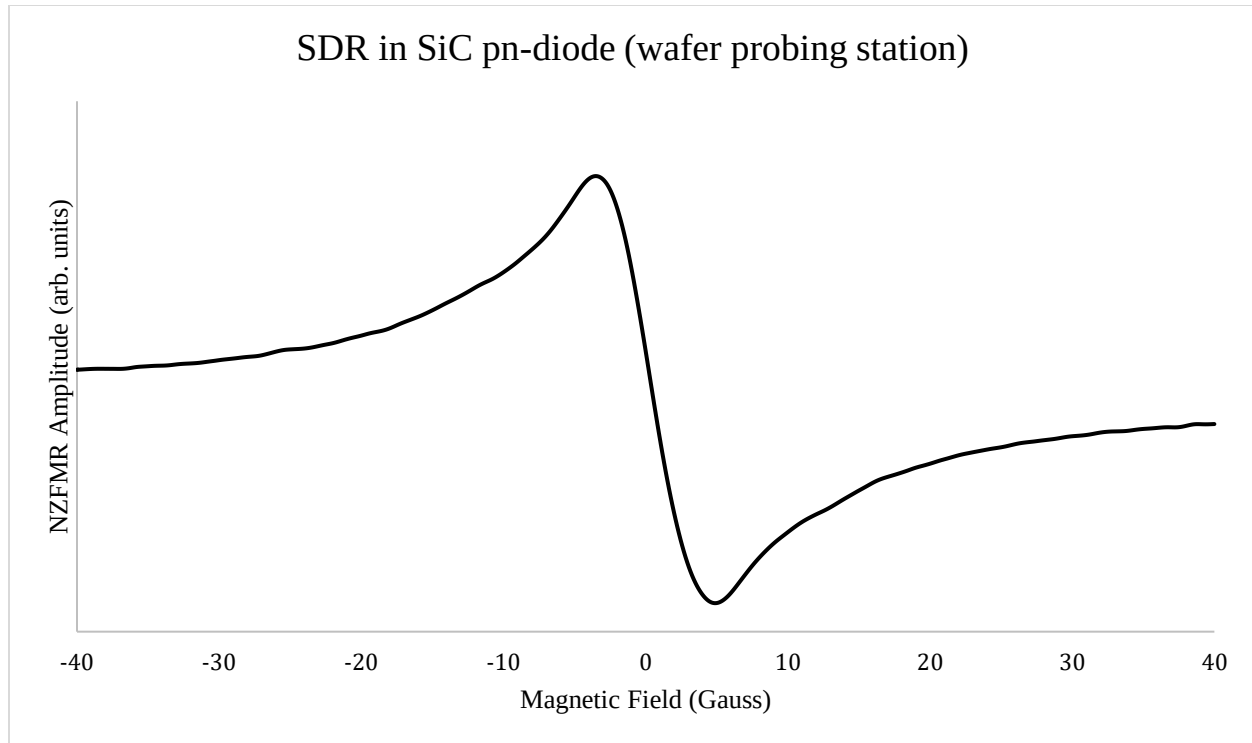


Figure B.1: SDTAT NZFMR spectrum in SiC pn-diode utilizing wafer probing station.

A 5nm a-SiC:H sample was studied using the NZFMR spectrometer incorporated into the wafer probing station. Instead of dicing the sample and mounting it to a PCB T with wire bonds for electrical connection, the sample was placed directly onto a conductive sheet of aluminum foil. The sample on the conductive sheet is shown in Figure B.2. The electromagnet used to sweep and modulate the magnetic field was removed for this image.

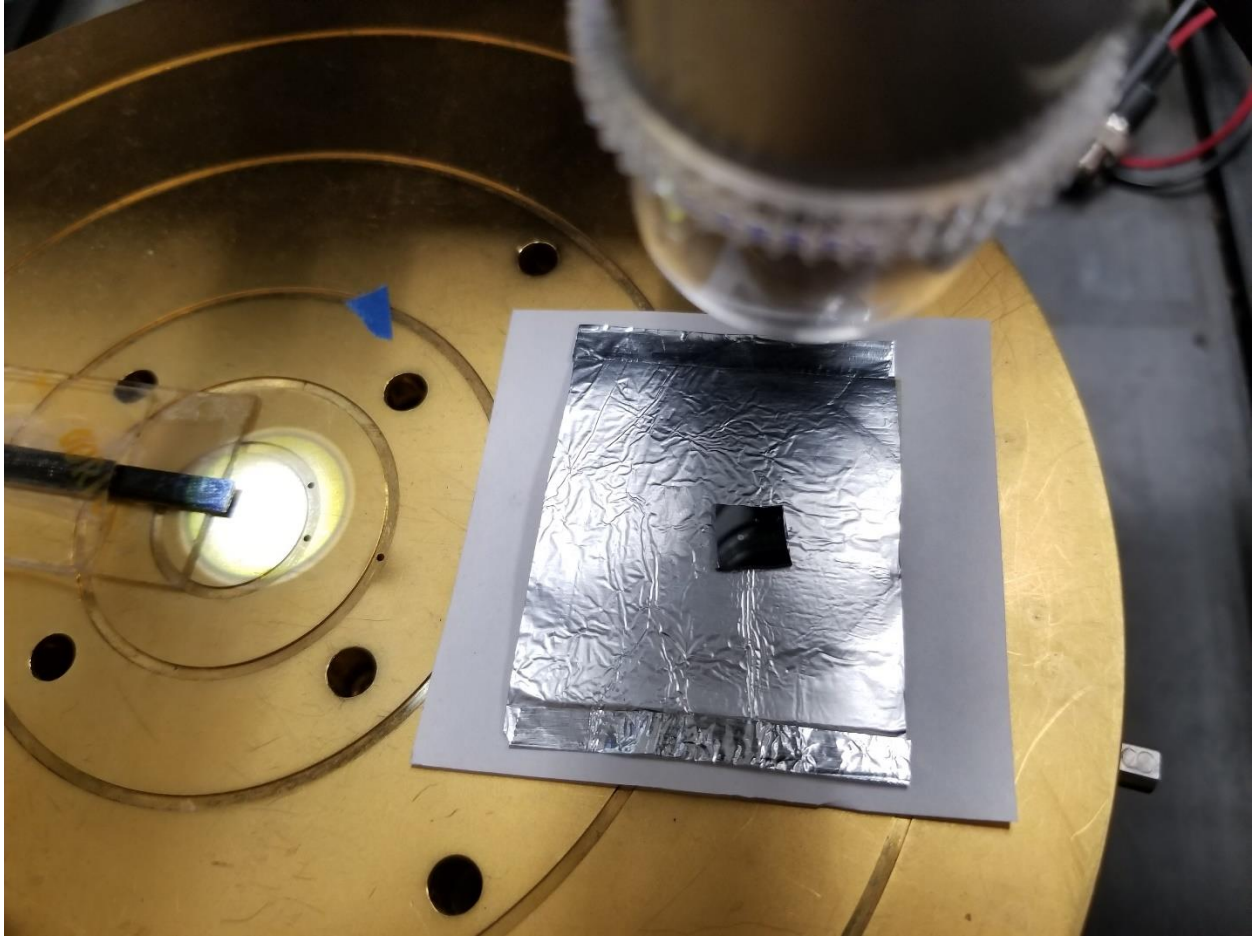


Figure B.2: 5nm a-SiC:H sample on conductive sheet with electromagnet and probe contacts removed.

Once the sample was placed onto the conductive aluminum sheet, the electromagnet was positioned such that the portion of the wafer sample under test was centered in the electromagnet. The coil for the electromagnet was wound around a 3-D printed blue PLA frame. Electrical contact was made to the sample by dropping one of the probes onto the top contact, and the other onto the conductive aluminum sheet the sample is resting on. This setup is shown in Figure B.3. Note that this portion of wafer was not diced for this measurement. There are two usable capacitors on this portion of wafer and the total sample size is approximately 1.5cm by 1.5cm, much larger than the samples used in the traditional EDMR and NZFMR spectrometers.

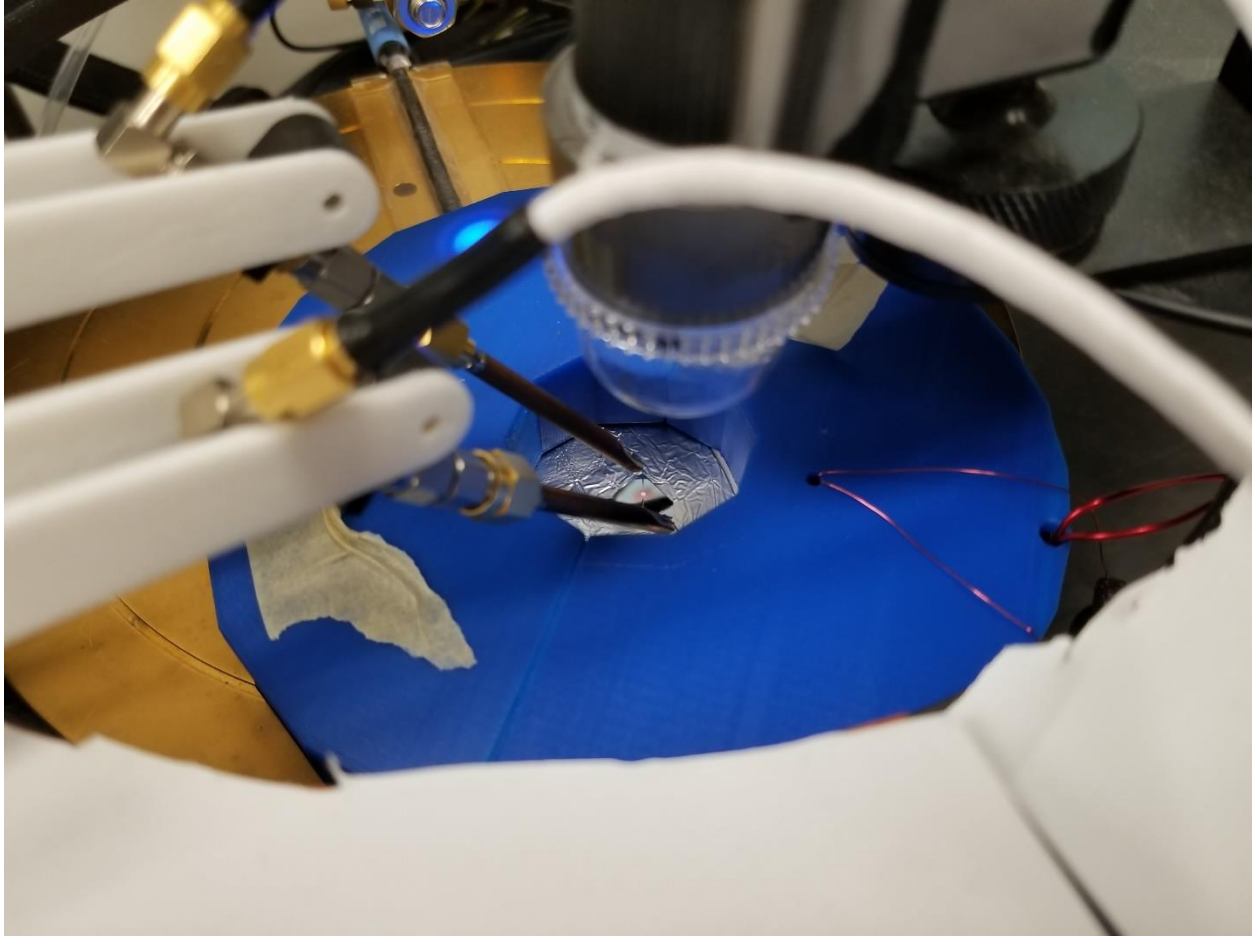


Figure B.3: 5nm a-SiC:H sample in NZFMR spectrometer electromagnet with electrical contact made using probes pictured. A digital microscope was used to ensure proper contact was made. Electrical contact to the back of the sample was made by contacting the conductive aluminum sheet.

The sample was biased at +3V and NZFMR was measured using a magnetic field modulation with an amplitude of 5 Gauss at 1000Hz. The measured spectrum has a very good signal to noise ratio and is shown in Figure B.4 alongside an NZFMR spectrum measured on a similar sample in a traditional spectrometer in which the sample had to be diced and mounted onto a PCB T. The conductive aluminum sheet was later replaced with a conductive copper sheet in order to reduce the series resistance of the measurement. The spectra are very similar

and the slight discrepancies can most likely be attributed to slight non-uniformities that arise due to the small nature of the single coil electromagnet used in wafer level system.

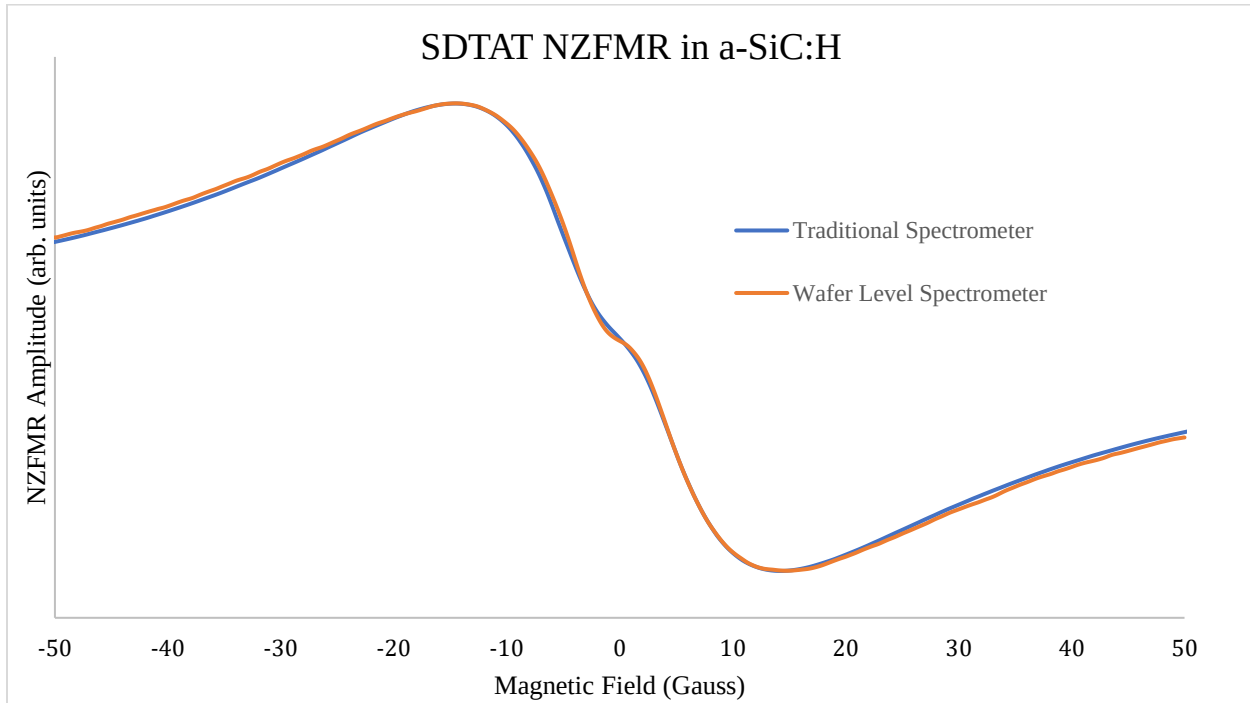


Figure B.4: SDTAT NZFMR spectra in wafer level and traditional spectrometers. The measurements were made under an applied bias of +3V using a magnetic field modulation of 5 Gauss at 1kHz.

This wafer level system looks promising for future work allowing more samples to be studied in a non-destructive manner. The faster data acquisition and higher throughput will allow for more data to be acquired in order to inform the development of computation models for NZFMR and potentially EDMR, as the system could easily be modified to include an RF resonator which would provide the B_1 oscillating magnetic field needed for EDMR. Implementing these spectroscopic techniques into wafer probing stations would allow for these very powerful techniques to be applied to the study of more samples with different materials, and devices of different architectures in both academic and industrial research settings.

APPENDIX C

SDTAT NZFMR MATLAB CODES

Primary Code Used to Simulate Spectra

This code was written by me, utilizing functions written by Elias Frantz for the SDTAT NZFMR modified two site model discussed in Chapter 3. For best results, convert file into MATLAB Live Script.

```
%% SDTAT NZFMR Simulation MATLAB Code

%{
This section of code was written by Colin McKay
This code utilizes simulation functions written by Elias Frantz
%}
%% Initial Variables, Run First

% initialization
clear
clc

% -100 Gauss to 100 Gauss (10mT =100 Gauss
Field= linspace(-10,10,4096)'; %Units: [mT]

% g value of free electron
g= 2.0023; %Units: []

% These are the variables used, the default values are given here
% HF1= 1; %Units: [mT] %Hyperfine interaction strength at Site 1
% HF2= 1; %Units: [mT] %Hyperfine interaction strength at Site 2
% r_ab= 1; %Units: [GHz]
% r_ae= 0.1; %Units: [GHz]
% r_ea= 1; %Units: [GHz]

f=0;
font='Cambria Math'; % changes font on plots

% Changes title text on plots
% will take form:
% 'Simulated NZFMR Spectra (Vary 'variable')
TextString='Simulated NZFMR Spectra (Vary ';

% adds 'Normalized ' to the start of the titles for normalized spectra
Normalized='Normalized ';

normalizedYlim=1.2; % y axis limits on normalized data
narrowXlim=20; % x axis limits on narrow plots when needed
```

```

%% Changing r_ab

% Variable Initialization
Variable='r_{\alpha\beta}';
HF1= 1; %Units: [mT] %Hyperfine interaction strength at Site 1
HF2= 1; %Units: [mT] %Hyperfine interaction strength at Site 2
r_ab= 1; %Units: [GHz]
r_ae= 0.1; %Units: [GHz]
r_ea= 1; %Units: [GHz]

% Variable values, define in any way you want
x=.4:.4:2;

% Calculates and plots the raw spectra with offsets
f=f+1;
figure (f)
clf
hold on
var=" ";
for i=1:length(x)
    r_ab= x(i);
    I_ab(:,i)= SpectraProducer(Field,g,HF1,HF2,r_ab,r_ae,r_ea);
    if i==1
        I_aboffset(:,1)=I_ab(:,1);
    elseif i>1
        offset=(min(I_aboffset(:,i-1))-max(I_ab(:,i))-.001); % tries to
optimize offsets but this is easily changeable
        I_aboffset(:,i)=I_ab(:,i) + offset;
    else
    end
    plot(Field.*10,I_aboffset(:,i))
    text=string(num2str(x(i)));
    var(i)=text;
end
legend(var, 'FontName',font)
legend boxoff
titletext=[TextString, Variable, ' '];
title(titletext, 'FontName',font)
xlabel('Magnetic Field (Gauss)', 'FontName',font)
ylabel('Simulated NZFMR Amplitude (arb. units)', 'FontName',font)
yticks('')
% This one works on its own
% ytop=;
% ybot=;
% ylim([-1*normalizedYlim normalizedYlim])

% Calculates normalized data, both Broad and Inflection
% Normalizes to Broad line
f=f+1;
figure (f)
clf
hold on
for i=1:length(x)
    [pks,locs]=findpeaks(I_ab(:,i));

```

```

        scale1_ab(:,i)=I_ab(:,i)./pks(1); % Scales to broad signal, same as in
later portions of code
        scale2_ab(:,i)=I_ab(:,i)./pks(2); % Scales to narrow signal, same as in
later portions of code
        plot(Field.*10,scale1_ab(:,i))
end
legend(var,'FontName',font)
legend boxoff
titletext=['a) ', Normalized, TextString, Variable, ' '];
title(titletext,'FontName',font)
xlabel('Magnetic Field (Gauss)','FontName',font)
ylabel('Simulated NZFMR Amplitude (arb. units)','FontName',font)
yticks('')
ylim([-1*normalizedYlim normalizedYlim])
annotation('arrow',[0.3732 0.4036],[0.2929 0.8119])
annotation('textarrow',[0.375 0.5661],[0.281 0.1929],'String','Normalized
Peaks','FontName',font)

% Normalizes to Narrow line
f=f+1;
figure (f)
clf
hold on
for i=1:length(x)
    plot(Field.*10,scale2_ab(:,i))
end
legend(var,'FontName',font)
legend boxoff
titletext=['b) ', Normalized, TextString, Variable, ' '];
title(titletext,'FontName',font)
xlabel('Magnetic Field (Gauss)','FontName',font)
ylabel('Simulated NZFMR Amplitude (arb. units)','FontName',font)
yticks('')
ylim([-1*normalizedYlim normalizedYlim])
annotation('arrow',[0.4232 0.5125],[0.3167 0.8214])
annotation('textarrow',[0.4268 0.4893],[0.2786 0.1976],'String','Normalized
Peaks','FontName',font)

%% Changing r_ae

% Variable Initialization
Variable='r_{\alpha e}';
HF1= 1; %Units: [mT] %Hyperfine interaction strength at Site 1
HF2= 1; %Units: [mT] %Hyperfine interaction strength at Site 2
r_ab= 1; %Units: [GHz]
r_ae= 0.1; %Units: [GHz]
r_ea= 1; %Units: [GHz]

% Variable values, define in any way you want
x=.04:.04:.2;

% Calculates and plots the raw spectra with offsets
f=f+1;
figure (f)
clf
hold on

```

```

var=" ";
for i=1:length(x)
    r_ae= x(i);
    I_ae(:,i)= SpectraProducer(Field,g,HF1,HF2,r_ab,r_ae,r_ea);
    if i==1
        I_aeoffset(:,1)=I_ae(:,1);
    elseif i>1
        offset=(min(I_aeoffset(:,i-1))-1.1*max(I_ae(:,i))); % tries to
optimize offsets but this is easily changeable
        I_aeoffset(:,i)=I_ae(:,i) + offset;
    else
    end
    plot(Field.*10,I_aeoffset(:,i))
    text=string(num2str(x(i)));
    var(i)=text;
end
legend(var, 'FontName', font)
legend boxoff
titledtext=[TextString, Variable, ' '];
title(titledtext, 'FontName', font)
xlabel('Magnetic Field (Gauss)', 'FontName', font)
ylabel ('Simulated NZFMR Amplitude (arb. units)', 'FontName', font)
yticks('')
ytop=max(I_aeoffset(:,1))+.003;
ybot=min(I_aeoffset(:,end))-.001;
ylim([ybot ytop])

% Calculates normalized data, both Broad and Inflection
% Normalizes to Broad line
f=f+1;
figure (f)
clf
hold on
for i=1:length(x)
    [pks,locs]=findpeaks(I_ae(:,i));
    scale1_ae(:,i)=I_ae(:,i)./pks(1);
    scale2_ae(:,i)=I_ae(:,i)./pks(2);
    plot(Field.*10,scale1_ae(:,i))
end
legend(var, 'FontName', font)
legend boxoff
titledtext=['a) ', Normalized, TextString, Variable, ' '];
title(titledtext, 'FontName', font)
xlabel('Magnetic Field (Gauss)')
ylabel ('Simulated NZFMR Amplitude (arb. units)', 'FontName', font)
yticks('')
ylim([-1*normalizedYlim normalizedYlim])
annotation('arrow',[0.3946 0.4125],[0.3905 0.8357])
annotation('textarrow',[0.4071 0.5982],[0.3476 0.1881], 'String', 'Normalized
Peaks', 'FontName', font)

% Normalizes to Narrow line
f=f+1;
figure (f)
clf
hold on

```

```

for i=1:length(x)
    plot(Field.*10,scale2_ae(:,i))
end
legend(var,'FontName',font)
legend boxoff
titletext=['b) ', Normalized, TextString, Variable, ' '];
title(titletext,'FontName',font)
xlabel('Magnetic Field (Gauss)','FontName',font)
ylabel ('Simulated NZFMR Amplitude (arb. units)','FontName',font)
yticks('')
ylim([-1*normalizedYlim normalizedYlim])
annotation('arrow',[0.375 0.5125],[0.4024 0.8357])
annotation('textarrow',[0.3857 0.4911],[0.3643 0.1929],'String','Normalized
Peaks','FontName',font)

%% Changing r_ea

% Variable Initialization
Variable='r_{e\alpha}';
HF1= 1; %Units: [mT] %Hyperfine interaction strength at Site 1
HF2= 1; %Units: [mT] %Hyperfine interaction strength at Site 2
r_ab= 1; %Units: [GHz]
r_ae= 0.1; %Units: [GHz]
r_ea= 1; %Units: [GHz]

% Variable values, define in any way you want
x=0.4:.4:2;
r_ab=.3;
r_ae=.1;

% Calculates and plots the raw spectra with offsets
f=f+1;
figure (f)
clf
hold on
var=" ";
for i=1:length(x)
    r_ea= x(i);
    I_ea(:,i)= SpectraProducer(Field,g,HF1,HF2,r_ab,r_ae,r_ea);
    if i==1
        I_eaoffset(:,1)=I_ea(:,1);
    elseif i>1
        offset=(min(I_eaoffset(:,i-1))-(1.1*max(I_ea(:,i))))); % tries to
optimize offsets but this is easily changeable
        I_eaoffset(:,i)=I_ea(:,i) + offset;
    else
    end
    plot(Field.*10,I_eaoffset(:,i))
    text=string(num2str(x(i)));
    var(i)=text;
end
legend(var,'FontName',font)
legend boxoff
titletext=[TextString, Variable, ' '];
title(titletext,'FontName',font)
xlabel('Magnetic Field (Gauss)','FontName',font)

```



```

ylabel ('Simulated NZFMR Amplitude (arb. units)', 'FontName', font)
yticks('')
ytop=max(I_eaoffset(:,1))+.07;
ybot=min(I_eaoffset(:,end))-.01;
ylim([ybot ytop])

% Calculates normalized data, both Broad and Inflection
% Normalizes to Broad line
f=f+1;
figure (f)
clf
hold on
for i=1:length(x)
    [pks,locs]=findpeaks(I_ea(:,i));
    scale1_ea(:,i)=I_ea(:,i)./pks(1);
    if i==1
        scale1_eaOffset(:,1)=scale1_ea(:,1);
    elseif i>1
        offset=-.03*(i-1);
        scale1_eaOffset(:,i)=scale1_ea(:,i) + offset;
    else
    end
    %scale2_ea(:,i)=I_ea(:,i)./pks(2); %perfect match, not needed
    plot(Field.*10,scale1_eaOffset(:,i))
end
legend(var, 'FontName', font)
legend boxoff
titletext=[Normalized, TextString, Variable, ''];
title(titletext, 'FontName', font)
xlabel('Magnetic Field (Gauss)', 'FontName', font)
ylabel ('Simulated NZFMR Amplitude (arb. units)', 'FontName', font)
yticks('')
ylim([-1*normalizedYlim normalizedYlim])

%% Changing HF1

% Variable Initialization
Variable='HF_{\alpha}';
HF1= 1; %Units: [mT] %Hyperfine interaction strength at Site 1
HF2= 0; %Units: [mT] %Hyperfine interaction strength at Site 2
r_ab= 1; %Units: [GHz]
r_ae= 0.1; %Units: [GHz]
r_ea= 1; %Units: [GHz]

% Variable values, define in any way you want
x=.4:.4:2;

% Calculates and plots the raw spectra with offsets
f=f+1;
figure (f)
clf
hold on
var=" ";
for i=1:length(x)
    HF1= x(i);
    I_HF1(:,i)= SpectraProducer(Field,g,HF1,HF2,r_ab,r_ae,r_ea);

```

```

    if i==1
        I_HF1offset(:,1)=I_HF1(:,1);
    elseif i>1
        offset=(min(I_HF1offset(:,i-1))-(1.1*max(I_HF1(:,i)))); % tries to
optimize offsets but this is easily changeable
        I_HF1offset(:,i)=I_HF1(:,i) + offset;
    else
    end
    plot(Field.*10,I_HF1offset(:,i))
    text=string(num2str(x(i)));
    var(i)=text;
end
legend(var, 'FontName', font)
legend boxoff
titletext=[TextString, Variable, ' '];
title(titletext, 'FontName', font)
xlabel('Magnetic Field (Gauss)', 'FontName', font)
ylabel ('Simulated NZFMR Amplitude (arb. units)', 'FontName', font)
yticks('')
ytop=max(I_HF1offset(:,1))+.0025;
ybot=min(I_HF1offset(:,end))-.0005;
ylim([ybot ytop])

% Calculates normalized data, both Broad and Inflection
% Normalizes to Broad line
f=f+1;
figure (f)
clf
hold on
for i=1:length(x)
    [pks,locs]=findpeaks(I_HF1(:,i));
    scale1_HF1(:,i)=I_HF1(:,i)./pks(1);
    try
        scale2_HF1(:,i)=I_HF1(:,i)./pks(2);
    catch
        disp('Only 1 peak')
        scale2_HF1(:,i)=I_HF1(:,i)./pks(1);
    end
    plot(Field.*10,scale1_HF1(:,i))
end
legend(var)
legend boxoff
titletext=['a) ', Normalized, TextString, Variable, ' '];
title(titletext, 'FontName', font)
xlabel('Magnetic Field (Gauss)', 'FontName', font)
ylabel ('Simulated NZFMR Amplitude (arb. units)', 'FontName', font)
yticks('')
ylim([-1*normalizedYlim normalizedYlim])
annotation('arrow',[0.3911 0.4125],[0.3071 0.7857])
annotation('textarrow',[0.4 0.5643],[0.2905 0.1881],'String','Normalized
Peaks','FontName', font)

% Normalizes to Narrow line
f=f+1;
figure (f)
clf

```

```

hold on
for i=2:length(x)
    plot(Field.*10,scale2_HF1(:,i))
end
legend(var(2:end))
legend boxoff
titletext=['b) ', Normalized, TextString, Variable, ' '];
title(titletext, 'FontName', font)
xlabel('Magnetic Field (Gauss)', 'FontName', font)
ylabel('Simulated NZFMR Amplitude (arb. units)', 'FontName', font)
yticks('')
xlim([-1*narrowXlim narrowXlim])
ylim([-1*normalizedYlim normalizedYlim])
annotation('arrow',[0.3339 0.6036],[0.1881 0.8071])
annotation('textarrow',[0.3357 0.4268],[0.169 0.1643], 'String', 'Normalized
Peaks', 'FontName', font)

%% Changing HF2

% Variable Initialization
Variable='HF_{\beta}';
HF1= 1; %Units: [mT] %Hyperfine interaction strength at Site 1
HF2= 1; %Units: [mT] %Hyperfine interaction strength at Site 2
r_ab= 1; %Units: [GHz]
r_ae= 0.1; %Units: [GHz]
r_ea= 1; %Units: [GHz]

% Variable values, define in any way you want
x=.4:.4:2;

% Calculates and plots the raw spectra with offsets
f=f+1;
figure (f)
clf
hold on
var=" ";
for i=1:length(x)
    HF2= x(i);
    I_HF2(:,i)= SpectraProducer(Field,g,HF1,HF2,r_ab,r_ae,r_ea);
    if i==1
        I_HF2offset(:,1)=I_HF2(:,1);
    elseif i>1
        offset=(min(I_HF2offset(:,i-1))-(1.1*max(I_HF2(:,i))))); % tries to
optimize offsets but this is easily changeable
        I_HF2offset(:,i)=I_HF2(:,i) + offset;
    else
        end
    plot(Field.*10,I_HF2offset(:,i))
    text=string(num2str(x(i)));
    var(i)=text;
end
legend(var, 'FontName', font)
legend boxoff
titletext=[TextString, Variable, ' '];
title(titletext, 'FontName', font)
xlabel('Magnetic Field (Gauss)', 'FontName', font)

```

```

ylabel ('Simulated NZFMR Amplitude (arb. units)', 'FontName', font)
yticks('')
ytop=max(I_HF2offset(:,1))+.003;
ybot=min(I_HF2offset(:,end))-.0005;
ylim([ybot ytop])

% Calculates normalized data, both Broad and Inflection
% Normalizes to Broad line
f=f+1;
figure (f)
clf
hold on
for i=1:length(x)
    [pks,locs]=findpeaks(I_HF2(:,i));
    scale1_HF2(:,i)=I_HF2(:,i)./pks(1);
    try
        scale2_HF2(:,i)=I_HF2(:,i)./pks(2);
    catch disp('Only 1 peak')
    end
    plot(Field.*10,scale1_HF2(:,i))
end
legend(var, 'FontName', font)
legend boxoff
titletext=['a) ', Normalized, TextString, Variable, ' ');
title(titletext, 'FontName', font)
xlabel('Magnetic Field (Gauss)', 'FontName', font)
ylabel ('Simulated NZFMR Amplitude (arb. units)', 'FontName', font)
yticks('')
ylim([-1*normalizedYlim normalizedYlim])
annotation('arrow',[0.35 0.4107],[0.2905 0.8071])
annotation('textarrow',[0.3536 0.5821],[0.2714 0.1881], 'String', 'Normalized
Peaks', 'FontName', font)

% Normalizes to Narrow line
f=f+1;
figure (f)
clf
hold on
for i=1:length(x)
    plot(Field.*10,scale2_HF2(:,i))
end
legend(var, 'FontName', font)
legend boxoff
titletext=['b) ', Normalized, TextString, Variable, ' ');
title(titletext, 'FontName', font)
xlabel('Magnetic Field (Gauss)', 'FontName', font)
ylabel ('Simulated NZFMR Amplitude (arb. units)', 'FontName', font)
yticks('')
xlim([-1*narrowXlim narrowXlim])
ylim([-1*normalizedYlim normalizedYlim])
annotation('arrow',[0.3464 0.5804],[0.3405 0.8048])
annotation('textarrow',[0.3643 0.4482],[0.2976 0.2119], 'String', 'Normalized
Peaks', 'FontName', font)

%% Changing HF1 and HF2 together

```

```

% Variable Initialization
Variable='HF_{\alpha} and HF_{\beta}';
HF1= 1; %Units: [mT] %Hyperfine interaction strength at Site 1
HF2= 1; %Units: [mT] %Hyperfine interaction strength at Site 2
r_ab= 1; %Units: [GHz]
r_ae= 0.1; %Units: [GHz]
r_ea= 1; %Units: [GHz]

% Variable values, define in any way you want
x=.4:.4:2;

% Calculates and plots the raw spectra with offsets
f=f+1;
figure (f)
clf
hold on
var=" ";
for i=1:length(x)
    HF1= x(i);
    HF2= x(i);
    I_HF12(:,i)= SpectraProducer(Field,g,HF1,HF2,r_ab,r_ae,r_ea);
    if i==1
        I_HF12offset(:,1)=I_HF12(:,1);
    elseif i>1
        offset=(min(I_HF12offset(:,i-1))-(1.1*max(I_HF12(:,i)))); % tries to
optimize offsets but this is easily changeable
        I_HF12offset(:,i)=I_HF12(:,i) + offset;
    else
    end
    plot(Field.*10,I_HF12offset(:,i))
    text=string(num2str(x(i)));
    var(i)=text;
end
legend(var, 'FontName', font)
legend boxoff
titletext=[TextString, Variable, ' '];
title(titletext, 'FontName', font)
xlabel('Magnetic Field (Gauss)', 'FontName', font)
ylabel('Simulated NZFMR Amplitude (arb. units)', 'FontName', font)
yticks('')
ytop=max(I_HF12offset(:,1))+.005;
ybot=min(I_HF12offset(:,end))-.0005;
ylim([ybot ytop])

% Normalizes to Broad line
f=f+1;
figure (f)
clf
hold on
for i=1:length(x)
    [pks,locs]=findpeaks(I_HF12(:,i));
    scale1_HF12(:,i)=I_HF12(:,i)./pks(1);
    scale2_HF12(:,i)=I_HF12(:,i)./pks(2);
    plot(Field.*10,scale1_HF12(:,i))
end

```

```

legend(var, 'FontName', font)
legend boxoff
titletext=['a] ', Normalized, TextString, Variable, ')];
title(titletext, 'FontName', font)
xlabel('Magnetic Field (Gauss)', 'FontName', font)
ylabel ('Simulated NZFMR Amplitude (arb. units)', 'FontName', font)
yticks('')
ylim([-1*normalizedYlim normalizedYlim])
annotation('arrow', [0.3768 0.3893], [0.269 0.8071])
annotation('textarrow', [0.3875 0.5643], [0.2476 0.2], 'String', 'Normalized
Peaks', 'FontName', font)

% Normalizes to Narrow line
f=f+1;
figure (f)
clf
hold on
for i=1:length(x)
    plot(Field.*10, scale2_HF12(:, i))
end
legend(var, 'FontName', font)
legend boxoff
titletext=['b] ', Normalized, TextString, Variable, ')];
title(titletext, 'FontName', font)
xlabel('Magnetic Field (Gauss)', 'FontName', font)
ylabel ('Simulated NZFMR Amplitude (arb. units)', 'FontName', font)
yticks('')
xlim([-1*narrowXlim narrowXlim])
ylim([-1*normalizedYlim normalizedYlim])
annotation('arrow', [0.3554 0.5732], [0.2214 0.8167])
annotation('textarrow', [0.3554 0.4643], [0.2071 0.2], 'String', 'Normalized
Peaks', 'FontName', font)

```

Functions Needed to Run Primary Code

The primary code used to simulate the SDTAT NZFMR spectra in this study calls upon four functions written by Elias Frantz for the SDTAT NZFMR simulation code referenced in chapter 3. The four function names are “DiracDelta”, “Pauli”, “SpectraProducer”, and “SpinSystem”. To use these codes together, save each function into its own .m file named with the exact case sensitive characters inside the quotations marks in the section heading.

“DiracDelta” Function

```

function [Out] = DiracDelta(m,n)
%DiracDelta is a numeric equivalent of the MATLAB built in: kroneckerDelta.
% [Out] = DiracDelta(m,n)
if m==n

```

```

        Out= 1;
else
        Out= 0;
end
end

```

“Pauli” Function

```

function [Out] = Pauli(Spin,xyz)
%Pauli2 creates a Pauli spin matrix (vector) given a given Spin
(S=1/2,S=1,S=3/2,...) and
%the vector direction (x,y, or z).
% [Out] = Pauli2(Spin,xyz)
iterations= 2*Spin+1;
m= linspace(-Spin,Spin,iterations);
Out= zeros(iterations,iterations);
switch xyz
    case 1
        for j= 1:iterations
            for k= 1:iterations
                Out(j,k)= (DiracDelta(j,k+1) +
DiracDelta(j+1,k))*0.5*sqrt((Spin*(Spin+1)-m(j)*m(k)));
            end
        end
    case 2
        for j= 1:iterations
            for k= 1:iterations
                Out(j,k)= (DiracDelta(j,k+1) -
DiracDelta(j+1,k))*0.5i*sqrt((Spin*(Spin+1)-m(j)*m(k)));
            end
        end
    case 3
        for j= 1:iterations
            for k= 1:iterations
                Out(j,k)= -DiracDelta(j,k)*m(j);
            end
        end
end
end
end

```

“SpectraProducer” Function

```

function [I] = SpectraProducer(Field,g,HF1,HF2,r_ab,r_ae,r_ea)
%PassSpectraFit is passed into fminsearch to extract the
%spectroscopic components of the NZFMR Spectra.
% [I] = SpectraProducer(Field,NuclearSpin,g,HyperfineValue,ks,kd,Gen,I0)
%% Conditions for fminsearch
HF1= abs(HF1);
HF2= abs(HF2);
ks1= abs(r_ab);
kd1= abs(r_ae);
Gen1= abs(r_ea);

```

```

%% Spin System
[sx,sy,sz]= SpinSystem([1/2,1/2,1/2,1/2]);
%% Spin Operators
S_tot= sz{1}+sz{2};
% Electron-Electron Spin Product Projection Operators
S1S2= sx{1}*sx{2} + sy{1}*sy{2} + sz{1}*sz{2};
% Electron-Nuclear Spin Product Projection Operators
IS1= sx{3}*sx{1} + sy{3}*sy{1} + sz{3}*sz{1};
IS2= sx{4}*sx{2} + sy{4}*sy{2} + sz{4}*sz{2};
% Singlet & Triplet Projection Operators
Ident= eye(size(S1S2));
Ps= (1/4)*( Ident-4*S1S2);
Pt= (1/4)*(3*Ident+4*S1S2);
%% Constants
uB= 5.788e-8; %eV/mT
h_bar= 6.582E-7; %eV*ns
%% Hamiltonians + Solve
Hhf1= g*uB*HF1*IS1 + g*uB*HF2*IS2;
I1= zeros(length(Field),1);
G1= (1/16)*Gen1*Ident;
for k= 1:length(Field)
    B0= Field(k)-0.037851;
    H01= g*uB*B0*(S_tot);
    H_NZFMR1= H01 + Hhf1;
    A= -( (1i)/h_bar)*H_NZFMR1-(ks1/2)*Ps-(kd1/2)*Pt;
    B= ((1i)/h_bar)*H_NZFMR1-(ks1/2)*Ps-(kd1/2)*Pt;
    P1= lyap(A,B,G1);
    I1(k,1)= (kd1/ks1).*real(trace(Ps*P1));
end
I1= I1-I1(1);
I= gradient(I1)./gradient(Field);
end

```

“SpinSystem” Function

```

function [sx,sy,sz] = SpinSystem(spins)
%SpinSystem creates cell arrays of sx, sy, and sz in the appropriate spin
%bases given a vector of nuclear spins.
% [sx,sy,sz] = SpinSystem(spins)
iterations= length(spins);

spinMat= [];
for l= 1:iterations
    spinMat= vertcat(spinMat,spins);
end

spinCellx{iterations,iterations}= [];
spinCelly{iterations,iterations}= [];
spinCellz{iterations,iterations}= [];

for m= 1:iterations
    for n= 1:iterations
        if n==m
            spinCellx{m,n}= Pauli(spinMat(m,n),1);

```



```

        spinCelly{m,n}= Pauli (spinMat (m,n) ,2);
        spinCellz{m,n}= Pauli (spinMat (m,n) ,3);
    else
        spinCellx{m,n}= eye (size (Pauli (spinMat (m,n) ,1)));
        spinCelly{m,n}= eye (size (Pauli (spinMat (m,n) ,2)));
        spinCellz{m,n}= eye (size (Pauli (spinMat (m,n) ,3)));
    end
end
end

sx{iterations,1}= [];
sy{iterations,1}= [];
sz{iterations,1}= [];

for j= 1:iterations
    x= spinCellx{j,1};
    y= spinCelly{j,1};
    z= spinCellz{j,1};
    for k= 2:iterations
        x= kron(x,spinCellx{j,k});
        y= kron(y,spinCelly{j,k});
        z= kron(z,spinCellz{j,k});
    end
    sx{j,1}= x;
    sy{j,1}= y;
    sz{j,1}= z;
end
end

```

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